

Vladimir Petrov

Talented Thinking

TRIZ



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ISBN 978-3-031-15504-8 ISBN 978-3-031-15505-5 (eBook)
<https://doi.org/10.1007/978-3-031-15505-5>

Translation from the Russian language edition: “Петров Владимир. Талантливое мышление: ТРИЗ” by Vladimir Petrov, © Ridero 2018. Published by Владимир Петров. [б. м.]: Издательские решения. All Rights Reserved.

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Preface

This is the textbook on talented thinking, which is also called creative, strong or inventive thinking.

It is significantly expanded and in-depth, which makes it possible not only to deepen knowledge on this topic, but also to acquire inventive thinking skills.

The book contains many examples and problems, as well as assignments for independent work. An analysis of most of the tasks is given in the appendix.

Inventive thinking consists of the following components: system thinking, evolutionary thinking, thinking through contradictions, thinking with the use of resources, thinking with the use of models and development of creative thinking.

This book describes all components of this thinking. The main attention is given to improvement of skills of inventive thinking.

The book is intended for a broad circle of readers from children to grown-ups of various specialties (businessmen, politicians, managers, designers, TRIZ trainers, etc.). It will be useful for all those, who wants to quickly get new ideas and have a developed inventive thinking.

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Acknowledgements I am deeply thankful to Genrich Altshuller, author of theory of inventive problem solving—TRIZ, my teacher, colleague and friend for his having created this fascinating theory. I am indebted to him for that unforgettable time, for his having changed my life and having made it more various and more interesting. Some of the materials from this book were discussed with Genrich Altshuller.

Introduction

This textbook consists of an introduction, seven chapters, conclusion and appendices.

“Introduction” describes the message and the structure of the book.

Chapter 1 describes the qualities and methods of development of inventive thinking.

Chapter 2 is dedicated to system thinking.

Chapter 3 considers two components of evolutionary thinking: identification of regularities of evolution and the use of trends of system evolution.

Chapter 4 describes thinking through contradictions, which mostly uses the logic of solving non-standard problems.

Chapter 5 is dedicated to improvement of skills for using resources.

Chapter 6 presents methods of modeling and the use of them for solving problems.

Chapter 7 teaches complex use of all components of inventive thinking.

“Conclusion” contains recommendations on improvement of skills for inventive thinking.

The Appendices contain author’s analysis of problems.

The textbook is written in such a sequence, in which it is recommended to study it.

Theoretical part is illustrated with a large number of examples, problems and graphic material (about 200 examples and problems and about 80 illustrations). Tasks for independent work are quoted at the end of each chapter.

The book is intended for a broad circle of readers. It could also be useful for businessmen, students, postgraduates, university professors, engineers, inventors, scientists and persons who solve creative problems and teach TRIZ to students of senior forms.

I wish you success, DEAR READER!

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Chapter 1

The Notion of Talented Thinking



Always try first to think and then better keep silence.

Isaak Asimov

"I, robot"

The thought needs regularizing.

Emmanuel Mounier

To think – that is the most difficult work and therefore, very few people take it up.

Henry Ford

One has to develop one's mind by reading much, not by reading many authors.

Marcus Fabius Quintilian

The mind is a capability, giving us the principles of a priori knowledge.

Immanuel Kant

1.1 Qualities of Talented Thinking

This kind of thinking is also called Strong, Creative, Inventive or TRIZ thinking.

It includes the following components:

1. **System thinking.**
2. **Evolutionary thinking.**
3. **Thinking through contradictions.**
4. **Thinking through resources (resource thinking).**
5. **Thinking through models.**
6. **Development of creative imagination (DCI).**

System thinking is based on system approach.

System thinking is understood by the author as an ability to see the components of the system, its *elements, hierarchy of the system, mutual influence* of system elements, system with supersystem and environment, taking into account *changes in time and according to condition, historical evolution*, the chain of *goal statement, identification of needs*, creation of *functional model*, tree of *operation principles, system level*. System thinking is considered in Chap. 2.

Evolutionary thinking (Chap. 3) has two constituents:

- (a) *Identification of regularities of evolution* (trends) in any phenomena, for example, as it is done in tests on logics or IQ (for example, such sequence: triangle, square, pentagon... what next?).
- (b) *Use of regularities of artificial systems evolution* for the evolution of a particular system.

Thinking through contradictions presupposes identification and resolution of contradictions (Chap. 4).

Resource thinking is an ability to identify and use resources (Chap. 5).

Modeling (Chap. 6)—is an ability to solve problems through modeling. Mental modeling is often used. In TRIZ modeling is effected with the aid of Su-Fields, smart little men. Also, there is component structural and functional modeling. Apart from different kinds of mental modeling it is desirable to embody simplest models of cardboard, plasticine, etc. It is desirable to use different kinds of mathematical and computer modeling.

DCI is aimed at *management of psychological inertia*. All principles and methods of TRIZ are used for development of creative imagination, the application of which individually or jointly will enable to significantly broaden the scope of creative imagination of the human. The materials of DCI are not going to be quoted in this book. They are described in greater detail in Refs. [1, 2].

1.2 Method of Talented Thinking Development

Talented thinking develops with the aid of constant application of each of the described kinds.

System thinking develops through the use of *system approach* (Chap. 2):

- Ability to see *hierarchy* of systems;
- *Interconnections and mutual influences between: individual parts of the system and the system, system and supersystem, system and environment, reverse influence;*
- Taking into account any *changes in time and according to condition*, caused by *influence and mutual influence;*
- Historical evolution;
- Statement of goals;
- Identification and forecasting of *needs*;
- Creation of *function model*;
- Identification of *operation principle* of the system;
- Creation of *structural and flow model*;
- Identification of *workability and competitive ability* of the system.

Evolutionary thinking develops through *identification of regularities* in different phenomena, systems, processes and sequences, as well as through the use of *trends of artificial systems evolution* (Chap. 3) for forecasting of evolution of these systems.

Thinking through contradictions develops through identification and resolution of contradictions (Chap. 4).

Resource thinking develops through identification and use of resources (Chap. 5).

Modeling develops through creation of mental, computer and material models for solving definite problems.

Creative thinking develops with the aid of special principles and methods of DCI, reading science fiction and evaluating science fiction texts.

1.3 Control Questions

1. Describe the qualities of inventive thinking.
2. What is system thinking?
3. What is evolutionary thinking?
4. What is thinking through contradictions?
5. What is resource thinking?
6. What is modeling?
7. How does creative thinking develop?

References

1. Genrich Altshuller, A. B. Seliutsky. 1980. Wings for Icarus: How to solve inventive problems.—Petrozavodsk: Karelia. 224 p. (in Russian).
2. Vladimir Petrov. 2018. Development of Creative Imagination: TRIZ/Vladimir Petrov. [s. l.]: Publishing solutions. 104 p. ISBN 978-5-4490-8547-4. (in Russian).

Chapter 2

System Approach



When you look into the abyss for long time, the abyss starts to look at you.

Friedrich Nietzsche

The biggest stupidity is to do the same thing and to hope for a different result.

A. Einstein

2.1 Non-system Approach

Let us quote some examples of non-system approach.

Example 2.1 The Parable about the Blind Men

An elephant was led in turn to several blind men. They were asked to describe, what it was like.

The first of the blind men touched the leg of the elephant and said that it was something round and massive, like a column.

Another touched the elephant's trunk and said that it is something flexible, like a snake.

The third man touched the tail and said that it was something thin, like a rope.

The fourth touched the side and said that it was like a wall.

This is a typical example of a non-system thinking.

Let us recollect the myth about Midas.

Example 2.2 King Midas

The king Midas with great respect hosted in his palace Silenus, the teacher of Dionysus, who fell behind Dionysus. Dionysus proposed that Midas could choose any gift as a prize.

Midas shouted:

- Oh, the great god Dionysus, make it so that everything that I will touch, would be converted into pure glittering gold!

Midas could never expect that the food and even his relatives would also be converted into gold.

Example 2.3 Fine

The driver was fined for crossing solid white line. The information about that was sent to the road police from the video camera.

It appeared that the solid white line was crossed by the shadow of the car, not by the car itself.

2.2 Main Notions of System Thinking

Very often, while solving problems or exploring a certain system, we are like these blind men. We consider only a small part of the problem or a part of the system, and this, as a rule, is insufficient. We even don't always know its constituents—subsystems, to say nothing of parts of these constituents—sub-subsystems, we don't see, where this system belongs. All this is a manifestation of absence of a system approach.

Below we shall quote the main notions and constituents of system approach.

2.2.1 *System Thinking*

System thinking is thinking, which uses *system approach* and is one of the elements of *inventive thinking*.

System approach is understood as analysis of the object as a whole multitude of elements with the entire assembly of interrelations and connections between them, i.e., the analysis of the object as a **system**.

The main object of consideration in a system approach is the **system**.

2.2.2 *System*

System (from the Latin “*systema*”, and from Greek “σύστημα”—“composed”, a whole composed of parts, combination) is a multitude of **elements**, which are interconnected and which interact, forming a single *whole*, which possesses **properties**, which are not characteristic of its constituents taken separately.

Such property is called **system effect** or **emergence**.

Emergence (from the English word “*Emergent*”—appearing, unexpectedly originating) in systems theory means the presence of special properties with the system, which are not characteristic of its subsystems and units, as well as the multitude of

elements, which are not interconnected by special system-forming bonds; impossibility to reduce the properties of the system to the sum of properties of its components; the synonym is “system effect”.¹

Example 2.4 Aircraft

An aircraft consists of wings, fuselage, motor, chassis, etc.

None of these elements possesses such property as the ability to fly. When united into a *single system*—an aircraft, the elements acquired a new property—to fly—**system effect**.

Example 2.5 Sentence (in language)

The sentence consists of *words* and *the method of creating a sentence—grammar*.

None of these elements possesses the property of expressing a thought. When united into a single system the words acquire a new feature—*thought, which is a manifestation of a system effect*.

The sentence is a *single whole*. It is autonomous and has its own regularities of evolution—development of grammar.

The interconnection of individual words is shown in a sentence as well as their properties, which are identified as related one to another.

Example 2.6 Telephone

The telephone consists of *microphone, earphone, keyboard, display, memory*, etc.

None of these elements possesses the property of being able to transfer sound at a distance. When combined into a single system—telephone, the elements acquired a new property—*being able to transfer sound at a distance—system effect*.

Example 2.7 Algorithm

An algorithm is a certain order of performing different operations, which leads to a definite result.

Algorithm consists of individual *steps*, which are performed in a certain *order*.

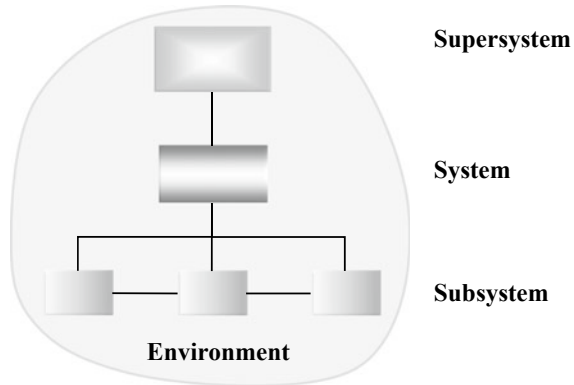
Each of the steps and the order of performing each of them will not lead to the necessary result. Combined into a single system they form an algorithm, which acquired a new property—*particular result—system effect*.

2.2.3 Hierarchy

Let us describe the hierarchy of the system (Fig. 2.1):

- *system* proper;
- its *subsystems*;
- *supersystem*;
- *external environment*.

¹ Emergence—material from Wikipedia (In Russian).

Fig. 2.1 System levels

It is possible to consider many levels of subsystems and supersystems. It is extremely important to know adjacent systems and external environment. Thus, system thinking should consider **hierarchical system levels**.

Subsystem—constituent parts of the system.

Supersystem—an object, which includes the system as a subsystem.

Hierarchy can have higher ranks, for example, super-supersystem, and lower ranks, for example, sub-subsystem.

Super-supersystem is an object, which includes a supersystem, while sub-subsystem is a number of elements, of which the subsystem consists. The number of ranks can be fairly high.

Example 2.8 Computer

System: personal computer.

Subsystems: system unit and input-and-output devices (for example, keyboard, mouse, monitor, printer, scanner, camera, etc.).

Sub-subsystems of the system unit are: processor, motherboard, image acquisition board, random access memory, hard disk, disk drive, sound card, network card, power unit, etc.

Supersystem: computer networks, etc.

Super-supersystem: world web, Internet.

External environment is the environment, in which the compute is located, for example, the room, air, etc.

Example 2.9 Telephone

System: telephone.

Subsystems: microphone and earphone, keyboard, display, memory, etc.

Sub-subsystems are elements, of which consist microphone, earphone, keyboard, display, memory, etc.

Supersystem—telephone exchange, telephone networks, etc.

Super-supersystem for ATS is regional and world telephone network.

External environment is most often the room and the air.

Example 2.10 Car

System—car.
Subsystems: wheels, engine, petrol tank, steering system, etc.
Sub-subsystems of the engine: piston and cylinder, piston rod, spark plug, valves, crankshaft, crankcase, etc.
Supersystem is traffic on roads to which relate: roads, filling stations, parking spaces, system of traffic circulation, garages, maintenance services, manufacturing plants, etc.
Supersystem: regional and world network of road movement.
External environment—open space and atmospheric phenomena.

Example 2.11 Tree

System—tree (Fig. 2.2).
Subsystems: tree trunk, crown and roots.
Sub-subsystems of crown—branches.

			Supersystem
			System
			Subsystem
			Sub-subsystem

Fig. 2.2 Hierarchy of a tree

The branches have their own subsystems: leaves and fruits.

The leaves have their subsystems: stalk, fibers, tissues of the leaf.

Supersystem is wood.

External environment for roots is soil; for tree trunk and crown—air and atmospheric phenomena.

2.2.4 Mutual Connections and Mutual Influence

However, the knowledge of these levels is not sufficient. It is necessary to take into account **the influence** of subsystems on a system, of a system upon a supersystem and external environment and the reverse influence of supersystem and external environment upon the system and subsystems. Without taking these influences into account we shall not only create a badly functioning system (or completely unworkable system), but can exert negative influence upon subsystems, adjacent systems, supersystem or environment.

We shall demonstrate the mutual influence of subsystems upon system, of system upon supersystem and environment based on the example of a tree.

Example 2.11 Tree (continued)

Type of the tree and of its subsystems largely depends upon external environment. Thus, for example, dwarf trees grow in the north and in the high mountain regions, plants, which are able to store moisture (Succulent plant), for example, cactuses, storing moisture in the stems, aloe, in which moisture hides in juicy leaves grow in the desert.

Subsystems of plants also largely depend upon the conditions of external environment. Succulents have pulpy and juicy stems, leaves and roots, bulbs, which are able to store water and economically use saved water for a long time. The skin of stems and leaves of succulents is covered with elastic varnish-like film—cuticula, which well reflects the sunbeams. The cactus also collects moisture from the air as a result of its condensation on hairs and prickles (areoles), the overall area of which appears to be rather large.

In their turn the plants influence external environment, emitting or absorbing oxygen or carbon dioxide from the atmosphere at different time of the day.

Example 2.12 Marine iguana

Sea iguana inhabits exclusively Galapagos islands. It eats seaweeds and has a unique (among modern lizards) ability to spend about an hour under water. Iguanas learned to hold their breath for this period, decelerate their heart rate and let the blood stream approach only the vital organs. It is the result of evolutionary changes in the diet—the food is found in the water (sea weeds), but not on land. This is an example of adaptation to external environment.

On land and in the water sea iguanas have little helpers—crabs and fishes *abudedefduf*.² These are cleaners, which eat parasites, which create numerous problems for sea lizards.

This is an example of a self-organizing system.

Such bonds have reasons and consequences, they are called.

Cause-and-effect link.

Example 2.11 The Tree (continued)

Why do low (dwarf) decumbent trees grow in the north and in high mountain region? Because the period, when they can grow, is rather short, in winter it is very cold and the winds are rather strong. *The reason* is strong frost and the winds, while *the consequence* is that the trees are low and stalk along the surface of the earth. That is what is called cause/effect connection.

The trees survive at such strong frost, since they are completely covered with snow. As a result of the action of frost, the plants get stronger, become frost-resistant. The content of sugar and the ratio of its components—saccharose and glucose varies. Long before strong frost, already during the autumn time, the cells of the trees get dehydrated, saved starch nutritious substances are converted into sugars, bonding water, which prevents the tearing of cells, which could take place at strong winds.

2.3 Systematicity

2.3.1 General Notions

The notion of systematicity is based on system approach.

Systematicity is a property, consisting in coordination of all interacting objects, including external environment. Such interaction should be completely balanced.

The object will be embodied as a system one if and only if it matches its intended purpose, is viable and does not negatively influence the objects located nearby and the external environment. Thus, in order to be embodied systematically, the object should match certain requirements.

System requirements

1. A system should match its **intended purpose**.
2. A system should be **viable**.
3. A system **shouldn't negatively influence** the objects located nearby and the external environment.
4. When creating a system, it is necessary to take into account **the regularities of its evolution**.

² *Abudedefduf* also known as the **sergeant-majors** is a genus of fish in the family Pomacentridae—Wikipedia.

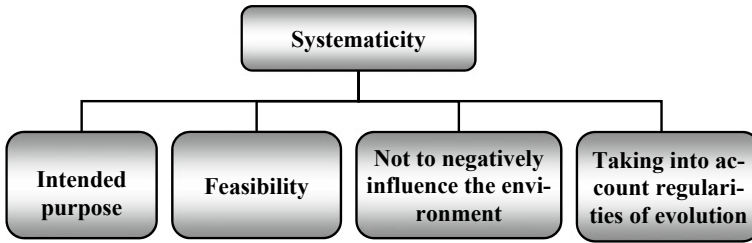


Fig. 2.3 Structure of systematicity

System requirements are constituents of the trend of increasing degree of systematicity (see Fig. 2.3).

Intended purpose of the system is described by *the main function of the system*, to fulfill the *main goal of the system*, to satisfy certain *need*.

Viability of the technical system is defined by its **workability** and **competitiveness**.

The system will be **viable**, if it is **workable** and **competitive**.

Workability is the ability to perform the assigned function with parameters, stipulated by technical requirements, during the expected service life.

In other words, **workability** is a high-quality functioning of the system, i.e., high-quality performance of the main function of the system.

Ergonomic parameters (characterize the correspondence of the marketable goods to the properties of the human body) can also be related to parameters of workability apart from high-quality functioning of the system (including those of *reliability* and *longevity*).

Workability is defined by the presence of necessary **elements** with required quality, presence and quality of necessary **connections** between elements, organization of necessary **flows** with required quality.

Competitiveness of the goods—is the ability of the goods to be attractive in comparison with other products of similar look and intended purpose, due to better correspondence of their quality and cost characteristics to the requirements of the given market and consumers' evaluations.

Competitiveness of the particular system is defined in comparison with the competing system. The competition depends upon:

- the number and quality of performed functions;
- the cost of this system;
- the timeliness of its appearance at the market.

Apart from technical functions, it is also necessary to take into account **aesthetic** and **psychological** functions. One of the main *aesthetic parameters*—is the design of product and its pack, including color scheme. To *psychological parameters* one should relate *prestige value*, *attractiveness*, *accessibility*, etc.

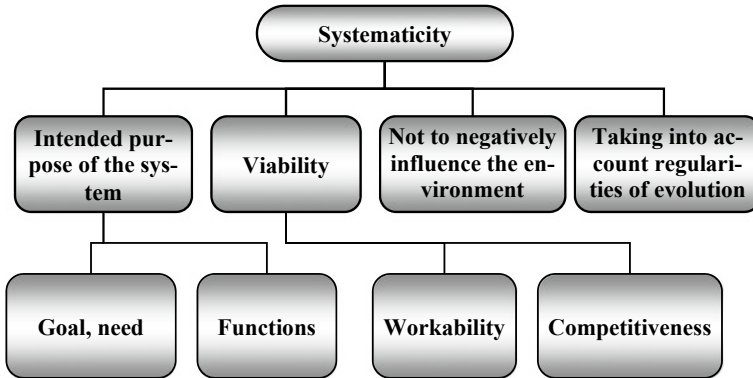


Fig. 2.4 Structure of the trend of increasing the degree of systematicity

Now it is possible to imagine a more detailed diagram reflecting the structure of systematicity (Fig. 2.4), which is a structure of the **trend of increasing the degree of systematicity**.

A system is workable, when it performs its main function. A workable system matches its intended purpose and has a definite **structure**.

Structure of the system should perform its *main*, all *basic* and *auxiliary functions*, constituting a set of interrelated **elements** and **connections**.

Workability depends not only upon *the structure of the system*, but also upon free passing of necessary *internal* and *external flows*.

2.3.2 Absence of Systematicity

Example 2.13 Telephone

Electromagnetic emission, emerging during the talk over mobile phone, harmful influences the surrounding devices, therefore on the airplanes and in the hospitals, it is not permitted to talk over mobile phones.

Antennas of transmitters of mobile communication harmful act upon the surrounding people.

Example 2.14 Car

Cars throw exhaust gasses into the atmosphere, contaminating the environment.

The road harmfully acts upon the tires, wearing them away.

The atmosphere harmfully acts upon the body of the car—corrosion appears.

2.3.3 Evolutionary Development

Systematicity also takes into account the regularities of **historical development of studied object—evolutionary development**. This is the last requirement of systematicity. It is taken into account in forecasting the development of studied object through taking into account the identified tendencies of historical and logical development of the given object, as well as taking into account the general trends of system evolution. As a result, they obtain the general tendency of evolution of the studied object and the conceptual idea of its next generations.

2.4 System Operator

System operator was developed by TRIZ author G. S. Altshuller.

Its structure is presented in Fig. 2.5.

A person with a routine thinking considers only the system proper. A deeper approach considers only the system itself. A deeper approach is to identify and to study the parts, of which the system consists—subsystems. Experienced people also identify, where the system belongs—they identify supersystem and environment. This is a **hierarchical structure** (cl. 2.2.3, Fig. 2.1).

This is the first constituent of system operator.

The second constituent of system operator is taking into account **the dynamics of evolution of the system**, its *subsystems* and *supersystems*. It is necessary to consider historical evolution of the system, its subsystems and supersystem. We shall call this constituent **evolutionary** or **genetic development**. To achieve this, they find out, which systems, subsystems and a supersystem existed *in the past*, and forecast their development for the *future*.

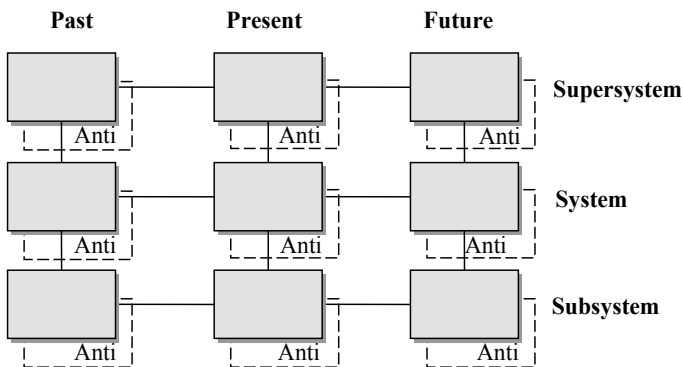


Fig. 2.5 System operator

The last constituent of the system operator is the identifying of **antisystem** at all levels and using them with regard to dynamics of development.

Antisystem is a system, which performs the opposite function as compared to the function being studied. Such consideration enables to broaden the idea of a system.

Thus, system operator has the following constituents:

- 1. **Structure of the system** and its **hierarchical levels** (system, subsystems, supersystem and environment);
- 2. **Dynamics of system evolution** at all levels is **evolutionary development** (present, past and future);
- 3. Taking into account and use of **anti-system, anti-functions** and **anti-actions**.

Let us quote the examples of using a system operator.

Example 2.11 Tree (continued)

System is a tree (Fig. 2.6). Subsystems of the tree were considered by us above in Example 2.11. In this example we shall choose the fruit, for example, pome. Supersystem is wood. Thus, we have considered *the hierarchical line*.

The past of the tree is the seed. The past of the fruit is the flower and its DNA. The past of the wood is earth. Now let us consider the future. One of the variants of the future of the tree is timber. One of the variants of the future of the pome (fruit) is a pie. One of the variants of the future of the wood is coal. This is *an evolutionary constituent*.

Example 2.15 Car (automobile)

System—a car (Fig. 2.7). Supersystem can be: highway, system of road traffic, including the system for regulating street traffic (marking on the road, street signs, traffic lights, road police, etc.), filling stations, repair workshops, plants manufacturing cars, etc.

The past of the car is a coach. The past of the engine is a horse. The past of a highway is a country road. Past of street traffic regulation is its absence. Everybody

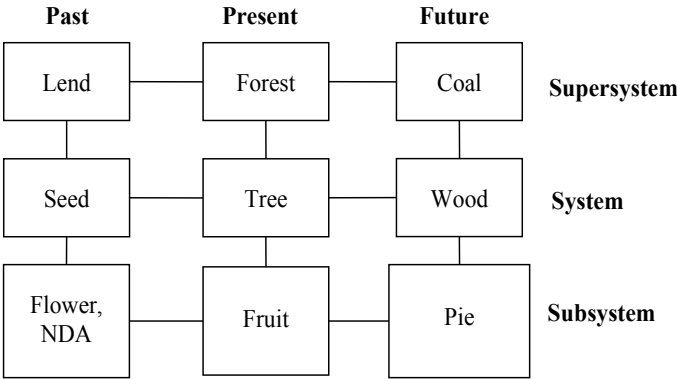


Fig. 2.6 System operator—tree

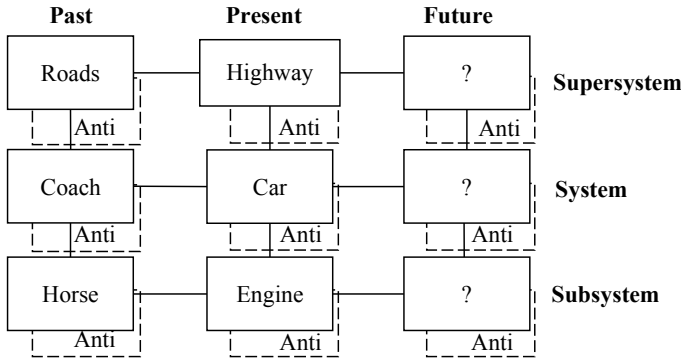


Fig. 2.7 System operator—car

could drive as he pleased and where he liked. The past of filling stations is post stations, where the carriages with horses could be replaced, where horses had rest and were fed with oats. Repair workshops in the past had the form of a smithery, while car manufacturing plants were coach-building workshops and farms where horses were bred.

Everyone can have his own vision of a future car, its subsystems and its super-system. First of all, the future of a car depends upon what systems it will consist of and with what supersystem it is going to be included. For example, already now the cars with electric engines have been developed, there are engines, operating on hydrogen and even on compressed air. All this will lead to the change of a super-system. In future there will be no road police—everything will be automated. The cars will “communicate” one with another, making road accidents impossible. The roads will be able to pass under the ground or above the ground, without occupying expensive place on earth.

Let us now consider *ANTI*component.

The function of a car is to transport (move) the passenger. Antifunction is to restrain the passenger (making him or her remain where they are). An example of such a system can be a prison, home arrest for the passenger or “arresting” the car (its evacuation) by the police, for example, for wrong parking.

The subsystem of engine has another function—motion of piston. Antifunction is braking (fixing). This system can be embodied as any clam, for example, vice clamps; fishing tackle, for example, sweep net; butterfly net, etc. with a car it can be wheel lock-up because of wrong parking (Fig. 2.8).

If we take “gas” as a subsystem, the function of which is to increase rotation speed of the engine (acceleration of motion), the antifunction—to decrease the rotation speed (deceleration of motion)—is performed by the brake.

The subsystem of wheel has two functions: motion of the car and supporting it at a certain distance from the road. Antifunction of the motion is locking. This function is also implemented by the wheel in the mode of braking. Antifunction of support

Fig. 2.8 Locking of a wheel

is attraction or repelling. Attraction to the road is performed by an antiwing. The function of repelling can be implemented by air cushion or a balloon (dirigible, etc.).

Supersystem highway has the functions of support and indication of direction of motion. Antifunction of support is repulsion (see above). Antifunction of indication of direction of motion is the absence of indication of direction. With airplanes, rockets, vessels, submarines and torpedoes there is no indication of direction in the form of a road. The indication is effected virtually with the aid of control system.

Supersystem of road movement control has the function of the same name. Antifunction is the absence of road movement control. This is a system, in which all elements are absent (See above). There should be a self-controlling system. Each car is associated with another car. All together they form a self-organizing system (like ants or bees).

Please, consider other antielements yourselves.

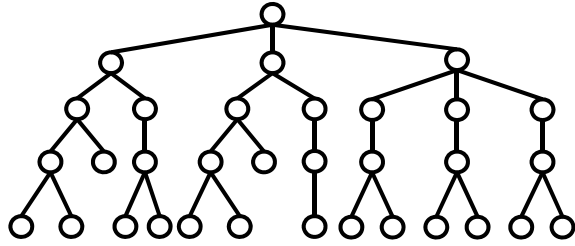
2.5 Impact Accounting

System approach implies **taking into account any changes and their influences** upon the system. The changes can occur **in time** and **according to condition**.

Example 2.16 Changes in time

Typical changes in time are change of day and night and of the seasons. Such changes are taken into account, for example, by turning on and off the light, heating and cooling of the rooms, etc.

Fig. 2.9 Graphic rendering of dynamic programming



Example 2.17 Changes according to condition

Typical changes according to condition are phase transitions, for example, at temperature of 0 °C at atmospheric pressure ice is converted into water. Great depth is characterized by high pressure. In space there is weightlessness, etc.

Each of us encountered with changes according to condition in road movement. When the traffic light is red, there is no motion, but when it is green, the motion takes place.

Each change should be taken into account in creation of new systems.

Taking all changes into account is one of the important constituents of system approach.

System thinking should be applied to any object, to any phenomenon and to any process.

2.6 Dynamic Programming

System approach is especially important for solving complicated (big) problems, for example, such ones as space exploration, development of complicated state and inter-state programs, etc.

Solving complicated problems directly is often impossible, therefore, the problem is **split into sub-problems**. It is used in dynamic programming.

Dynamic programming in mathematics and in theory of calculation systems is a method of solving complicated problems by splitting them into simpler sub-problems. It is applicable to problems with optimal substructure, which look like a set of overlapped sub-problems, the degree of complexity of which is a little bit less than that of the source one. In this case the time of calculation as compared to “naive” methods can be significantly reduced (see Fig. 2.9).³

Let us consider an example on dynamic programming.

³ Dynamic programming—material from Wikipedia.

2.7 Examples

Example 2.18 Curing mental patients

A new system of curing mentally deceased people started to be used in recent times.

Before that they were treated only in specialized medical institutions. Nowadays the treatment more often takes place at home. Specialized group of doctors trains the family members in terms of how to treat the patients. This group works not only with the family members, but also with environment. If the person is employed, the group works with the employees, with whom the patient is in regular contact. If the person studies, the group works with the team, where he studies, etc.

All levels of hierarchy, with which the patient comes into contact, are taken into account systematically in this case.

Such treatment not only showed significantly higher efficiency of returning the human to normal life, but is also characterized by significantly lower cost, since not all resources of specialized medical institutions are involved.

Practically all components of system approach are used here.

1. **Hierarchy of system**—work not only with the *system* (the patient), but also with:
 - 1.1. *supersystem* (his family, colleagues and all other people, with whom the patient is associated);
 - 1.2. *subsystem* (specific features of the patient);
 - 1.3. *external environment* (special conditions are created at home and abroad).
2. All **interconnections** of the system with supersystem and external environment are taken into account.
3. **Cause-effect connections**. What should be done by everybody, with whom the patient has contacts, how the external environment will influence him, etc.
4. **Taking into account the changes and their influences**. The smallest changes in the patient are constantly monitored and the corresponding decision is taken concerning the correction of behavior of the surrounding people and, if it is needed, the change of the environment takes place.
5. **System effect**. All these actions lead to a definite result—convalescence of the patient and his return to normal life.

Example 2.19 Nuclear weapons

Nuclear weapons is a restraining factor in starting a war, implying this kind of arms. The leaders of the states, who start a nuclear war cannot be in safety themselves, which is the case with conventional weapons.

Example of *cause-effect connections, taking into account the changes and their influences*.

Example 2.20 Glasnost (publicity)

Development of glasnost makes it almost impossible to hide significant “sins” by the politicians. In democratic countries it makes the politicians strictly abide the moral

norms, however, does not allow people, whose reputation is stained in this or that way, or who is able to be blackmailed, etc., to go into politics.

Example on *cause-effect connections, taking into account the changes and their influences*.

Example 2.21 Interconnected vessels

Engineer V. Moskalyov asserts that the law of equality of fluid levels in interconnected vessels is formulated incompletely⁴:

“Here are the main reasons due to which the equality of levels in interconnected vessels will be significantly violated:

- (1) The fluid in one vessel is significantly cooler (or warmer) than in the other vessel⁵;
- (2) In one vessel the walls are moistened by fluid, while in another they are not moistened, the size of cross-section of the vessels is not great;
- (3) Each of the vessels in the area of fluid meniscus is a capillary and the diameters of capillaries are different;
- (4) The system of interconnected vessels moves along a curvy line and the axis of instantaneous rotation is located at different distance from the vessels. If the interconnected vessels are connected to the pipeline, in which the fluid moves, the levels in them can be significantly different because of different ratio of static to dynamic charges and also quite a number of “ifs” could be mentioned. So cunningly looks in practice the application of a law, which could seem simplest...”⁶

Example on changing stereotypes (psychological inertia) with the aid of *taking into account the influences* upon the system and the obtainment of new knowledge.

Example 2.22 Railway track

Creating railway track is a typical example of a system approach.

It is necessary to create not only the locomotive (steam engine), but also the railway track. One is impossible without another.

Further on, we shall speak only about the track, or about its width, to be exact.

For two first British railways creator of steam engine George Stephenson accepted the width of the track equal to 1435 mm (4 feet 8½ in.), which got the name of *Stephenson* or *European track*.

⁴ Tekhnika—molodyozhi, No. 5, 1976. (in Russian).

⁵ This is a violation of condition of uniform fluid. The extended formulation of the law reads that “the ratio of fluid levels in interconnected vessels is in inverse proportion to the ratio of their density values”.

⁶ N. N. Latypov. Engineering euristics. (In Russian).

At first, he took the streetcar track as a pattern, which was equal to 1372 mm (4 feet 6 in.) and was selected out of the average size of the horse's croup. However, it wasn't possible for him to place a boiler and cylinders with pistons on this platform. He increased the width of the track only by 2.5 in. Thus a very strange size appeared and remained for long time, if not forever: 4' 8½". In a metrical system it is equal to 1435 mm.

Further on, in different countries in different years and for different purposes tracks of various size were created: from 3000 mm in Germany at the end of 1930-ies to 1269 mm in Great Britain (Rudyard Lake Steam Railway).

The broader the track, the more stable is the railway car and the more amount of cargo at assigned height of the vehicle it is able to carry.

In the USA a track appeared, which is 5' or 1524 mm wide. This width was carried to Russia by the American railway engineer George Washington Whistler. He completely designed a two-track railway, the length of which is 685 km. He also designed all necessary edifices and bridges. Thus, *the Russian track* appeared.

In the process of unification of track width, they don't always accept the most profitable (from the technical viewpoint) width of the track. They rejected the track of 2140 mm in England in 1840-ies, since the laying of a new track implied the broadening of elevated approaches and gutters, changing the bridges, tunnels, etc. Therefore, the track, which was 1435 mm⁷ wide was the winner in England, USA and Europe.

Please, indicate yourselves, what elements of system thinking are used in the present example.

Example 2.23 Aircraft designer

"An aircraft is such a device, in which two principles are necessarily struggling: strength and weight. The machine should be made strong and light, while strength and lightness are in constant antagonism one with another",—writes in his book "Stories of an Aircraft Designer" A. Jackovlev, creator of glorious JACK-aircrafts.

The aircraft designer has many assistants (Fig. 2.10) and each of them imagines an ideal aircraft in his own way: for a specialist in aerodynamics the most important thing is utter streamlining; the specialist who calculates the strength of the aircraft sees an ideal aircraft as unbreakable; for a technologist the most important thing is the simplicity of manufacturing; a motor specialist thinks that the ideal aircraft is a big motor and small wings; the representative of Aeroflot agency dreams about a capacious flying compartment for passengers..."⁸

⁷ Materials from the following work are used in this example: V. N. Romanenko. Practice of work in the field of innovations: general information on laws and technique of creative work/Vladimir Romanenko, Galina Nikitina.—Saarbrücken: Lambertacad. publ. (LAP), cop. 2015.—90 p.: ill.; 22 cm.; ISBN 978-3-659-80864-7. (In Russian).

⁸ Altshuller [4].

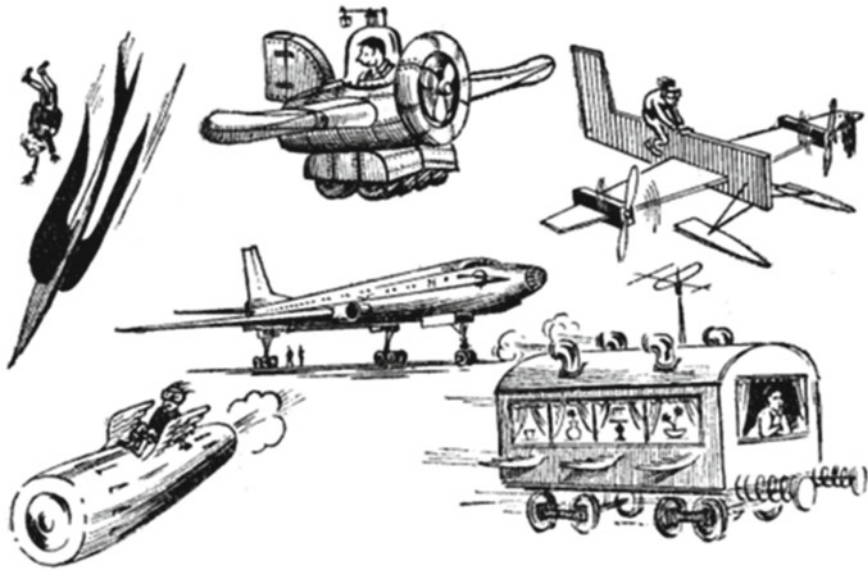


Fig. 2.10 Different ideas of an airplane

Indicate yourselves, which elements of system thinking are used in this example.

Example 2.24 Deburring

During the teething of a cogwheel with a milling cutter, the burrs are formed, which are later on removed.

It appeared that at the section of assembly the already manufactured cog-wheel is put on the shaft of the motor and is welded to it. The burrs appear to be inside the welding seam. After that the metal excrescences, formed by the welding, are taken off the butt of the cogwheel by polishing.

During the process of polishing the burrs are also eliminated automatically. The problem disappeared.

Indicate yourselves, which elements of system thinking are used in this example.

Example 2.25 “They use angry men for carrying water” (He who is angry is seldom at ease)

Peter the Great issues an ukase—yokes with water-filled buckets were hung on those people, who publicly expressed their anger or irritation and made him run around the water basin. The buckets swung, the water poured on “the angry man”, and he immediately calmed down.

Hence the Russian proverb “They use angry men for carrying water”.

Indicate yourselves, which elements of system thinking are used in this example.

Example 2.26 The legend of the origins of Tibet

In Eastern Tibet, where earthquakes are frequent, there is a legend, saying that Tibet rests on a gigantic fish called Niadinba-ba. The head of the fish is located exactly

under the Eastern Tibet. Sometimes the fish gets tired of holding the land of Tibet, it stirs and turns its head, which causes earthquakes.⁹

Indicate yourselves, which elements of system thinking are used in this example.

Example 2.27 The Parable of King Solomon

It happened once that two women of loose behavior came to King Solomon, who was known for his wisdom, to ask for his judgement. They lived in one and the same house and were neighbors. Each of them recently gave birth to a child.

Last night one of them choked her kid and put his dead body aside another woman, taking her live kid to her. In the morning the women started to argue, each of them trying to prove that the live child belongs to her, while the dead baby belongs to her neighbors.

They continued this argument in the same way in front of the King. Having listened to them, King Solomon ordered to bring a sword.

The sword was immediately brought in. Without hesitating a moment, the King Solomon said:

- Let both women be satisfied. Dissect the live child in two halves and give one half to each woman.

One of the women, having heard his words, changed countenance and begged:

- Give the child to my neighbor, she is his mother, don't kill him!

Another woman, on the contrary, agreed with the solution of the King.

- Dissect him, let him be neither hers, nor mine, —said she decisively.

Here said King Solomon:

- Don't kill the child, but give it to the first woman: she is his real mother.

Indicate yourselves, which elements of system thinking are used in this example.

Example 2.28 Hospitals

The person, who gets into a hospital, spends there several days, until he undergoes all analyses and the medical workers state the diagnosis and perform procedures necessary for treatment.

Each day of staying in the hospital costs quite a lot.

The number of places in the hospital is restricted, therefore, during the periods of mass illnesses, for example, influenza, there is a lack of places in the hospital and such situations emerge, when the patients are placed in the corridor.

The patient gets into the admission ward and spends a lot of time there, until they identify, to which section it is necessary to send the patient.

During the periods of mass illnesses there is a lack of medical personnel. These medical personnel have to work overtime. They get tired and make mistakes, which

⁹ Ye. I. Kychanov, B. N. Melnichenko. History of Tibet from ancient times to the present day.—M: Vost.lit., 2005.—351 p.—ISBN 5-02-018,365-2.—P. 9. (In Russian).

might also lead to lethal outcomes. The lack of highly qualified specialists is especially problematic.

In the hospitals there are very many microbes and often the people get infected with other illnesses in the hospitals. The microbes are brought not only by patients, but also by visitors.

In polyclinics, as a rule, the specialists are less highly qualified. The polyclinics are not equipped with specially complicated and expensive equipment, for example, CT and MRT. In connection with that it is impossible to state an exact diagnosis in polyclinics. The process of identifying the illness is spread in time and the absence of high-quality specialists can lead to statement of the wrong diagnosis and wrong treatment.

How to avoid these disadvantages?

Let us use system approach.

It is proposed to create specialized medical centers with all necessary equipment and highly qualified specialists. These centers will be able to come in contact with any specialists of the world, with a supercomputer, containing large data arrays in the field of medicine and an artificial intellect.

The patient will come to the center by a certain time, so that he or she will not have to wait. Preliminary analyses will be made. This procedure will occupy not more than two hours. The data of analyses are directed to a supercomputer, which in a few minutes will issue a preliminary list of diagnoses and, if necessary, will prompt, which additional analyses should be made. Having all data and a refined list of diagnoses and methods for treating these diseases, a highly qualified specialist defines the final diagnosis and method of treatment. All procedure even in most complicated cases will not occupy more than 5–10 h. In the most complicated cases the patients will be sent to specialized hospitals.

In order to significantly reduce the probability of infecting such centers with microbes, the patients will be changed into sterile clothes. Express analysis of patients in terms of identification of viruses with the patients and if such viruses are found, corresponding measures will be taken (masks, special rooms with special sterilization, etc.). Besides, there will be no visits of relatives and acquaintances to the patients, since they could bring additional viruses. All rooms will be subject to specialized sterilization.

Such an approach will allow the following:

1. To reduce the time of stating the diagnosis and the time of patient's staying in medical institutions;
2. The quality of treatment process will be enhanced:
 - 2.1. The diagnosis will be stated much more accurately and much quicker;
 - 2.2. Many illnesses will be prevented, since the tendencies will be identified, which can lead to illnesses.
3. The patient will stay under home conditions and won't have to interrupt work during the stay at the hospital.
4. Expenditures on treatment will be significantly reduced:

- 4.1. The patient will basically stay out of the hospital;
- 4.2. The number of in-patient beds will be reduced;
- 4.3. The number of medical personnel in the hospital will be reduced.
5. The level of servicing patients at the hospital will be enhanced, since:
 - 5.1. The flow of patients will be significantly reduced and the medical workers will have more time for each of the patients;
 - 5.2. An opportunity will appear to show practically each patient to a highly qualified specialist.
6. The possibility of infecting hospital rooms with viruses, since the flow of patients and visitors will be significantly reduced.

On the whole such system will enable even a small country to save billions of dollars and to make the population much healthier.

2.8 Conclusions

Let us do the summing up and list the main elements of system approach:

1. System;
2. Hierarchy;
3. Interconnections and mutual influences;
4. Cause/effect connections;
5. Systematicity and system requirements;
6. Evolutionary development;
7. Taking into account the changes and their influences;
8. Goal-setting;
9. Approach “Anti”.

For a detailed acquaintance with a system approach see [1–3].

2.9 Independent Work

2.9.1 Control Questions

1. Give a definition of system thinking and system approach.
2. List the main notions of system approach.
3. Give a definition of a system.
4. Describe the hierarchy of a system. Name the hierarchical levels of a system.
5. Quote the notions accompanying the notion of a system.
6. Describe kinds of changes.

7. What is a system operator?
8. Who is the author of system operator?
9. Name the main axes of a system operator.

2.9.2 Topics of Reports and Reference Papers

1. History of origination of the term system. Review and analysis of existing definitions of system.
2. Analysis of notion system thinking and system approach with different authors.
3. Analysis of non-system approach to nature, artificial systems and, in particular, to technology in the history of development of mankind.
4. The significance of mutual influences in different fields of knowledge.
5. The influence of system approach in science, technology, business and life.

2.9.3 Fulfill the Tasks

1. **Quote examples of technical systems**
2. Quote examples of system approach
3. Use system operator for
 - 3.1. Lamp
 - 3.2. Computer
 - 3.3. Any system and/or process
4. Show impact accounting in:
 - Nature.
 - Technology.
 - In the family, society, etc.
5. **Solve the problems, using system approach.**

Problem 2.1 Captain Blood

Condition of the problem

Captain Peter Blood, character of the novel “Odyssey of Captain Blood” by Raphael Sabatini, goes to the enemy’s ship for negotiations.

How can he provide for safety of his own life and of the lives of his comrades?

Problem 2.2 Cigarettes

Condition of the problem

How to get a large sum of money for smoked-out cigarettes?

Problem 2.3 Inquisition

Condition of the problem

Many centuries ago, in Spain a newly suspected person is called to the inquisitor. As a rule, the death sentence was the result.

The smiling inquisitor says to the suspected: "It's my lucky day and I'd like to leave you a chance for life. Here are two rolled pieces of paper. On one it is written "Life", while another is inscribed "Death". Draw your lot".

The suspected turned pale, he immediately understood that the word "Death" stood on both pieces of paper.

How can he save his life?

Problem 2.4 The Bear Hunting

Condition of the problem

A young hunter and an experienced one once went to hunt a bear.

When the experienced hunter saw the bear, he wounded it insignificantly. The enraged animal sprang at the hunters. They ran to the village. The young hunter says to the experienced one that the bear will soon catch up with them, but the experienced hunter does not answer him, but runs further.

Then the young hunter turned and shot the bear.

The experienced hunter started to scold the young one. Why?

Problem 2.5 A Cruel Law

Condition of the problem

Once upon a time there lived a cruel ruler, who did not let anyone into his domain. Near the bridge across the boundary river there stood a watchman, who was armed from head to foot. He was ordered to ask every traveler:

– Why do you come here?

If the traveler told the lie, the watchman had to seize him and to hang him immediately. But if the traveler told the truth, he was still destined to die: the watchman had to drown him in the river immediately.

Thus, was the cruel law of the stone-hearted ruler and no wonder that nobody ever dared to approach his domain.

But in spite of all that once a peasant calmly came to the guarded place near the prohibited border.

– Why do you come here?—asked the watchman in a rude way, getting ready to execute the bold man, who was carelessly walking towards his certain ruin.

But the answer was such that the stultified watchman, strictly abiding the cruel law, could not do anything with the quick-witted peasant.

What was the answer?

Problem 2.6 Teacher and pupil

Condition of the problem

What is described below is said to have happened in Ancient Greece. The teacher of wisdom, sophist Protagoras, took up the task of teaching all techniques of soliciting art to Quantlus. It was agreed between the teacher and the pupil that the teacher had to pay the remuneration to his teacher immediately after he revealed his first success, i.e., immediately after the first legal case won by him.

The time came when the course of Quantlus' training was completely over. Protagoras was waiting for the payment, but his disciple did not hurry to be a solicitor in court. What's to be done? Finally, Protagoras decided to recover his debt through court and started a legal case against his disciple. His reasoning was as follows: if the case was won by him, the money should be charged based on the sentence of the court; however, if the case was going to be lost, which means: won by his disciple, the money still would have to be paid by Quantlus according to their agreement—the condition was to pay after the first lawsuit won by the disciple.

However, the disciple, on the contrary, considered the legal case, started by Protagoras, to be completely hopeless. He really learned something from his teacher and his reasoning was as follows: if the court obliges him to pay, he does not have to pay because of the agreement—he lost the first case; however, if the case would be decided in his favor, he still wouldn't be obliged to pay—based on the court sentence.

The day of the court came. The judge was in great trouble. However, after thinking for a long time he finally found the outcome—such a sentence, which, without in any way violating the conditions of the agreement between teacher and disciple, at the same time gave the teacher an opportunity to get the agreed remuneration.

What was the sentence of the judge?

Problem 2.7 The Preacher

One preacher told his night dream to the congregation. He had a dream that he got into an old castle. He was seen into a room, where there stood a huge table with delicious dishes. People were sitting round the table. But they did not eat anything and quarreled. What was the matter?

And here the preacher understood what the matter was. The hands of the people did not fold because of absence of arm creases. They could not reach their arms.

Then he got into another room, in which there was a similar table, round which also the people were sitting and their arms also had no creases, but the people were very happy. Why?

Problem 2.8 Four Triangles

Condition of the problem

How to create four triangles with the help of six matches?

Problem 2.9 How is it done?

Condition of the problem

You see here a wooden cube composed of two pieces of wood (Fig. 2.11). The upper half of the cube has protrusions, which are inserted into the indents of the

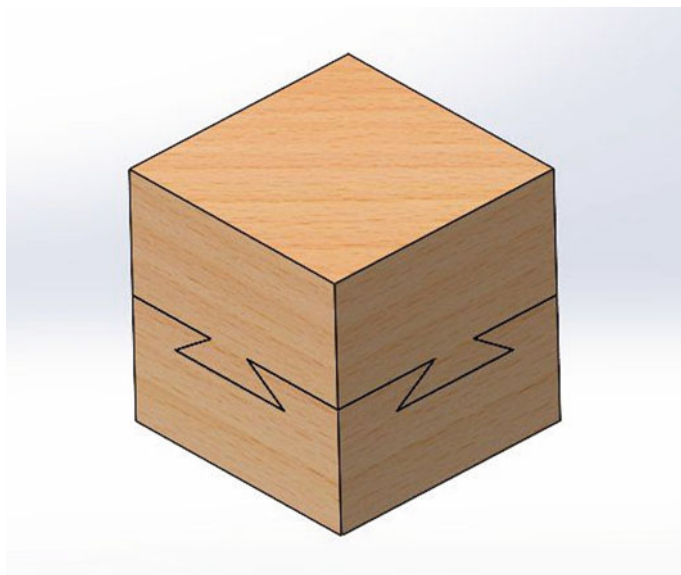


Fig. 2.11 An elaborate joint in assembled form

lower part. Pay attention to the shape and the positioning of protrusions and explain how the carpenter managed to join two pieces?¹⁰

Problem 2.10 Demidov

Condition of the problem

Akinfiy Demidov possessed business acumen, bravery, enterprise, range, ability to organize and to head big business. Historians think that Demidov's workers coined silver rubles in the secret cellars. When the empress Yelizaveta Petrovna learned about these illegal activities, she invited Demidov to a game of cards. Having won, she asked him: "Whose money have you lost to me today?"

What should Demidov do?

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Chapter 3

Evolutionary Thinking



If you don't think about the future, you won't have it.
John Galsworthy

3.1 Review

Evolutionary thinking has two constituents:

- (a) *Identification of trends of evolution in any phenomena*, for example, as it is done in tests on logic or IQ (for example, the sequence: triangle, square, pentagon...what comes next?). Identification of trends is a very important quality. It is necessary not only for scientists, but also for every person.
- (b) *Use of trends of system evolution* for the evolution of a particular system. Basically, a group of trends of system evolution is considered.

The following **trends of system evolution** are considered to be basic (Fig. 3.1):

- *trend of increasing the degree of ideality;*
- *trend of increasing the degree of controllability and dynamicity;*
- *trend of transition to supersystem;*
- *trend of transition to microlevel;*
- *trend of coordination;*
- *trend of trimming – deployment;*
- *trend of balanced system evolution.*

3.2 Identification of Trends of Evolution

It is very important to identify and to be able to prolong the trends. It could be tendencies in human behavior, in politics, economy, trade, natural phenomena, fashion,

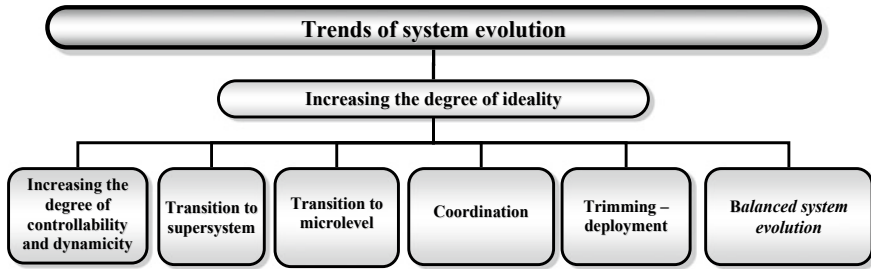


Fig. 3.1 Structure of trends of system evolution

everyday life, etc., in everything, what surrounds us. They could embrace different periods of life, from billions of years to portions of a second.

Example 3.1. Expansion of the Universe

There are different hypotheses of origination of Universe and galaxies. The processes of expansion of the Universe are usually counted from the Big Bang, which approximately took place 14 billion years ago. There are main stages of Big bang.¹

In 1926 famous American astronomer Edwin Powell Hubble proposed (and in 1936 modernized) the classification of galaxies according to their morphology. Because of a characteristic shape it is also called “Hubble Tuning-fork” (Fig. 3.2).

Example 3.2. Origins of Earth

The Earth originated approximately 4.5 billion years ago from particles of a proto-planet cloud, its mass was gradually increasing. The Earth gradually warmed up. Supposedly the core was formed during several hundred million years. During the process of cooling a hard core emerged. Other shells were formed for much longer periods of time. Air shell and oceans were gradually formed. First granite crust was formed about 3.8 billion years ago.

Atmosphere and hydrosphere of the Earth emerged in the process of evolution.

Life on Earth originated about 4 billion years ago, and the human being—about 200 thousand years ago.

What is in store for the Earth in future? This question can be answered only with a great degree of ambiguity, abstracting both from the possible external space influence and from the activities of humanity, transforming the environment and not always to the best.

At the end of it the earth’s depth will be cooled to such an extent that the convection in the mantle and, consequently, the motion of the mainlands (which means, the formation of mountains, eruptions of volcanoes and earthquakes) will gradually weaken and stop. Weathering will rub off the unevenness of earth crust and the surface of the planet will hide under water. Its further fate will b defined by the average temperature of the year. If it gets significantly lower, the ocean will freeze

¹ Chronology of the universe—material from Wikipedia. https://en.wikipedia.org/wiki/Chronology_of_the_universe.

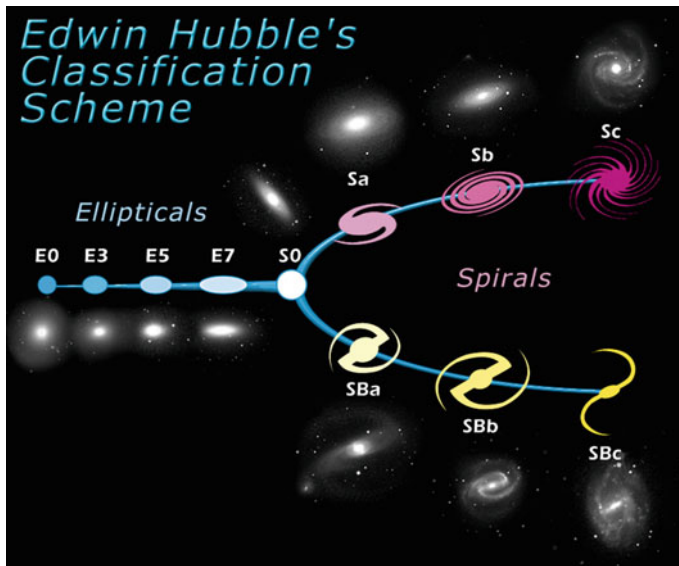


Fig. 3.2 Hubble tuning fork²

and the Earth will be covered with ice crust. If the temperature goes up (and most probably the growing luminosity of the Sun will lead exactly to that), water will evaporate, laying open the surface of the planet. It is obvious: both situations will lead to impossibility of life on Earth, at least, in our modern impression of life.³

Example 3.3. Evolution of the Society

Historical science considers different periods of evolution of society, starting with primeval (Stone, Copper, Bronze and Iron Ages), which starts 200 thousand years ago and finishes approximately in 1500 A.C.

It is interesting to trace the tendencies of evolution of social relations, forms of power, norms of behavior and origination of religions.

“The march of history was defined by three greatest revolutions. Everything started with the cognitive revolution 70 thousand years ago. The agrarian revolution, which took place 12 thousand years ago significantly facilitated the progress. Scientific revolution—it is only 500 years—is quite capable of putting an end to history and to lay the foundation of something different, which never existed before”.⁴

² URL: https://en.wikipedia.org/wiki/Hubble_sequence#/media/File:HubbleTuningFork.jpg.

³ Evolution of the Earth URL: <http://galspace.spb.ru/index17.html>. (In Russian).

⁴ Harari and Vintage [1].

The tendencies of evolution of different states embrace from several thousand years to several dozen years.

Example 3.4. Meteorological forecasting

Today we can get a fairly accurate weather forecast from one day to ten days. To achieve this, weather conditions are measured in many points of the Earth and also from space. The analysis of these data enables to identify the tendencies of changes in weather conditions.

Example 3.5. Instantaneous tendencies

Sometimes human life depends upon the ability to quickly see the trend of evolution.

It could be an unexpected situation on the road and the driver during fractions of a second should react in the right way. An experienced driver, constantly guarding the road, can foresee complicated situations.

Judging by the motion of legs, body and then the hands the boxer can foresee the blow, which his antagonist prepares. The same situation can occur in war actions.

In connection with the above it is very important to identify the tendencies both in large intervals of time and in fractions of a second.

Example 3.6. Evolution in fashion

History of costume is a marvelous example of studying changes in trends of evolution of this system.

There is an opinion that the first articles of clothes appear with the loss of hair coat on the human skin.

First the humans used the skins of animals, which they hunted, as clothes.

Further on, they started to sew animal skins together using sinews and hair.

Then the materials, of which clothes was made, its shape and manufacturing technology were changed.

Approximately 10 000 years ago goats and sheep became domestic animals, and since then woven woolen clothes came into existence.

Later on, they started to use cotton and flax and only in the twentieth century appeared synthetic fabrics.

In this example we shall trace only the changes in clothes' form.

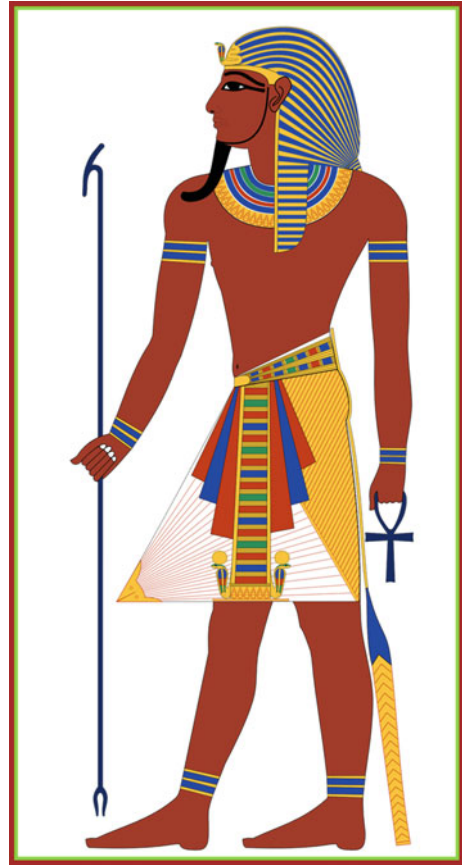
Let us start with the clothes of Ancient Egypt.

For many centuries Egyptian clothes was unchangeable, constant and undifferentiated. There were only two types of clothes: male and female. The difference in clothes between separate social classes was only in finishing and materials. However, here exact laws and accurate geometrical forms controlled.

The main male clothes were the apron (schenti)—the loincloth (Fig. 3.3). Female clothes covered a much larger portion of the body. Female slaves wore practically nothing.

Further on, clothes became more and more complicated. The top of complicity of female dress was crinoline (Fig. 3.4), which was fashionable since 1851 till 1867.

Fig. 3.3 Clothes of ancient Egypt⁵



Further on, the dresses became simpler and their volume decreased. In the twentieth century the ladies started to wear trousers. Nowadays the clothes fits the female body much more tightly.

Example 3.7. Evolution of bathing suits

First female bathing suits had the form of a dress. Further on, the surface area of a female bathing suit started to decrease. The general tendency is shown in Fig. 3.5.

⁵ URL: <https://www.clipartmax.com/max/m2H7Z5K9d3N4m2d3/>

Astonishing Ra Egyptian Pharaoh Clipart Image For Clothing–Ancient Egypt Male Clothing is a (908 × 1730) png clipart image which is manually selected and technically optimized. This transparent clipart, 153 KB, is about clipartmax.com. Astonishing Ra Egyptian Pharaoh Clipart Image For Clothing–Ancient Egypt Male Clothing clipart image can be downloaded and shared for free. If you like it, remember to share on your facebook, Twitter, Pinterest, etc.



Fig. 3.4 Crinoline⁶

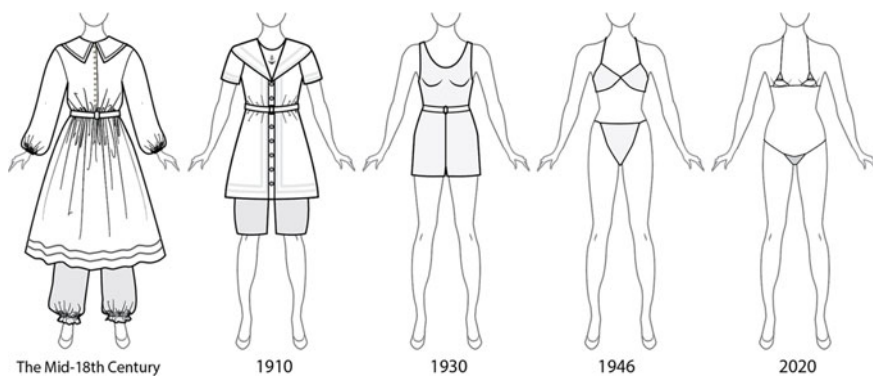


Fig. 3.5 Evolution of ladies' bathing suit⁷

⁶ This drawing was kindly provided to me by Ms. Helen Solodkina.

⁷ This drawing was kindly provided to me by Ms. Helen Solodkina.

Let us show an example of identifying economic trend.

Example 3.8. Variation of light intensity

Specialists in economics from the United States developed a new methodology, owing to which it is possible to measure the variation in financial state of the country by watching them from space. This methodology will help to watch the states, which don't publish and even conceal their official economic indicators. By the present moment the results of this work were not published by any of the scientific editions. Only short conclusions regarding the research were published in a corresponding press release at the official web-site of the Brown University.

To provide for indirect evaluation of parameters for economic growth or decline of state, the economists used the method of measuring the intensity of the night lighting of the cities. Authors of the research think that artificial light is necessary for any business activity during the darkness hours (whether it is trade or other services). The higher the intensity of the illumination, the higher is the level of business activity, and the higher is the purchasing power of citizens, which leads to the higher gross domestic product of the state.

The researchers came to a conclusion that the official statistics of standard parameters is strongly different from the data, which are obtained by a new "space" method. For example, with the Congo Republic the growth of gross domestic product was registered at the level of 2.4%. At the same time according to official parameters Congo is characterized by decline only, which amounts to 2.6% annually. The growth of Myanmar economy was registered as 3.4% annually, though according to official statistics it amounts to 8.6% annually.⁸

3.3 Use of Trends of System Evolution

Use of trends of system evolution can not only explain the evolution of existing systems, but also to forecast the development of the future ones.

Example 3.9. Development of printers

Let us briefly describe the history of printer evolution.

The necessity for printer emerged in 1951, when the first serial computer UNIVAC I (Universal Automatic Computer I) was created in the USA (it was developed by the American company Remington Rand). Each of the computers could produce from 400 to 2000 calculation operations per second, which at that time was considered to be tremendous rate. The calculations were retyped by a large team of typists.

Not only was it very expensive, but also mistakes were made.

In 1953 the corporation Remington Rand created the first printing device for computer UNIVAC 1. The device got the name of UNIPRINTER; part of this name (printer) became a proper noun.

⁸ Brown Economists Measure GDP Growth from Outer Space URL: <https://news.brown.edu/articles/2009/09/space>.

The operation principle was transferred from the electrical typing machine (analogy) with some difference.

The picture was formed as a result of a strike of the corresponding character against the paper through the dyeing band.

UNIPRINTER was a **drum printer** and functioned in the following way: behind the paper sheet there was a number of hammers, which were controlled by an electric magnet. In front of the sheet there was a dyeing band, while in front of the band a drum was rotating, the width of which was equal to the width of the entire page (120 symbols), and which carried 120 rings, each of which contained all letters of the alphabet. The drum rotated uninterruptedly and when the required letter in the required position of the line appeared to be above the paper, one of the hammers struck against it. Thus, during one turn of the drum it was possible to print the entire line. After that the paper was fed upwards for printing the next line, etc.

Passing over from electrical typing machine to the first printer took place because of using several trends at once.

The typists were substituted by the control system—this is *the trend of increasing degree of controllability*.

The carriage did not move—*trend of increasing the degree of ideality*.

The number of letters on the drum increased 120 times—this is a step aside from ideality. It can also be looked upon *as a trend of transition to supersystem*.

Besides, *the principle* the other way round was also used, not the character struck against the dyeing band, which left a trace on paper, but the paper with the aid of a hammer, located at the reverse side of the paper struck against the character.

Later on typesetting printers appeared. Symbols were not rigidly fastened in these printers, but were set up for each line. At the moment of printing one line the next one was set up, which increased the rate of printing up to 100 pages per minute.

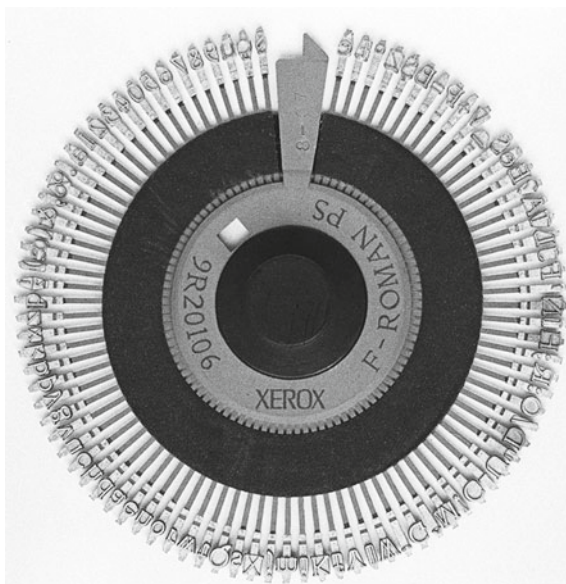
This solution implied the use **of trend of increasing the degree of dynamicity**. The symbols were not rigidly fastened to the drum, but were set up in advance. Principle of preliminary action was used—during the printing of one line another one was being prepared.

Daisy wheel printing (Fig. 3.6) had still more resemblance to a typing machine. The main detail of these printers was a metal printwheel, at the ends of the moveable petals of which embossed symbols were located. “The printwheel” was rotating around its axis, the hammer struck “the petal” of the wheel, pressing it to the paper through the dyeing band and leaving the print. Having changed “the printwheel”, it was possible to change the font or the alphabet. During one minute the machine printed up to 78 thousand symbols, which is hundreds of times quicker than the speed of the most skillful typist.

The following trends were used here: *the trend of increasing the degree of ideality*—the number of symbols reduced and was not repeated, and *the trend of increasing the degree of dynamicity*—each sign is moveable.

Dot Matrix printing was invented by the company Seiko Epson in 1964. The principle of printing is simple: a matrix—a set of several small needles, which form a

Fig. 3.6 Daisy wheel printing⁹



necessary symbol. The needles strike against the paper through the dyeing band and thus a picture consisting of points is formed.

This principle of printing implied the use of one of the tendencies of *trend of increasing the degree of dynamicity—trend of increasing the degree of fragmentation*. Each of the signs is split into constituent parts—needles. Besides, this is the use of *trend of increasing the degree of ideality*—the number of parts decreased, there are no separate parts with individual symbols—it's a universal device, which can create any symbol or picture.

Noteworthy is the tendency of decreasing the number of used symbols: in a *drum printer* there are *120 sets of symbols*; in a *daisy wheel printer* there is *only one set of symbols*; in a *dot matrix printer*—*not a single set*.

120 – 1 – 0.

Since that moment the printers started to depict not only symbols, but also any picture, however as yet only in black-and-white version or in any other monochrome embodiment, when a band of other color was used instead of black.

Inkjet printing was developed in parallel with the pin-cake principle. Scientific basis in this trend was laid down by the British physicist and Nobel-prize winner Lard Raleigh, who already in the nineteenth century studied the fluid jet disruption and the formation of drops, and the patent for this method of printing was issued to William Thomson in 1867.

⁹ https://en.wikipedia.org/wiki/Daisy_wheel_printing#/media/File:Xerox_Roman_PS_Daisywheel_-_mono.jpg.

The operation principle of inkjet printers is similar to that of dot matrix printers—the picture on the carrier material is formed of prints. The function of pins is performed by matrix, microscopic jets of ink.

There are two operation principles for shooting the jets of ink at the paper, piezoelectric one and the increase of temperature (gas bubbles technology).

In *piezoelectric technology* of printing each nozzle has a piezoelectric membrane, which under the action of the electric pulse bends pushing a drop of ink out of the nozzle. After that a new portion of ink is fed into the nozzle from the cartridge.

In technology of *gas bubbles* each nozzle is equipped with a microscopic heater, which instantaneously heats the ink up to the temperature of boiling (500 °C). The vapor which is generated hereby pushes a drop of ink from the nozzle. After shutting off the heating the ink cools down and a new portion of ink enters the nozzle from the cartridge.

The jets are shot from the micronozzles, the diameter of which is less than that of a human hair.

History of creation of jet printer is described in.¹⁰

This method of printing implied the use of one of the tendencies *of a trend of increasing the degree of controllability—change of field*. Mechanical field (strike) is substituted with a jet of ink, which is controlled either by heating (thermal field) or by piezo effect. In case of using piezo effect this is the use of *trend of transition to microlevel*.

Laser printer. At the end of 1938 the American Chester Carlson obtained the first xerographic picture, using static electricity for transferring toner (dry ink) to paper. This principle of copying machines also forms the basis for functioning of a laser printer.

A negative charge is fed to an aluminum cylinder (photo-drum), covered with a thin film of photo-sensitive semi-conductor, and then the laser beam takes this charge off, where the printing is necessary. Further on, pulverized ink is applied over the drum, which sticks to it in “disinfected” places. And when the drum comes in contact with the paper, a print is left on it, which owing to high temperature reliably sticks to the surface (Fig. 3.7).

This principle of printing implies the use of one of the tendencies of *trend of increasing the degree of controllability—change of field*. The jet of ink was substituted by the most controllable optical field.

LED printer—one of the kinds of printers, the operation principle of which is similar to that of laser printer.

The main difference of LED printer from laser printer is based on the system of lighting the photosensitive roller. Laser printer does this via one source—laser, which, using the scanning system of prisms and mirrors, runs along the entire roller. LED printer uses a line of light diodes, located along the entire roller surface in its lighting system. The line may include from 2.5 to 10 thousand diodes. In all the rest it is identical to the system of laser printing.

¹⁰ How a jet printer works URL: <https://castleink.com/blogs/printer-help/inkjet-printer-history>, <https://www.seapdigitalprinter.com/news/inkjet-printers-from-scratch.html>.

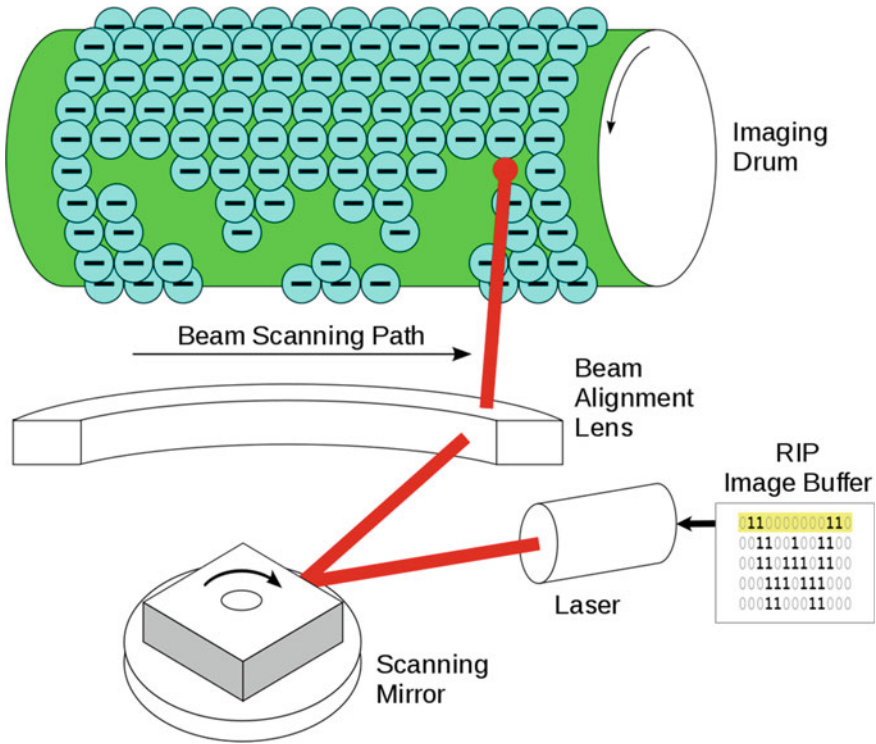


Fig. 3.7 Laser printer¹¹

The advantage of these printers is in the compactness of light diode line. The dimensions of color LED printers are two times less than those of laser printers. They are simpler and more reliable, since there are no moveable parts in forming pictures. All LEDs give similar light spots, which correct the distortion of light spots at the edges of the photo drum. The data to the line can be fed in parallel, data capture is performed using a radio scanner, but such type of data feeding makes the data capture more complicated. Most often they use continuous feeding of data, which makes the data capture significantly easier.

In case of diverging of lighting brightness with some LEDs stripes appear along the line of paper motion with high or low density of printing quality, which is a significant disadvantage of such equipment. In contrast to laser printer, where only one laser LED is corrected, it is impossible to fully correct all LEDs in a LED line, since the line can contain up to 5000 LEDs and more.

Another disadvantage of LED printer is the impossibility of shifting elementary dots. The possibility of shifting is needed for improvement of quality of picture in printing. They also use the variation of size of the dots, which is characteristic both for LED devices and for laser devices as well.

¹¹ URL: https://en.wikipedia.org/wiki/Laser_printing#/media/File:Laser_printer-Writing.svg.

Because of their compact character, LED printers are characterized by restricted focusing of light from some of the light diodes of the line. Besides, LD system is preconditioned by low printing rate—about 40 pages per minute as compared to laser technology—from 110 pages per minute.¹²

This principle of printing implied the use of *trends of increasing degree of ideality and of transition to supersystem*. Moveable parts of forming pictures, which existed in a laser printer are eliminated – the system became more ideal and is characterized by transition to supersystem—use of multiple identical LEDs.

Solid-ink printer uses briquettes of solid ink for printing. The technology was developed in 1986 by Tektronix company. In 2000 Xerox company acquired the corresponding subdivision of Tektronix together with the rights for all developments in the field of solid ink printing.

Part of solid ink is melted, and it is fed into the immobile printing head, which applies the picture over a rotating drum made of anodized aluminum, covered with a silicon lubricant. Slightly warmed sheet of paper is pressed to the drum with a special roller. The picture is transferred during one run, therefore, it is effected at a high speed.

Expendable materials can be additionally loaded without interrupting the printing process.

Reliability of these printers is much higher than those of laser ones due to design and minimum of moveable parts.

This principle of printing implied the use of *trend of increasing the degree of ideality*—minimum of mobile parts, which existed in a laser printer.

Example 3.10. Evolution of locking devices

First there were door catches and bolts. Then they were equipped with cross-bars and were locked with keys.

The first locks were invented in China, Egypt and Mesopotamia. They were made of wood. The key was also wooden. Keys were heavy and they were carried on the shoulder.

The rulers locked their food supplies, jewellery, the doors of their bedrooms.

Transition from bolt to lock is a *jump in system evolution*.

Wooden designs of locks of different types existed in Europe up to the second half of the nineteenth century.

After that they started to manufacture hybrid systems—the body was wooden, while the crossbar, key, etc. were made of iron.

In Ancient Greece such a lock was spread, which was locked by a string tugging, while it was unlocked by a long bronze key.

After that first completely metal structures appeared in Europe: stationary (rim locks) and detachable (padlocks). Probably, padlocks were borrowed from China and India.

¹² URL: https://en.wikipedia.org/wiki/LED_printer.

Use of different materials in a lock is a manifestation of *trend of coordination*.

Further on they started to use springs in the design of locks.

A revolutionary design was a cylinder-shaped lock, which was patented by the Englishman Joseph Bramah in 1784. It was a lock with a flat key. Inside the lock there were spring-mounted pins. The cylinder was turned only in such case, when the pins coincided with cuts in the key. Further on, this lock was improved and as of today, these are the most widely spread locks.¹³

After that they started to manufacture mechanical numeric locks with rotating disks.

Further on, numeric locks became electromagnetic ones. Correctly dialed code included an electric magnet, which pulled aside the tongue of the lock, opening the door. Until today many locks of front doors operate according to this principle.

The next stage is the use of magnetic keys in the form of a card for a dead lock.

Further on, they started to use various electronic locks.

Locks became readjustable.

The key started to take the form of a fingerprint, pattern of eye retina print, voice spectrum, etc.

Nowadays an electronic key contains a chip and even a Bluetooth, therefore, it can be readjusted remotely. A smartphone can also be a key.

Since the moment of appearance of springs in the lock and during the next stages of evolution this was a manifestation of *trend of increasing the degree of controllability*.

It is possible to foresee further steps in the evolution of locking devices—this is the full line of using this trend. The lock will become smarter. It will be self-adjustable, self-organizing, self-developing and, finally, self-reproducible.

Example 3.11. Evolution of religions¹⁴

The scientists think that the primary form of religion is **animism**.¹⁵ Animists worshipped material objects and forces of nature, through belief in existence of spirits, which control these forces. They believed that animals, plants, inanimate nature (for example, rocks, rivers, etc.), natural phenomena, demons, fairies and ghosts could influence their life. In each region there were their own objects of worship. According to their opinion each of the objects selected by them, was in charge of something of his own. Thus, a certain group of people thought that they could ask a source of water for something, while they could address the rock for something else, the rocky goat could be addressed for some third thing, and the wind could be addressed for another thing, etc. The number of such objects was rather large and one tribe could worship one group of objects, while another tribe chose

¹³ https://www.youtube.com/watch?v=_7am-ad2NGs.

¹⁴ This example is based on the following works:

Classics of world religion theory. Anthology. M.: Canon. 1996.– 496 p. ISBN: 5–88,373-036–1. Harari and Vintage [1].

¹⁵ **Animism** (from Latin *anima*, “breath, spirit, life”) is the belief that objects, places and creatures all possess a distinct spiritual essence (*Material from Wikipedia*).

another group. Thus, for example, one tribe could consider a certain tree to be sacred and it was forbidden to cut it, otherwise, the spirit of this tree would be enraged and will take its revenge.

The tribes never imposed their opinion upon the others and never sent missionaries to other places. On the contrary, they thought that “the spirit of my object shouldn’t help my neighbor or enemy”...

These religions were very narrow and local, they were incorporated with the local landscape, climate, natural phenomena, etc.

From the standpoint of trends of system evolution it was a *polysystem* with such parts, which were absolutely not connected.

While people lead a non-migratory way of life at sufficiently small territories (several hundreds, maximum several thousand square kilometers) local spirits could fairly provide for all their needs. With the growth of their domains (kingdoms) and with the development of trade people needed the patronage of somebody, whose might embraced the entire kingdom or a trade region.

In connection with that the need for **polytheistic religion**¹⁶ appeared.

These religions presupposed that the world is ruled by a group of almighty gods: the goddess of fertility, god of rain, god of war. People addressed these gods and they, if they remained satisfied with the rituals and sacrifices, sent rain, victory or health.

Animism did not disappear completely with the advent of polytheism. The demons, fairies, ghosts, sacred rocks, sources and trees appeared to be integrated by the majority of polytheistic religions. These spirits were given much smaller importance than the gods, but in everyday life of a simple man they were rather needed. While the king in the capital city sacrificed dozens of fat lambs to the god of war, praying for the victory over the barbarians, a peasant in his hut lit a candle dedicated to the spirit of a particular tree and prayed for preserving the life of his child, who had fallen ill.

Thus, from the point of view of trends of system evolution, the transition from animism to polytheism is *a transition to supersystem*. Individual spirits were grouped into mightier gods, who were responsible for a separate “branch” each. Besides, this is an *increase of the degree of controllability*—control became simpler: people addressed one, not a hundred of gods (spirits). It is possible to speak also about *trimming* the religious system. They use *system approach* in polytheism. Apart from hierarchy, they speak about mutual connection and mutual influence between people and gods. Polytheists thought that prayers and sacrifices, good and evil deeds tell upon the fate of the entire ecosystem, for example, people can infuriate the gods and they will cause an awful flood, in which many insects and animals will die, from the smallest ones to the biggest. Polytheism elevated the status not only of gods, but also of people.

Polytheism was unable to formulate its own universal message. In connection with that a need for the new religion appeared.

¹⁶ **Polytheism** (from Greek πολύς, “many” + Greek θεός, “god”—“**plurality of gods**”)—religious philosophy based on belief in several gods, usually united into a Pantheon of gods and goddesses (*material from Wikipedia in Russian*).

The next stage in the evolution of religion was **monotheism**.¹⁷

There is an opinion that some individual adherents of polytheism started to love one of their gods and started to believe that this is the only god who embodied the higher force of the Universe and it would be possible to come to terms with him. Thus, monotheistic religions were born.

Some scientists think that the first monotheistic religion originated in Egypt about 1350 B.C., when the pharaoh Akhenaten announced that one of the junior gods of the large Akhenaten pantheon—Aton in reality was the supreme force, which rules the world. Akhenaten converted the cult of Aton into a state religion and decided to eliminate or minimize the worshipping of other gods. Religious reform failed. After the death of pharaoh its celestial patron was forgotten and people returned to old gods.

The first monotheistic religion which has survived till the present day, was *Judaism*.¹⁸ This religion was spread only among Jews and did not presuppose missionary activities.

A breakthrough in monotheistic religion was *Christianity*.¹⁹ This religion started with the Jewish sect, which recognized the longed-for Messiah in Jesus from Nazareth. One of his first disciples, Paul from Tars successfully practiced missionary activities, which were addressed to all peoples. Jewish sect converted mighty Roman Empire into their faith and then did the same with other countries.

Another monotheistic religion—*Islam*²⁰—appeared later, in the seventh century, at the Arabian Peninsula. Islam, like Christianity, started with a small sect in a remote provincial place, spread even quicker and conquered a huge territory from the Atlantic Ocean to India. Since that moment monotheism became the main driving force of history.

Monotheism is the further step in *transition to supersystem* and *increase of degree of controllability*. Individual gods were united in one almighty God (transition to supersystem). He could rule the entire Universe and addressing one God only made the interaction simpler—all this is a manifestation of the trend of increasing the degree of controllability.

With the advent of monotheism, a contradiction appeared. How can the god, who is so mighty could allow for so much evil and suffering?

¹⁷ **Monotheism** (lit. “Iatria”, from Greek μονος— one, θεος – God)—is the belief in one God. A narrower definition of monotheism is the belief in the existence of only one God that created the world, is all-powerful and intervenes in the world. —based on materials from Wikipedia.

¹⁸ **Judaism** (originally from Hebrew יהודה, *Yehudah*, “Judah”; via Latin and Greek) is the ethnic religion of the Jewish people, comprising the collective religious, cultural and legal tradition and civilization of the Jewish people. The oldest of religions currently existing in the world – based on materials from Wikipedia.

¹⁹ **Christianity** (from the Greek Χριστός—“the Lord’s anointed”, “Messiah”)—an Abrahamic monotheistic religion based on the life and teachings of Jesus of Nazareth, described in New Testament—based on materials from Wikipedia.

²⁰ **Islam** (араб. الإسلام—“submission”, “sacrificing oneself [to the God]”).

(pronounced: /ɪzlam/; Arabic: اَلْإِسْلَام, romanized: *al-Islām* [al-Islam] listen)) is an Abrahamic, monotheistic Abrahamic religion teaching that there is only one God (Allah), and that Muhammad is a messenger of God—based on materials from Wikipedia.

This contradiction was resolved by transition to **dualism**—appearance of two forces of the Good and the Evil, which struggle one with another. They are often presented in the form of God, who does Good and Satan, who does Evil. In theology the dualism is usually called ditheism or bitheism.

This is also *a transition to supersystem* by integration of *inverse systems*.

Detailed consideration of monotheism shows that it takes the form of a mixture of monotheistic, dualistic, polytheistic and animistic convictions under the uniting slogan of monotheism. A monotheist believes in a monotheistic one God, in a dualistic devil, in polytheistic saints and animistic ghosts. The researchers of religions speak about coexistence of contradictory and even mutually exclusive ideas, combination of rituals and practices from different cultures.

From the standpoint of trends of system evolution this is also *a transition to supersystem* through integration of different systems.

3.4 Independent Work

3.4.1 Questions for Self-check

1. What is evolutionary thinking?
2. List the trends of evolutionary thinking.
3. Name the main elements of hierarchy of system.
4. What trends are used for development of evolutionary thinking?

3.4.2 Topics of Reports and Reference Papers

1. Evolution of religions.
2. Evolution of painting.
3. Evolution of system selected by you.

3.4.3 Fulfill the Tasks

1. Identification of trends of evolution

Constantly identify trends and tendencies in evolution.

1.1 Of your achievements in:

- 1.1.1 Work;
- 1.1.2 Creation;
- 1.1.3 Sports;
- 1.1.4 etc.

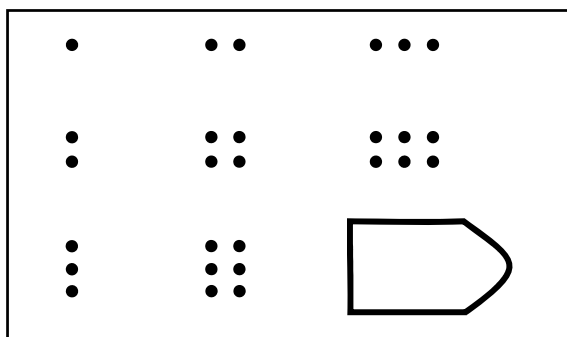


Fig. 3.8 Find the regularity

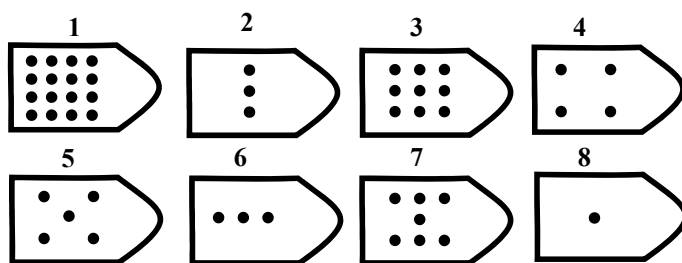


Fig. 3.9 Variants of answers

1.2 In the family budget:

- 1.2.1 Income;
- 1.2.2 Expenditures;
- 1.2.3 Balance.

1.3 Expenditures of your time, for example. Using the system of Lyubishchev.²¹

1.4 Changes in life, which take place around you.

1.5 etc.

2. Solve the problems on identification of regularities

2.1 Problem 3.1. Find the regularity

Condition of the problem

It is necessary to identify the regularity in the position of the figures (Fig. 3.8) and to select the missing figure from the presented variants (Fig. 3.9).

²¹ G. S. Altshuller. On the system of A. A. Lyubishchev URL: <http://www.improvement.ru/bibliot/altshull.shtm>.

2.2 Problem 3.2. Find the regularity and continue the row of numbers

Condition of the problem

101, 112, 131, 415, 161, 718 ...

2.3 Problem 3.3. Find the regularity

Condition of the problem

It is necessary to identify the regularity in the arrangement of figures inside the squares (Fig. 3.10) and to choose the variant of the answer from the tables (A–E) in Fig. 3.11, in order to continue the sequence.

2.4. Problem 3.5. Find the regularity

Condition of the problem

Find the regularity in the sequence, presented in Fig. 3.12, and place necessary figures instead of interrogation marks.

2.5 Problem 3.6. Find the regularity

Condition of the problem

Arrange the arithmetic signs (+ – × ÷) in empty boxes (Fig. 3.13) in such a way so that the equation should be true. In this case it is possible to indicate only one of the signs two times. The arithmetic counting goes one by one, from left to right, without taking the priority of signs into account.

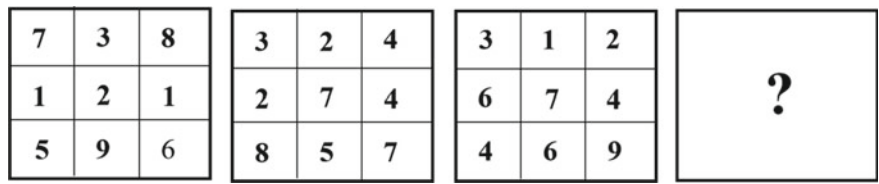


Fig. 3.10 Find the regularity

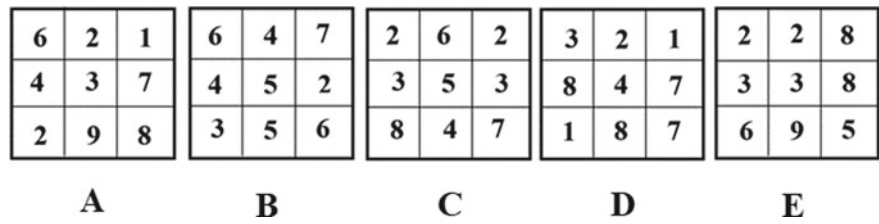


Fig. 3.11 Variants of answers

10	3	6	7	?
1	?	5	4	9

Fig. 3.12 Sequence

6		3		5		7		4		8	=	13
---	--	---	--	---	--	---	--	---	--	---	---	----

Fig. 3.13 Sequence

10	11	9	10	8
?	5	6	4	9
11	?	6	8	7
13	12	?	13	15

Fig. 3.14 Sequence

2.6. Problem 3.4. Find the regularity

Condition of the problem

Figure 3.14 presents one definite sequence of numbers (not necessarily only from left to right). Identify it and insert the missing figures instead of interrogation marks.

2.7. Problem 3.9. Find the regularity

Condition of the problem

Arrange 12 cards, as it shown in Fig. 3.15. The rules are as follows: the sum of all 12 cards is equal to 84 points; all twelve cards are of different value; the points of junior cards are equal to their value; ace = 1, jack = 11, queen = 12, king = 13; the cards of equal color cannot be located one near another neither in vertical, nor in horizontal position; in each horizontal row there should be 4 different suits; each vertical column should contain 3 different suits. Additional conditions are as follows: six is near and higher than ten, which is nearby, and higher than two of spades; the value of card C is three points lower than that of F, which, in its turn, is three points less than that of the card L, which is higher in value than that of A; ace of clubs is near and higher than the card, the value of which is 3 points less than that of the card H, which coincides in suit with the card C; jack of diamonds is in one horizontal row with a card of clubs, the value of which is 2 points higher than that of the card B.

2.8. Problem 3.10. Identify the regularity (Fig. 3.16) and continue the sequence

Condition of the problem

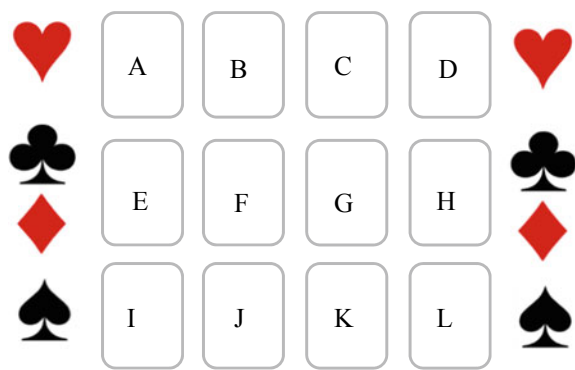


Fig. 3.15 Cards

O	T	T	F	F	S	S	E	N	T
---	---	---	---	---	---	---	---	---	---

Fig. 3.16 Find the regularity

2.9. Problem 3.11. Find the regularity

Condition of the problem

Find the regularity in numbers presented in the figure and insert the missing number instead of the interrogation mark.

97.263
25.298
13.452
?
3.420

2.10. Problem 3.12. Find the regularity

Condition of the problem

Guess the sequence (Fig. 3.17) and add the required letter instead of interrogation mark.

2.11. Problem 3.13. Find the regularity

Condition of the problem

Guess the sequence (Fig. 3.18) and identify, what number should complete the sequence?

Fig. 3.17 Sequence

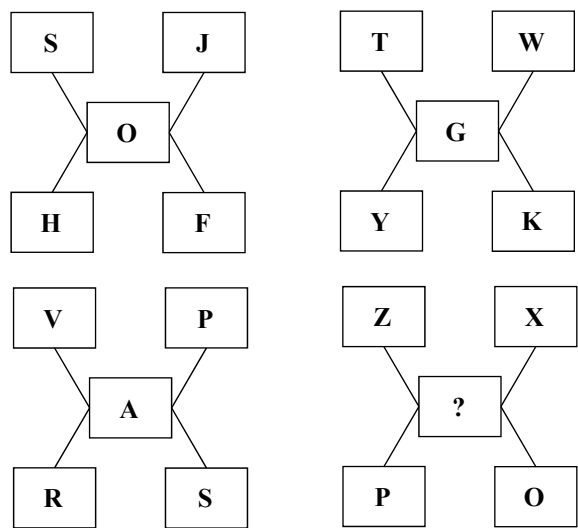


Fig. 3.18 Sequence

3	8	15	22
32	42	51	58
70	79	88	?

2.12. Problem 3.15. Identify the regularity

Condition of the problem

It is necessary to identify the regularity in the arrangement of figures (Fig. 3.19) and to select the missing figure out of the presented variants (Fig. 3.20).

2.13. Problem 3.16. Find the regularity

Condition of the problem

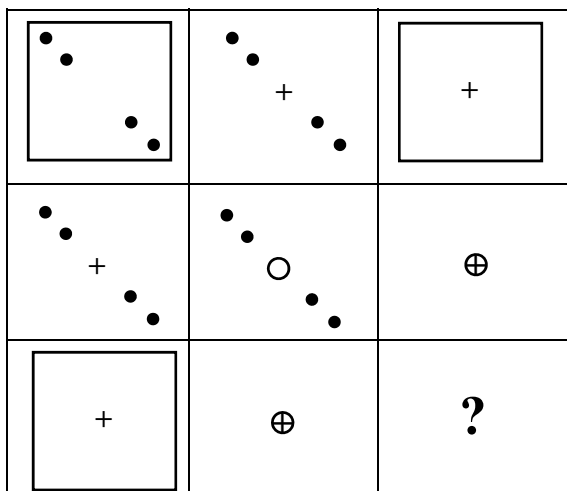
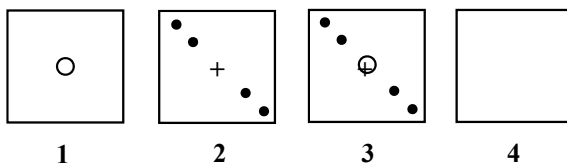
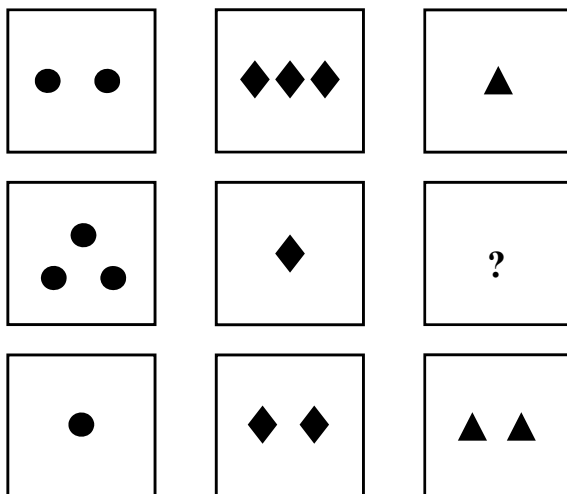
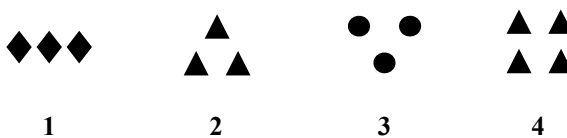
It is necessary to identify the regularity in the arrangement of figures (Fig. 3.21) and to select the missing figure out of the presented variants (Fig. 3.22).

3. Use of trends of system evolution

Trace the history of evolution of a particular system and, using the trends of system evolution, propose, how this system is going to evolve in future.

Systems:

- 3.1 Telephone
- 3.2 TV set

Fig. 3.19 Sequence**Fig. 3.20** Variants of answers**Fig. 3.21** Sequence**Fig. 3.22** Variants of answers

- 3.3 Computer
- 3.4 Lighting
- 3.5 Transport
 - 3.5.1 Car
 - 3.5.2 Train
 - 3.5.3 Ship
 - 3.5.4 Aircraft
 - 3.5.5 etc.
- 3.6 Writing instruments
- 3.7 House
- 3.8 City
- 3.9 Village
- 3.10 Kitchen
 - 3.10.1 Food
 - 3.10.2 Methods of cooking it
 - 3.10.3 Cooking utensils
- 3.11 Shop
- 3.12 Clothes
- 3.13 Medicine
- 3.14 Sports
- 3.15 School
- 3.16 University
- 3.17 Science
- 3.18 Book
- 3.19 Theater
- 3.20 Cinema
- 3.21 Money
- 3.22 The man
- 3.23 Society
- 3.24 Means and methods of amusement and passing free time
- 3.25 Any system, selected by you.

Reference

1. Harari, Yuval Noah; Vintage. 2015. Sapiens: A brief history of Humankind. (Harper), pp. 464.
ISBN-10: 0062316095, ISBN-13: 978-0062316097.

Chapter 4

Thinking Through Contradictions



In order to obtain a product, one should resolve the contradiction: "Having done something, one can win in this and that. But how to arrange it all that you won't lose in something else?"

G. Altshuller

To think creatively (in particularly, to fantasizing) means to identify contradictions and to resolve them.

G. Altshuller

4.1 Review

Thinking through contradictions is an ability to identify and resolve contradictions.¹

Notions and definitions of contradictions were given in textbooks [1–5]. Let us recollect them.²

Creation of a new object most often implies the improvement of parameters of the system.

Complicated inventive problems (of unknown types) require a non-trivial approach, since the improvement of some parameters of the system leads to inadmissible deterioration of other parameters. **A contradiction** appears.

Contradiction is one of the *basic concepts of TRIZ*.

Three kinds of contradictions are considered in TRIZ. TRIZ author Genrich Altshuller called them:

- **Administrative contradiction (AC);**
- **Technical contradiction (TC);**
- **Physical contradiction (PC).**

¹ G. S. Altshuller. Letter of 51.09.12.1981. URL: <http://www.altshuller.ru/corr/correspondence1.asp#tc>.

² Idem.

These terms were selected by Altshuller for solving technical problems, however, they are not always suitable for solving problems from other fields, therefore, we called them as follows:

- Administrative contradiction—**surface contradiction (SC)**;
- Technical contradiction—**requirements contradiction (RC)**;
- Physical contradiction—**contradiction of properties (CP)**.

Surface contradiction (SC)—*contradiction between the need and the possibility to satisfy it.*

It is rather easy to identify. It is often assigned by administration or by the customer and is formulated in the following form: “*It is necessary to perform this and that, but how to do it, is not known*”, “*Some parameter of the system is bad, it has to be improved or some disadvantage is to be eliminated, however, it is not known, how to do it*”, “*There are faulty goods in the manufacturing of products, while the reason for that is not known*”, etc.

Thus, SC is expressed in the following form:

- Undesirable effect (UE)—something is bad;
- Improvement—it is necessary to create something new, but it is not known how.

Problem 4.1. Bus

Condition of the problem

The bus should transfer many passengers. How to arrange for it?
This is a typical AC.

Requirements Contradiction (RC) is a *contradiction between certain parts, features or parameters of the system.*

RC appears as a result of improving certain parts (parameters) of the system due to inadmissible deterioration of other parameters.

This is the reason for appearance of *surface contradiction (SC)* by deepening it. Most often several RC are found in the depth of one SC.

RC appears as a result of disproportion of evolution of different parts (parameters) of the system. In case of significant quantitative changes of one of the parts (parameters) of the system and dramatic “lagging behind” of its other part (parts) a situation emerges, when the quantitative changes of one of the sides of the system come into contradiction with other sides.

Let us continue to consider the problem concerning the bus.

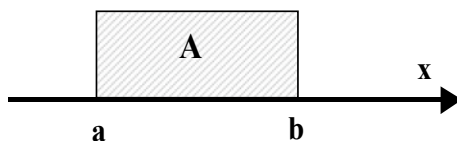
Problem 4.1. Bus (Continued)

Analysis of the problem

In order to carry many passengers, the bus should be capacious, i.e., it has to be of great dimensions. However, a big bus is characterized by poor maneuverability. Thus, it is possible to formulate RC.

RC: contradiction between **capacity** of the bus and **maneuverability**.

Fig. 4.1 Graphic representation of an interval



Contradiction of properties (CP) is conferring diametrically opposite features (*for example, physical*) or conditions of a certain part of a system.

It is necessary for identification of reasons, which generated *requirement contradiction*, i. e., is the further deepening of it. Refinement (deepening) of contradictions may continue further for the sake of identifying the primal cause.

For a person, who is not acquainted with TRIZ, the wording of CP sounds unusual and even wild. The requirement to the wording of CP sounds as follows: **some part of the system should stay in two mutually excluding conditions at once**: be *cold* and *hot*, *mobile* and *immobile*, *long* and *short*, *flexible* and *rigid*, *electrically conductive* and *non-conductive*, *exist* and *not exist*, etc.

One of the features meet some of the parameters of CF, while another feature meets another parameter.

FC is a kind of unusual inequation. Usually **a distance** (or **an interval**) is indicated with the aid of inequations, for example,

$$a < x < b \quad (4.1)$$

An interval is graphically presented in Fig 4.1.

This interval is the zone “A”. An “optimal” solution is usually sought for in this interval.

TRIZ acts in quite a different way.

It is identified, what property “c” the system should possess, so that the parameter “A” (in RC) should be the best. Let us mark this feature as “P”. Farther on, it is identified, what property “p” the system should possess, so that the parameter “B” should be the best. Let us mark this property as “anti-P”.

This conflicting action is presented in the form of an inequation (4.2).

$$P > p > \text{anti-P} \quad (4.2)$$

For the sake of visualization let us present this inequation in Fig. 4.2.



Fig. 4.2 Picturing-contradiction of properties

The wording of RC requires that “C” should concurrently stay in the zone “A” and in the zone “B”, which is *impossible*, judging by the diagram (Fig. 4.2).

Problem 4.2. Power supply unit

Industrial grid of alternating current is used for powering many electronic units, though the majority of their units, for example, amplifier, generator and others need constant powering voltage. Because of this at the output of the feeding unit an element is necessary, which has contradictory physical features.

CP: Element of a power supply unit **should pass alternating current**, since there is no other source of current, and **should not pass alternating current**, since the device needs direct current.

Resolution of contradiction

This contradiction of properties is resolved in **structure** due to passing of only positive half-wave and non-passing of the negative one. It is achieved with the aid of a rectifier, embodied on diodes, which possess indicated physical properties and implementing the function of transducing alternating current into direct current.

Let us continue to analyze the problem concerning the bus.

Problem 4.1. Bus (continued)

Analysis of the problem.

Let us formulate CP for the given problem.

CP: The bus should be *small*, in order to be **maneuverable** and *big*, in order to **be able to transport** many passengers.

To be more exact, these are the requirements, which are given not to the whole bus, but only to its *passenger compartment*.

In order to identify the necessary properties of the system (CP), it is necessary to know the direction, which corresponds to the ideal solution—**ideal final result (IFR)**. The identification of IFR and its main properties were described in the textbook of Level 1 [2]. Let us recollect them.

Main features of IFR:

1. Improve a bad parameter without deteriorating the good one.
2. Improve parameters without making the system more complicated.
3. Improve parameters without causing harmful actions.
4. Improve parameters at a required moment.
5. Improve parameters in a required place.
6. Improve parameters according to the necessary condition.
7. All actions should be performed independently.

The main line of solving problems takes the form of a chain:

$$SC \rightarrow RC \rightarrow IFR \rightarrow CP \rightarrow SOLUTION \quad (4.3)$$

Connection between the elements of this chain is described in the logic of solving non-standard problems (Fig. 4.3).

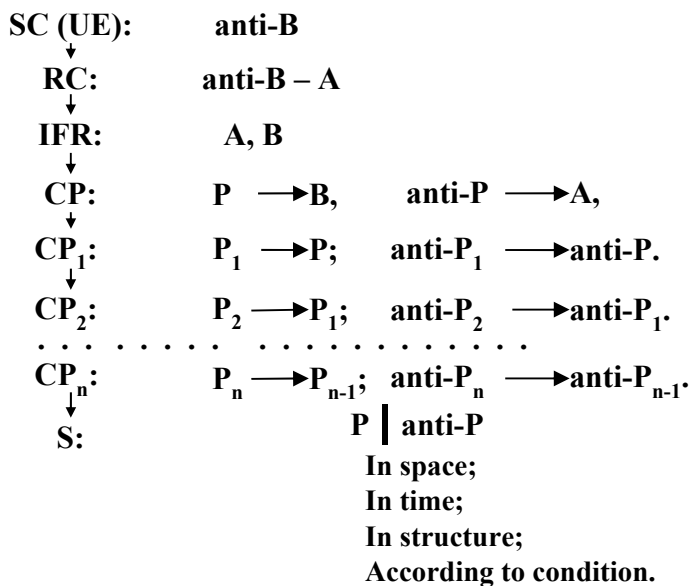


Fig. 4.3 Logic of solving non-standard problems

Notation:

SC—surface contradiction.

S—solution.

RC—requirements contradiction.

A, B—properties or parameters of the system.

CP—contradiction of properties.

P, P₁, P_n—required properties or conditions of the system.

IFR—ideal final result.

|—sign of separation of contradictory properties.

Thus, thinking through contradictions could be described in the form of logics of solving non-standard problems.

4.2 Practice

4.2.1 Conditions of the Problems

Problem 4.3. Beryllium

Beryllium is one of the hardest metals and has a high elastic modulus. Construction materials based on beryllium are characterized concurrently by lightness, hardness and resistance to high temperature. Being 1.5 lighter than aluminum, these alloys

are at the same time harder than many specialized steels. In connection with that it is profitable to manufacture especially rigid and light structures.

However, beryllium is rather toxic, it causes hard lung diseases, which are often finished with a lethal outcome, therefore, beryllium does not find extensive use in technology.

A compromise outcome is beryllium alloys. However, this is associated with the violation of crystal structure of the metal, which creates exactly those properties, which should be used.

What's to be done?

Problem 4.4. Odo Nobunga

Military and political leader of Japan Odo Nobunaga, one of the most outstanding samurais in history of Japan, was cruel, had the appearance of a monkey and could not stand lies. Everyone, who spoke lies, was killed by him.

– Is it true that I resemble a monkey?—he asked once one of his attendants.

What would you answer, if you were this attendant?

Problem 4.5. Pharaoh

The pharaoh had unrestricted uncontrolled power.

What should be done to make him rule without abusing his people and without committing crimes?

Problem 4.6. School inspector

The inspector, who was inspecting some school, noticed that each time he asked the class a question, all pupils raised their hands, being eager to answer. Moreover, though the teacher each time chose different pupils, the answer was always correct.

How could it be done?

Problem 4.7. Travilo (from the Russian word “*travleniye*”, meaning etching)

In the novel by Yu. Bryder and N. Chadovich “*Bastions of Dith*” the citizens of the city of Dith have special guns intended to shoot at different evil forces. These guns shoot with balls made of travilo. **Travilo** is a substance, which corrodes or etches practically everything: metal, clothes, flesh... How can one charge such a gun with travilo? It will etch through the gun itself, and through the hands of the shooter.

Problem 4.8. Repair of tug-boats

Dudinsk seaport is situated in the north of Russia in Krasnoyarsk region and is created for servicing Norilsk combine. The ships are brought to this region by tug-boats.

Every year it is necessary to examine the underwater part of tug-boats and, if necessary, to repair it.

For this purpose, the tug-boats are transported to repair plant or a special structure—dockage facility is used.

The nearest repair plant is in Murmansk, which is rather far, while there is no dockage facility in the port and it costs quite a lot.

What's to be done?

Problem 4.9. The duel

In the film “New Don Juan” the servant, pretending that he is Don Juan, gets challenges to the duel from half a hundred noblemen.

What should he do?

Problem 4.10. Lost nuts

Changing the wheel of his car, a man dropped four nuts of its fastening, so that they fell through the grill of the sewage runoff and it was impossible to get them. He was sure that he would have to remain there for long, but the boy who was passing by, gave him rather a good advice, which enabled him to drive further. What did his idea consist in?

Problem 4.11. Penstock

While building a penstock at the Oust-Ilim hydro-power plant in Siberia it was necessary to lower down a 4000-tonnes pipe on a slope.

The pipe is built vertically (Fig. 4.4a), and then is lowered down on the slope (Fig. 4.4b). Corresponding devices did not exist at that time, and one had to develop a special system of lifting cranes or leveling jacks. It requires much time and money.

What's to be done?

Problem 4.12. Overheating of tire casings

Long-distance lorry drivers often drive along the highways at a great speed and without stopping. It leads to the overheating of tires due to rubbing against the asphalt, especially in summer, when the asphalt is heated by sunbeams. The temperature of the tire may attain the value of over 130 °C. The drivers have to stop and to wait until the tires get cooled.

What's to be done in order to avoid that?

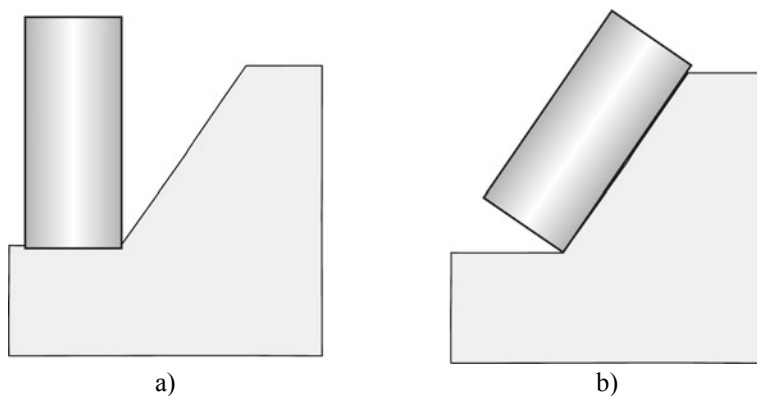


Fig. 4.4 Lowering penstock to the slope

4.2.2 Analysis of Problems

Let us perform the analysis according to logic of solving non-standard problems.

Problem 4.3. Beryllium

SC: anti-B

SC. Beryllium is harmful for human beings. Undesirable effect (UE)—*harmful action upon the human (anti-B)*.

RC: A – anti-B

IIT. Beryllium is a very *strong* and *light metal* (A), however, it *harmfully acts* upon the human (**anti-B**).

IFR: A, B

IFR. Beryllium *does not harmfully act upon the human* (B) and is characterized by *high hardness and lightness* (A).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP. It is *necessary* for the human being *to be in contact* (P) with beryllium, in order to service structures, made of it, which are characterized by *high hardness and lightness* (A), and *the humans should not be in contact* (anti-P) with beryllium, in order *not to do harm to the human health* (B).

Resolution of CP

Separation of contradiction **in space**.

Solution

Beryllium structures are used for creating objects, with which the human does not come in contact, for example, for artificial sputniks of the Earth.

Specialists from the company British Aircraft performed research, which showed that the application of beryllium in supporting structures of artificial sputniks of the Earth enables to reduce the weight of these structures by 35%. What concerns the

properties of beryllium, which are harmful for the human, they are insignificant for sputniks, which operate in automatic mode.

Problem 4.4. Odo Nobunaga

SC: A

SC. How to remain alive? Positive effect (PE)—to remain alive (**A**).

RC: A – anti-B

RC. Contradiction between the necessity *to remain alive* (**A**) and the necessity *to tell the truth*, that the samurai resembles a monkey (**anti-B**).

IFR: A, B

IFR. The attendant remains *alive* (**A**) and *tells the truth* (**B**), that the samurai resembles a monkey.

CP: P → A, anti-P → B

CP. The attendant *should tell the truth* (**P**) to the samurai, that he resembles a monkey, since he did not like lies, and *shouldn't tell the truth* (**anti-P**), the samurai resembles a monkey, in order to remain alive.

Resolution of CP

Subdivision of contradiction **according to condition**.

Solution

– God bless you, Your Excellency! — he shouted. —How could it be?! On the contrary, the monkey has the honor to resemble you!

It caused the laughter of the samurai and set him into an excellent mood.

Principle 13 is used. *Other way round*.

Problem 4.5. Pharaoh

SC: anti-B

SC. How to avoid the abusing of people by the pharaoh and his crimes? Undesirable effect (UE)—*pharaoh's crimes* (**anti-B**).

RC: A – anti-B

RC. Contradiction between *efficiency of ruling the state with unrestricted power* (**A**) and *crimes* (**anti-B**).

IFR: A, B

IFR. The pharaoh has *unrestricted power* (**A**) and *does not commit crimes* (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP. The pharaoh *shouldn't be under the control of the people* (have uncontrollable power), in order to *efficiently rule a big state* (**A**), and *should be under the full control of the people*, in order to rule without *disgrace and crimes* (**B**).

Resolution of CP

Separation of contradiction in **time**.

Solution

In Ancient Egypt during the life of the pharaoh all his orders (including criminal ones) were unquestionably fulfilled, but after the death of the pharaoh the people judged their ruler. And if at least one man had claims to the pharaoh and did not forgive him his guilt, the pharaoh was sentenced to the highest punishment: he was denied the burial. And without the burial the soul, according to beliefs, was destined to experience eternal tortures... The mummy of non-buried pharaoh was installed—as edification—in the house of “the punished”, so that it should be an eternal reproach to the inheritors.³

Problem 4.6. School inspector

SC: A

SC. How did the pupils give a correct answer each time? Positive effect (PE)—all give *a correct answer* (**A**).

RC: A – anti-B

RC. Contradiction between the manifestation of activity of pupils (all of them raise their hands) (**anti-B**) and their knowledge (correct answer) (**A**).

IFR: A, B

³ Oppel [6].

IFR. 100% activity of pupils (*all of them raise their hands*) (**B**) and *all of them give correct answers* (**A**).

CP : $P \rightarrow A$, anti- $P \rightarrow B$

CP. In order to *manifest activities* (**B**) *all pupils should raise their hands* (**P**), in order to give *correct answer* (**A**), *not all pupils should raise their hands* (anti-**P**).

Resolution of CP

Separation of contradiction in **structure**.

Solution

The teacher agreed with his pupils in advance that they would volunteer to answer irrespective of if they know the answer or not. However, those, who know the answer, should raise the right hand, and those who don't know it – the left hand. Each time the teacher chose a new pupil, but always the one, who raised his right hand.

Principle 10 is used. *Principle of preliminary action and resources*.

Problem 4.7. Travilo

SC: anti-B

SC. How to shoot with a travilo, which can etch through everything? Undesirable effect (UE) – travilo *brings harm to arms and to people servicing it* (**anti-B**).

RC: A – anti-B

RC. Contradiction between the necessary property of travilo *to etch through everything* (**A**) and bringing *harm to the arms and servicing people* (**anti-B**).

IFR: A, B

IFR. Travilo retains the features *to etch through everything* (**A**) and *not to bring harm to the arms and people* (**B**).

CP : $P \rightarrow A$, anti- $P \rightarrow B$

CP. Travilo *should etch through everything* (**P**), since this is its *basic feature* (**A**), and *shouldn't etch through everything* (anti-**P**), since it will bring *harm to arms and people* (**B**).

Resolution of CP

Separation of contradiction in **time**.

Solution

Preparation of balls made of travilo takes place directly inside the gun of components, which are not dangerous if taken separately and which are put inside the gun.⁴

There is a small disadvantage of the idea described in the novel. These components cannot be united inside the gun. Travilo will etch through the gun itself.

Binary weapons are known—two or more safe components are united in a shell during the flight. In the air the interstices between the components are broken, for example, due to rotation of the shell and the substances are mixed.

Problem 4.8. Repair of tug-boats

SC: anti-B

SC. Absence of dock in Dudinsk port. Undesirable effect. Undesirable effect (UE)—absence of dockage facility (**anti-B**).

RC: A – anti-B

RC. Contradiction between the necessity of *examination of tugboats bottoms* (**A**) and *absence of docking facility in the port* (**anti-B**).

IFR: A, B

IFR. *Examination and repair of tugboat bottom* (**A**) is performed *without the acquisition of the docking facility*. (**B**).

CP : P → A, anti-P → B

CP. *The docking facility is necessary* (**C**), in order to *perform examination and repair of the bottom of tugboats* (**A**), and *the docking facility is not necessary* (**anti-P**), in order *not to acquire it* (**B**).

Resolution of CP

Separation of contradiction in **structure**.

Solution

The docking facility was “built” of ice (Fig. 4.5).

⁴ Yu. Bryder, N. Chadovich. Bastions of Dith//Bryder and Chadovich [7].

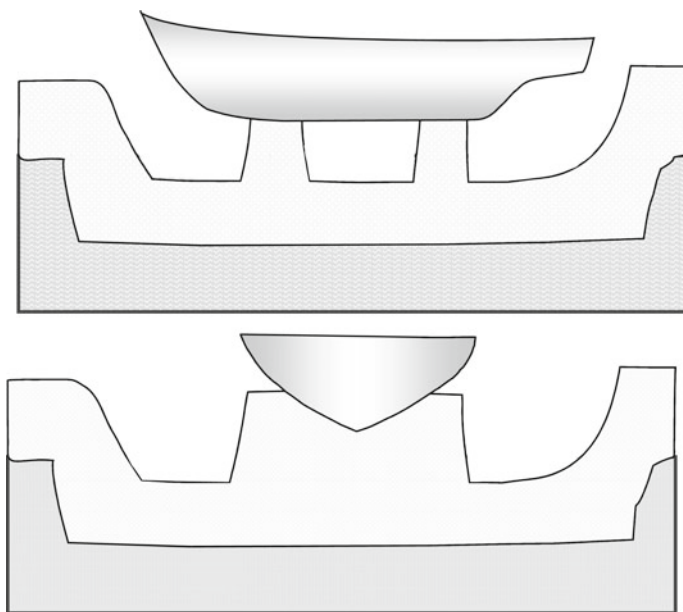


Fig. 4.5 Ice docking facility

In autumn the tugboats were left on the river.

When the river started to freeze, the ice was gradually cut away, thereby “creating” an ice docking facility.

Problem 4.9. Duel

SC: anti-B

SC. How to remain alive after a duel. Undesirable effect (UE)—*to perish as a result of a duel* (**anti-B**).

RC: A – anti-B

RC. *He fulfills the promise (A) to take part in the duel*, since that is the matter of honor for a nobleman, however, he will *die* as a result of it (**anti-B**).

IFR: A, B

IFR. *He fulfills the promise (A) to take part in the duel, and does not die (B).*

CP : $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP. The servant *has to fight the duel* (**P**), in order to *fulfill the promise* (**A**) to take part in the duel, and *does not have to fight the duel* (**anti-P**), in order to *remain alive* (**B**).

Resolution of CP

Separation of contradiction according to **condition**.

Soution

The servant proposes them to resolve the dispute themselves and to decide whose turn is to fight first. They decide that in a usual way—through duels. As a result all are killed.

Resources *of knowledge* of the human nature are used.

Problem 4.10. Lost nuts

SC: anti-B

SC. Absence of nuts for fastening the wheel on a car. Undesirable effect (UE)—*absence of fastening of the wheel* (**anti-B**).

RC: $A - \text{anti-B}$

RC. Contradiction between the necessity *to drive* (**A**) and *absence of fastening of the car wheel* (**anti-B**).

IFR: A, B

IFR. The car *moves* (**A**) *without corresponding nuts*, which fasten the wheel (**B**).

CP : $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP. Nuts *should exist* (**P**) for *fastening the wheel*, so that one could (**A**), and *should not exist* (**anti-C**), since *they are absent* (**B**).

Resolution of CP

Separation of contradiction in **structure**.

Solution

The boy proposed to unscrew one nut from each of three wheels and to fasten the fourth wheel with them. Having done it, the man managed to drive up to the nearest garage on firmly fastened wheels.

Resources of the machine are used.

Problem 4.11. Water duct

SC: anti-B

SC: Absence of equipment for lowering the water duct on the slope. Undesirable effect (UE)—absence of equipment (**anti-B**).

RC: A – anti-B

RC: *Necessity for lowering down the equipment on the slope (A)* and the absence of equipment (**anti-B**).

IFR: A, B

IFR: The water duct is lowered down on the slope (**A**) without special equipment (**B**).

CP : $P \rightarrow A$, anti- $P \rightarrow B$

CP: *It is necessary to have equipment (P)*, in order to *lower down the water duct on the slope (A)*, and *it is not necessary to have the equipment (anti-P)*, since *it is absent (B)*.

Resolution of CP

Resolution of contradiction **in structure**.

The water duct should lower down independently, without equipment.

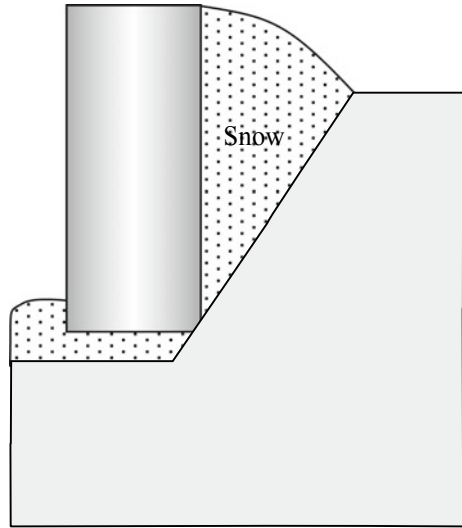
Solution

Under the pipe and between the pipe and slope snow was thrown (Fig. 4.6). When it melted in spring, the pipe lowered down.

Problem 4.12. Overheat of tires

SC: anti-B

Fig. 4.6 Lowering down the water duct on the slope



SC: Car tires get overheated as a result of long-time driving. Undesirable effect (UE)—*tire overheating (anti-B)*.

RC: A – anti-B

RC: *Necessity for quick and long-time driving (A) leads to overheating of tires and to their damage (anti-B).*

IFR: A, B

IFR: Quick and long-time driving (A) does not lead to overheating and damage of tires (B).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: *The tire gets heated (P) as a result of quick and long-time driving (A), and should not get heated (anti-P), in order not to damage the tire (B).*

Resolution of CP

Resolution of a contradiction **in structure**.

The tire should get cooled itself.

Solution

A small amount of water is pooled inside the tires. It starts to boil at the temperature of approximately 100 °C, gets converted to vapor, absorbing heat from the tire. Further

on, the vapor gets on cold surface (for example, the rim of the wheel). Water vapor gets condensed, being converted into water. A cycle of such phase transitions is repeated several times.

4.3 Independent Work

4.3.1 *Questions for Self-control*

1. What is a contradiction? Give a definition.
2. Describe types of contradictions in TRIZ.
3. What is a surface contradiction? Give a definition.
4. What is a requirements contradiction? Give a definition.
5. What is a contradiction of properties? Give a definition.
6. Describe the methods for resolution of requirements contradiction. Give examples.
7. Describe a chain of contradictions, used in TRIZ. Give examples.
8. What is logic of solving non-standard problems?

4.3.2 *Topics of Reports and Reference Papers*

1. Examples of contradictions in your field of knowledge and methods for resolving them.
2. Use of a chain of contradictions in solving problems in your field of knowledge.
3. Use of logic of solving non-standard problems for solving problems in your field of knowledge.

4.3.3 *Fulfill the Tasks*

1. **Give examples of solving a physical contradiction.**
 - 1.1. In space.
 - 1.2. In time.
 - 1.3. In structure.
 - 1.4. According by condition.
2. **Solve problems using the logic of solving non-standard problems**
- 2.1 **Problem 4.13. The castle of Montaigne**

Conditions of the problem

France. sixteenth century. Civil war. Philosopher Michel de Montaigne had a castle of his own. His neighbors—noblemen locked their castles and armed their peasants. Montaigne had neither servants, nor arms ...

What trick did he use, so that his castle would not be robbed?

2.2 Problem 4.14. Artificial leg**Conditions of the problem**

In prosthetics it is very important that the artificial leg should have the same shape as the actual leg. It might seem to be simple: to make a plaster cast of an actual leg and then to cast an artificial leg. However, it does not work, because nobody needs two left or two right legs.

What's to be done?

2.3 Problem 4.15. Taking off the paranja in Iran**Conditions of the problem**

The last shakh of Iran Mokhammed Reza Pekhlevi was a progressive man, and he did not like that the women wear paranja only in his country, that is the first point. The second and the most important point: he wanted to weaken the influence of Islamic clergy in the country, since wearing the paranja is a commandment of Islam and to take it off means to dishonor the woman.

But how is it possible to make the women take the paranjas off? If an order is issued for the women to take the veils off their faces, it will be an abuse of religious feelings of the majority of population. It cannot be ordered, because civil disturbances could start.

2.4 Problem 4.16. Conveyer of Ford**Conditions of the problem**

In due time, when Henry Ford created his conveyor, he was very proud of it and organized guided tours to the workshops, demonstrating the functioning of the conveyor and acting as a guide himself. However, crowds of visitors created hindrances for the workers, distracting them from their work, which decreased productive efficiency. The workers complained to Ford about that and he invented a trick, which eliminated the problem. What did he do?

2.5 Problem 4.17. Ancient sculpture**Conditions of the problem**

Michelangelo wanted to prove his geniality and to falsify the sculpture "Sleeping Cupid".

According to the story written by the artist and historian of Renaissance epoch Giorgio Vasari, the sculpture was acknowledged to be an authentic Ancient statue and sold to the Cardinal San Giorgio Raffaello Riario for 200 ducats.

How did Michelangelo manage to do that?

2.6 Problem 4.18. Chinese braids

Conditions of the problem

In 1641 the Manchurians established their ruling in China (Tsing dynasty). As a sign of faithfulness, the Chinese had to shave their heads and to leave a tress on the top of the head. First the barbers were commanded to execute the order of the Emperor. With a razor in one hand and sword in the other they patrolled their territories, offering each Chinese a choice: either to sacrifice the head or the haircut. The majority agreed to undergo an operation, though it was shameful, but some people were indignant and even rebelled. However, the second Emperor of Manchurian dynasty Kan Shi chose a wittier and more efficient measure, as a result of which the Chinese themselves, without any compulsion started to shave their heads and carry braids on the tops of their heads.

What did the wisdom of Kan Shi's decree consist in?

2.7 Problem 4.19. Evacuation of a manufacturing plant

Conditions of the problem

During the Great Patriotic War one of the most urgent tasks of industry was the evacuation of manufacturing plants located near the front line. One of such plants was the metallurgical combine "Severonickel". A decision was taken concerning the dismantling of the combine and transporting it to Norilsk. Trying to capture a strategically important manufacturing plant of non-ferrous industry as a single whole and intact, the Germans did not bombard it, which, of course, made the evacuation of equipment easier. However, the employees of Severonickel understood that it was enough for the Germans to know about the dismantling of the combine, which was in full swing, and hundreds of bombs will be thrown on the territory of the manufacturing plant. It means that one had to deceive the enemy.

But how?

2.8 Problem 4.20. The Bat

Conditions of the problem

As it is known, the bats are oriented in space through ultrasound. In order "to see" small objects by hearing them, the bats should have a rather sensitive hearing apparatus, while in order to be oriented at long distances, they should issue an ultrasound of large amplitude. However, in this case the mouse would stun itself: the sound issued by it would damage the auditory nerve endings.

What outcome did nature find?

2.9 Problem 4.21. Children from nonboring houses

Conditions of the problem

One American farmer was annoyed at the children of his neighbors. They chose a lawn in front of his house as a favorite place for their noisy games. Constant running and shouting threw the farmer out of his gear. The first thing that he did was to shout at the boys. The result was zero, “thuggish behavior” continued.

Then the farmer changed his tactic. He invited the children to his house, treated them to some candy and placed an agreement with them. In a few days there was peace at the farm.

What did the farmer propose to the children?

2.10 Problem 4.22. Fishing

Conditions of the problem

Since old times the citizens of the southern France fished in the mouth of the Rhone river. With the tide hoards of fish come into the river, while with the low tide they go back into the sea. It is not possible to place a fishing net on their way: it will be carried away by the low tide.

What's to be done?

2.11 Problem 4.23. A Wise servant

Conditions of the problem

There lived a Khan, who was very angry, all people were afraid of him. The Khan loved only his son Khusain. Khusain every day went hunting. It happened once that he did not return home by night. The Khan got angry and called all his servants. He ordered them to seek for his son, but the man, who would bring bad news will have a melted lead thrown into his throat. After long war ambling through the woods, the servant finally found Khusain, but he was dead: the wild animal killed him.

How can the servants tell Khan about the death of the beloved son, but at the same time remain alive?

2.12 Problem 4.24. A False picture

Conditions of the problem

One of the museums of Germany acquired a picture by Rubens in a museum of France. The German experts, who examined the canvas in situ, acknowledged its authenticity and confirmed it with their signatures on the back side of the picture. After that the picture was packed and sent to Germany. When it was delivered to the indicated address, another examination was made (the picture might have been replaced during transportation) and it was discovered that it was a fake, while on the reverse side of fake picture there were authentic signatures of experts, certifying its authenticity!

2.13 Problem 4.25. Lorgnette**Conditions of the problem**

Catherine the Second did not like aristocrats carrying lorgnettes and long white gloves.

But how could one prohibit these things?

What's to be done?

2.14 Problem 4.26. Potatoes**Conditions of the problem**

In France potatoes were treated with incredulity for long time.

How to disseminate potatoes in France?

2.15 Problem 4.27. Sharks**Conditions of the problem**

A pneumatic gun is used for protection against sharks, which shoots explosive bullets. Such bullets are sure to kill the shark, however, the blood, which is discharged in large amount, attracts other flesh-eaters. What's to be done?

2.16 Problem 4.29. Burial in Tibet**Conditions of the problem**

There are neither cemeteries, nor individual graves in Tibet. An exception is made only for Dalai Lama and Panchen-Lama. Their bodies after their death are blamed and covered with gold. It is very difficult to dig graves in such rocky soil and the religious philosophy dictates not "committal to earth", but "committal to sky". However, since there is no fuel of any kind, cremation is impossible.

What do they do in Tibet in order to perform the ritual of transcending of the dead man to a different level of existence?

2.17 Problem 4.31. The Genie**Conditions of the problem**

A peasant once unearthed a jug on his field. He immediately opened it—he expected to find golden coins inside... However, an angry Genie rushed out of the bottle.

- For thousand years I prayed to God that somebody would let me out, – shouted he, shaking his fists. – I promised heaps of gold and diamonds to my savior... But nobody even listened to my prayers! Then I decided to kill the man, who would disturb my peace. One more thousand years passed and you decided to look for hidden treasures in your own kitchen garden!!! You are in your dying hours, poor man!

However, the peasant was not a coward...

- Let it be like this, Genie. But fulfill my last request.
- Speak it out,—said the Genie reluctantly. —I will fulfill it. But bear it in mind: I will not let you go, I will not do any harm to myself, and I'm not going to get back into the jug. I also read fairy tales and I will not let anyone deceive me.

The peasant expressed his request—and remained alive. What in particular did he ask for?

2.18 Problem 4.32. Vaccine

Conditions of the problem

Influenza is an infectious disease. It can be eliminated through vaccination. However, the virus of influenza is able to mutate quickly. By the time a serum is developed against a particular strain of virus, the next variant of it already exists in the world. In connection with this the struggle with epidemics of this illness is rather difficult.

What's to be done?

2.19 Problem 4.33. Ferryboat

Conditions of the problem

Ferryboats are intended for carrying people and means of transportation by water between two shores (of the river, lake, straight and sea). One of the most widely spread kinds of a ferryboat moves along the rope, stretched between two shores of a river. The ferryboat may take the form of a boat, several connected boats or a pontoon. Either an engine or the force of the human muscles are used for motion. In the latter case the ferryman rotates the pulling hoist, applying significant force (Fig. 4.7).

How to make the work of the ferryman, easier, if there is no opportunity to use the engine?

2.20 Problem 4.34. Chess-player

Conditions of the problem

One weak chess-player proposes a strong chess-player to have a chess game with him through letter-exchange for money. The condition is that in case of a loss the weak player pays the strong one a sum, which is two times less, because he plays worse. If the match ended in a draw, the strong player pays the full sum to his partner, but in case a weak player wins, he gets a double sum.

He has no chances to win, but nevertheless he gets at least a half of the sum. How did he manage to do it?

2.21 Problem 4.33. Blinding the driver

Conditions of the problem

Rays of the sun blind the driver. The drivers use protective visor, but it restricts the vision. Neither is the use of protective sunglasses always comfortable.

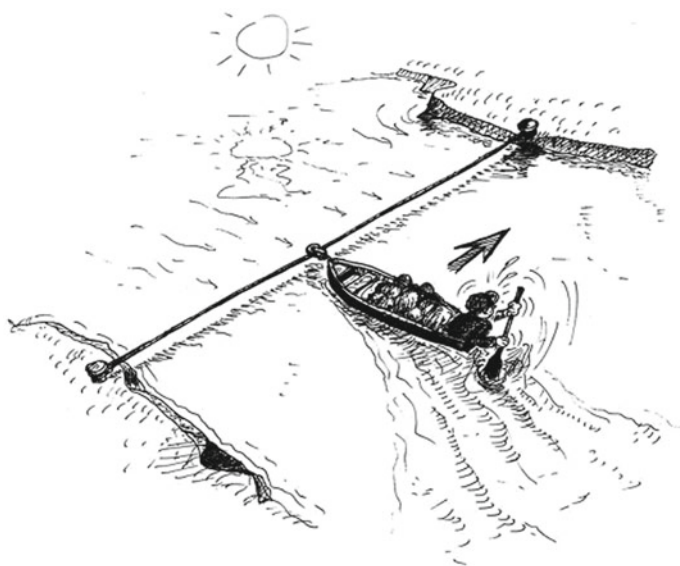


Fig. 4.7 Ferryboat crossing⁵

What's to be done?

2.22 Problem 4.34. Foldable umbrella

Conditions of the problem

Invention of a foldable umbrella significantly simplified its use. However, there are disadvantages:

- The umbrella creates a hindrance in entering the building or in getting into a car. It is necessary to close it prior to entering the building or getting into a car. During this time the owner of the umbrella will get wet under the rain.
- The outward surface of the umbrella gets wet under the rain and everything, which is touched with the umbrella, also gets soaked.

What's to be done?

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Chapter 5

Resource Thinking



*The lesser amount of resources we have at our disposal, the more rationally we use them.
The kite flows up only against the wind.
Winston Churchill
(1874–1965)
British statesman and politician*

5.1 Review

Resource thinking is an ability to identify and to use resources.

The notions and definitions of resources were given in textbooks [1–3]. Let us recollect them.

Resources is a very important section of TRIZ. Application of resources increases ideality of the system and of problem solving, therefore resources are used practically with every TRIZ tool.

Use of resources is a powerful tool for improvement of system and for making them less expensive.

Initially it is necessary *to identify resources* and then *to use them*. The general algorithm is shown in Fig. 5.1.

Structure of resources is presented in Table 5.1.

5.2 Practice

This section contains examples and problems on using different kinds of resources.

Example 5.1 Components of Ford

It happened once that Henry Ford ordered his engineers to explore all used car dumps of the USA, in order to understand, what components and with what frequency are

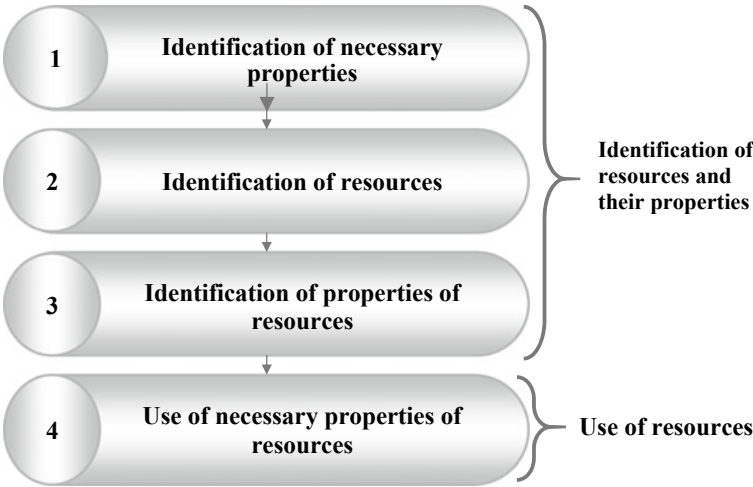


Fig. 5.1 General algorithm for identification and use of resources

Table 5.1 Structure of resources

Provision of resources			Subsystem	System	Supersystem	Environment
Type of resource	Function					
	Structure	Element				
		Connections				
		Shape				
	Substance					
	Energy					
	Information					
	Flow	Substance				
		Energy				
		Information				
	Space					
	Time					
	System effect	Process				
		Result				

broken in “Ford-T”. It appeared that everything could be broken but the central projecting pin of the front bridge, which never got broken.

Ford gave the order to reduce its quality.

Resource—excessive quality.

Example 5.2 Victories of sultan

Sultan Abdul-Ata won several battles.

Before the battle he made his wives open their faces and undress in front of the enemy's army. It so strongly shocked the Moslems that sometimes drove them to suicides.

Resource—psychological effect.

Example 5.3 Bombers

During World War II color-blind people were included with the crews of British bombers for the purpose of enhancing the efficiency of bombardments.

They could not be deceived by the German camouflage nets.

Resource—peculiarities of the eyesight of color-blind people (resources of knowledge of color-blind people are used).

Example 5.4 The Spy

During World War II British counterintelligence got a message that there is a German spy among the passengers of a railway car. The British detained all passengers and for 6(!) hours questioned them making special emphasis on their biographical data.

The agent gave himself away since he was the only one among the passengers who never got tangled.

Resource—psychological effect.

Example 5.5 Preservation of ecological purity

Ecological administration cannot monitor all enterprises.

A system of ecological monitoring has been created in the USA. A minimum unit, for which the limit of emissions has been stipulated, is a territorial production cell, which often includes several enterprises, which have to monitor one another and make one another reduce the emissions. Moreover, a permission to start new production or to enlarge the existing one is issued only on condition of reducing the total amount of emissions. If you want to enlarge your manufacturing—reduce the harmfulness!

Resource is a system effect.

Example 5.6 German intelligence

During the war the German intelligence planted a very interesting puzzle in the British center of development of weapons. Loss of time on guessing this puzzle braked the work of the center approximately for two weeks—no other explosion or any other kind of subversive activity could yield such effect.

Resource—system effect and time.

Example 5.7 Pavements of Barnaul

Barnaul silver melting factory, which was opened in 1744, yielded 90% of all Russian silver. A lot of slag remained after melting.

Later on a problem of paving roads emerged. At that time, it was a usual thing to pave the roads with stones, but they had to be carried to Barnaul, that's why a decision was taken to pave the roads with slag.

At that time the technological process of obtaining silver was imperfect and particles of precious metal remained in the slag.

Resource—wastes (resources of substance).

Problem 5.1 Army of the King

Conditions of the problem

One Kingdom was attacked by a large army, which defeated the entire army of the King. The King ran to his mother, but she did not let him in and ordered him to fight the honest battle.

What should the King do: he has no army, but the enemy has to be defeated. What's to be done?

Analysis of the problem

SC: anti-B

SC. The King lost his army. Undesirable effect (UE)—*Loss of the army (anti-B)*.

RC: A – anti-B

RC. It is necessary for the King *to defeat* the army of his enemy (A), but he *has no army of his own (anti-B)*.

IFR: A, B

IFR. The King *achieves victory over his enemy (A)* without the presence of army of his own.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP. The King *should have an army (C)*, in order to *achieve victory over his enemy (A)*, and *he shouldn't have an army (anti-C)*, since he *hasn't got it*.

Resolution of CP

Resolution of contradiction through **use of resources**.

What resources are at the disposal of the King? His subjects –peasants from the villages and their property, including animals.

Solution

The King called the shepherds of his kingdom and ordered them to lead all their herds towards their enemies. Clouds of dust from under the legs of the animals spread throughout many miles. And then the enemy's army got frightened with the invisible force of the enemy, since nothing was seen behind the dust, and they started

to run away from their enemy's land. Thus the King achieved a victory with his cunning, not with his force.

Principle 24 is used. The intermediary.

Problem 5.2 Collecting cherries

Conditions of the problem

In the Tsarist Russia the serf girls had to collect cherries. During this process the girls not only collected cherries but also ate them. It was not profitable for the landlord, since he couldn't provide a supervisor for each of the girls.

What's to be done?

Analysis of the problem

SC: anti-B

SC. Collecting berries, the girls ate them as well. Undesirable effect (UE) –*eating the berries* (**anti-B**).

RC: A – anti-B

RC. It is necessary to *collect berries* (**A**), but the collectors (serf girls) not only collect berries, but also *eat them* (**anti-B**).

IFR: A, B

IFR. The girls *collect berries* (**A**), but *have no opportunity to eat them* (**B**).

CP: P → A, anti-P → B

CP. The girls *should not eat* (**P**) *the berries*, in order to *collect them* (**A**), and they *should eat* (**anti-P**) the berries, since they want to eat them.

Resolution of CP

Let us separate these contradictory features in **space** and in **structure**.

Hands are needed for collecting berries, while mouth is needed for eating them.

It means that the hands should be free, while the mouth should be busy.

Let us use **resources**.

What resources could be used, in order to make the mouth busy in such a way, so that it might be checked by one supervisor controlling all girls?

Solution

The girls got the order to sing songs.

Problem 5.3 Signature

Conditions of the problem

Fake signatures are currently a menace. It brings significant losses to the economy of the state, to enterprisers and to private persons. People tried to eliminate this problem: they made signatures more and more complicated, they improved the qualities of signatures for hours. But all this has a number of disadvantages: first of all, a person cannot reproduce his or her signature exactly; secondly, technology is so highly developed nowadays that any signature can be copied with a high degree of accuracy.

What's to be done, in order to be able to identify the fake?

Analysis of the problem

SC: anti-B

SC. The signature can be faked. Undesirable effect (UE)—*falsification of a signature* (**anti-B**).

RC: A — anti-B

RC. Necessity for *signing the documents* (**A**) and *a possibility to falsify the signature* (**anti-B**).

IFR: A, B

IFR. *The signature is put on documents* (**A**) without any opportunity of falsifying it (**anti-B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

IIC. Individual peculiarities of a signature *should be visible* (**P**), in order to *make signing of the document possible* (**A**), and *should not be visible* (**anti-P**), in order to *make falsification of the signature impossible*.

Resolution of CP

Montblanc company developed ink, which enables to make the signature or text, written by the owner of the pen, completely individualized. It became possible owing to the ink Montblanc Personal Code Ink, containing components with *invisible unique vegetative code*, which is rather similar to the DNA code of the human. The authenticity of Montblanc Personal Code Ink could be checked by Montblanc

upon submitting a document, letter or souvenir, containing the handwritten text of the owner. After that the court examination will help to confirm the fact of using Montblanc Personal Code Ink.

Problem 5.4 A Peasant (a tale)

Conditions of the problem

Once a peasant was walking through the dark wood to the market, intending to sell an old horse's skin. He lost his way and left the wood only by darkness. He knocked at the door of the nearest house asking for bed, but the housewife did not let him: she had a guest—a sexton, whom she received secretly from her husband, treating him to delicious dishes and wine. However, all of a sudden, the owner of the house appeared and invited the peasant into the house. The housewife got frightened, hid all the dishes in the oven, but only the peasant noticed it. When the man asked his wife to feed them, she took only the porridge from the oven and put it on the table. However, the peasant knew about other dishes!

What should the peasant do in order to taste some delicious dishes and not to betray the housewife?

Analysis of the problem

SC: A

SC. How can the peasant taste the dishes? Positive effect (PE)—to taste the dishes (A).

RC: A — anti-B

RC. Contradiction between the necessity *to taste the dishes* (A) and *not to betray* the wife of the owner of the house (anti-B).

IFR: A, B

IFR. The peasant *got the dishes* (A) and *did not betray* (B) the wife of the owner of the house.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP. The peasant *has to tell the owner of the house about the hidden dishes* (P), in order to taste the dishes (A), and should *not tell the owner of the house about that* (anti-P), in order not to betray his wife (P).

Resolution of CP

Separation of contradiction **according to condition** and **in structure**.

Solution

The peasant put his sack with a horse's skin under the table and secretly stepped on it. Loud squeak was heard and the owner of the house was rather frightened and asked about the origin of this sound. The peasant answered that a magician was sitting in his sack, who fulfills all his wishes. And he wants to have a cheerful feast and the magician orders him to look inside the oven and to get delicious food therefrom. Thus, the peasant ate a lot and at the same time did not betray the wife of the owner of the house.

Resources and principles 24 are used. Principle of intermediary.

Problem 5.5 The Robbers**Conditions of the problem**

In the stalls of one king there was a horse which was known for its force and quickness. He was guarded night and day and the guards never closed their eyes. The horse ate only the grass from the Dark wood. The servants bought such grass from the poor people, who wanted to earn some money. Once three very cunning robbers wanted to steal this horse. They penetrated into the stalls and immediately escaped therefrom riding this horse.

How did they manage to get into the stalls unnoticed?

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

The robbers picked grass from the Dark Wood, rolled one robber into the grass and sold the grass to the servants. Thus, the robbers got into the stalls and stole the horse.

Resources and principle 24. Principle of intermediary.

Problem 5.6 Log house**Conditions of the problem**

If the house is built of damp wood, it won't stand for long. Later on, in case of irregular drying cracks will appear and fungus will grow. In order to prevent the rotting of the house, in order to make the house strong and provide for its standing for many years, the logs for construction were dried and then impregnated with a solution of tar. Drying under natural conditions required long time, but in Siberia the summer is short. Since times unknown the citizens of Transbaikalia built such houses, which can still be seen in old Transbaikalia villages. These houses are strong and exist for more than one century.

What is the secret of this?

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

The secret consisted in the fact that the logs for these houses were prepared in March. At that time the moisture did not go upwards along the tree trunk, while under the warmed bark (in Transbaikalia the days are sunny) tar is already formed. A tarred tree is more endurable, the wood of larch tree is in general rather endurable, that's why it was used for making first three crowns as well as gutters and roadway inlets. It was also used for roof tile crown, which held the ceiling. The pine tree was used for making the very ceiling and floor, since its wood is light and warm. There were never earth floors in Siberia, almost everywhere you step on deep frozen soil and the floors were paved with timber. In the villages standing on the banks of the Ylim river even the yards were paved.

Resources: time, condition, knowledge.

Problem 5.7 The Barge**Conditions of the problem**

In old times the merchants organized floating markets. On powerful Siberian rivers the way was long: since early spring till late autumn they floated to the Far North, visiting each rich village. The barges were equipped with sails. The stream course and the favorable wind quickly carried them to their destination. However, quick headwind was a hindrance for the floating: the barges, overcoming the force of wind, slowly moved to the far reaches of the river. However, the human experience knew the secret of struggle.

What does the secret consist in?

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

Water sails were lowered down into the river water from each side in the nose part of the barge: a big wooden frame covered with tarpaulin, was lowered into the water. One rib was fastened close to the board and another was braced to the nose and stern using ropes. The barge became similar to a fish from a fairy tale with fins, protruding under the right angle to the body. The headwind braked the barge, while the favorable current forcefully beat into the water sails and, counteracting the wind, carried the barge along the waves.

Resources: water flow, substance—water sail.

Problem 5.8 Tibet “Clocks”**Conditions of the problem**

The citizens of Tibet don't bake bread, from times unknown zamba is a substitution for bread with them: they throw a handful of flour made of dried barley, pour some tea with butter and salt over it, stir this tight and powdery dough with fingers and eat it fresh. The grains for zamba are fried, adhering to norms of fulfilling each stage:

first they heat a clay basin with sand on the fire, then they take the basin off the fire, throw a handful of barley into it, stir it and once again put it on the fire. Finally, they throw the sand with grains on the sieve.

How do Tibet women “count the time” thereby?

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

Not a single woman would dry grains for zamba without a song. This song is long and old. The song will prompt everything: when is it time to take the basin off the fire, when to throw a handful of barley, for how long should she stir the dish and when to throw the sand with grains on the sieve. The stanzas of this song is a wise counting of time.

Resources: bars and words of the song, time.

Problem 5.9 A Man Attracting Trouble

Conditions of the problem

You occupy one of the important positions in the state, for example, direct the security department. Once you learn that there is a man in your country, who is able to cause unpleasant things with people, whom he meets. Imagine that he passes you by, for example, you will fall down and break your leg. After that he will come to visit you at the hospital and after this visit the ceiling in the hospital room will collapse and in the house across the street a fire will start. At the same time nothing will happen to himself. Propose, how one can use such a person with topmost efficiency.

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

Make a subverter in a hostile country out of this man. Unexplainable technological accidents at industrial plants will immediately start there as well as other unpleasant things.

This man can also be used for elimination of mafia. Everybody knows who the authorities of mafia are. But nobody can get them into jail – more often than not there are “no proofs”. It is possible to introduce him into a criminal structure or make his stay in this zone still more horrible.

It is possible to conduct constant video recording of all his travels for the purpose of creating horror films, or to conduct the analysis of formed situations for the purpose of developing recommendations concerning labor safety.

Using people with such extraordinary abilities for industrial espionage is described in the novel by R. Zilasny and F. Saberhagen “Coils”.

To present him to an “especially friendly” country.

Resource—properties of the human being.

Problem 5.10 Safety of cows

Conditions of the problem

African farmers have great losses because of lions, attacking live stock. To protect their households, they install traps for beasts of prey, leave poisoned baits around and shoot the animals. All these measures appear to be useless: in case the number of lions is decreased at some territory, beasts of prey from neighboring areas come here.

How to preserve the life of livestock without killing lions?

Analysis of the problem

Fulfill the analysis of the problem yourselves.

Solution

Inexpensive and effective solution, enabling to preserve the life of both hoofed animals and lions was proposed by Neil Jordan—a professor of the University of New South Wales and active participant of ecological organization Botswana Predator Conservation Trust. Once during an excursion in an African natural park he paid attention to the fact that a lion, who was going to attack an antelope, changed his decision after the animal turned back and looked at him. It brought him to the idea of creating “artificial eyes”, drawn on the rear side of the object of hunting.

The implementation of the experiment started last year. With the aid of stencils and safe paints Jordan and his colleagues painted one third of the herd from the village situated in the North of Botswana. The idea proved to be perfectly fruitful: during the time of the experiment (10 weeks) the lions killed several out of 39 cows that had no “eyes” painted on their bodies, but did not touch a single one out of 23 painted cows.

This year the novelty will embrace a large number of farms and many hoofed animals. The opponents of this method say that in a long-term perspective it will hardly be useful: the lions are clever animals and learn quickly. Sooner or later they will understand that in reality their victim is not looking at them. The scientists, in their turn, remind that the beasts of prey can travel through large territories and each time a new lion meets a “four-eyed cow”.

Principle 26. Principle of Copying.

Resource—picture (imitation), knowledge

5.3 Independent Work

5.3.1 Questions for Self-check

1. What are resources? Give a definition.
2. Describe a classification of resources system.
3. What origins of resources exist? Where is it possible to take resources?
4. Describe types of resources. Give examples.
5. According to what criteria can one evaluate resources? Give examples of different evaluations.
6. Describe methods for changing resources.
7. How do they define the properties of resources?
8. Describe the sequence of application of resources for solving a problem.
9. Describe the sequence of identification and use of resources for application of the system according to new destination.

5.3.2 Topics of Reports and Reference Papers

1. Use of resources in inventive activities and in engineering.
2. Use of resources in information technologies.
3. Use of resources in everyday life.

5.3.3 Fulfill the Tasks

1. Give examples of different kinds of resources

- 1.1 Give examples of functional resources.
- 1.2 Give examples of structural resources (elements, connections and form).
- 1.3 Give examples of substance and field resources.
- 1.4 Give examples of flow resources.
- 1.5 Give examples of resources of time and space.
- 1.6 Give examples of system resource.

2. Solve problems using resources

At first try to analyze the problem from the standpoint of logics of solving non-standard problems or at least formulate the IFR.

2.1 Problem 5.11. Treasures of the Temple

Conditions of the problem

In Lhasa, in the prayer hall of the Jokan temple, which was erected by the Tibet King Sronzan Gambo (founder of Tubo dynasty) in 646, there is a statue of Shakyamuni, brought from China by the wife of the king Princess Veng-Cheng. The majestic

figure of Shakya-muni steps out from the darkness, lighted with huge lamps arranged around it. The statue of Shakya-muni is cast of gold and decorated with countless precious stones. It has facial resemblance with the young Buddha. During the entire period of existence of the temple not a single precious stone was carried out of this hall, though the temple is open not only for pilgrims. Nobody ever mentioned even to touch the stones unnoticed.

How do the priests of the temple provide for good preservation of the temple's treasures?

2.2 Problem 5.12. Tent

Conditions of the problem

A tent is a tissue, which is stretched by wires (stretchers). Usually a tent is fastened by attaching it to trees or to other structures standing nearby. There are also tents with supporting pillars. Imagine that you come to the beach and you have nothing to fasten the edges of the tent with. There are no trees around or other means for fastening the tent edges.

How can one fasten the stretches?

2.3 Problem 5.13. Fire on the island

Conditions of the problem

Imagine that one man is staying on the island. Because of long-time drought the grass and the bushes on the island have become extremely dry. All of a sudden a fire broke out at one end of the island and the wind drove the fire towards the man. He cannot escape to the sea, since there are numerous sharks near the shore. And there are no shores without plants on the island. What should this man do in order to escape death?

2.4 Problem 5.14. Copyright protection

Conditions of the problem

Some publishers of dictionaries and atlases copy them from other publishers as pirated editions. How to prove that the given book is a pirated edition?

2.5 Problem 5.15. Saucer

Conditions of the problem

A saucer, plate or cup should evenly stand on horizontal surface. For this purpose their bottom is polished, but it will be expensive to polish the entire surface of the bottom and it will take too much time.

What's to be done?

2.6 Problem 5.16. A Cucumber in the Bottle

Conditions of the problem

Everybody knows that there is a method to place a ship model inside the bottle. But how can one put an entire ripe cucumber inside the bottle without damaging the bottle?

2.7 Problem 5.17. Radiator

Conditions of the problem

One of the most widely spread methods for creating thermal comfort in the room is heating it with the aid of radiator, in which hot water circulates.

It is possible to increase the efficiency of the radiator functioning through enlarging the area of contact with the ambient air or through enhancement of radiating capacity of the radiator.

It is possible to increase the area of contact with the air by increasing the number of ribs of the radiator up to creating needle-type ribs, like it is done with radiators for cooling electronic devices, however, it requires changes in the design of radiators.

Another method of increasing the area of contacts with the ambient air is to increase the number of radiators, for example, by filling all walls with the radiators or by placing them in several rows.

If all walls are filled with radiators, there will be a hindrance with hanging something on the walls, for example, shelves. Arrangement of the radiators in several rows will occupy a lot of place and will decrease the area of the room.

It is possible to enhance the radiating capacity of radiators by conferring the shape of a parable to it, as it is done in electric heaters. It is difficult to manufacture radiators of parabolic shape and it is possible only, when new radiators are produced.

How to increase the efficiency of heat release of available radiators without increasing their number?

2.8 Problem 5.18. Ice in the pipes

Conditions of the problem

Because of condensation of moisture in the warm air leaving the room, ice crust is formed on the internal surface of ventilation pipes in winter. Ice crust can clog the entire cross-section of the pipe.

Many mechanical devices have been invented, which eliminate the ice crust from the internal surface of the air ducts. All of them are unreliable (they get iced themselves) and require servicing.

What's to be done?

2.9 Problem 5.19. Repair of pipes

Conditions of the problem

Leakages in water pipes take place due to different reasons. As a result, water leaks out. In order to do the repairing, one has to shut off water for long time and to drain it through pipes. The repairing could be done without shutting off water, however, much water will leak out and there may be a deluge in rooms. How to do repair work without shutting off water and without water leaking out?

2.10 Problem 5.20. Evaporation of water

Conditions of the problem

Institute for Air Dynamics and Hydro Dynamics (TsAGI) was created according to the initiative of Nickolay Yegorovich Zhukovsky, in December 1918.

At the institute, which started functioning in the building of the Technical school, the laboratory was heated with a small tile stove, but the amount of heat was insufficient. The employees of the laboratory put on the cooker a water boiler in order to obtain more heat. A stove with a water boiler on it more or less heated the laboratory. However, the boiler produced a lot of vapor, i.e., a lot of moisture was released into the air. Moisture in the class was worse than cold and it was utterly unpleasant to study. What's to be done?

2.11 Problem 5.21. The Sail

Conditions of the problem

How to create a sail, which will regulate the traction force itself, automatically, without any electronics?

2.12 Problem 5.22. Buoy

Conditions of the problem

A buoy is floating on the surface of fluid in a vessel. It is necessary that with the cooling of fluid, the portion of buoy immersed in it, should increase. What's to be done?

2.13 Problem 5.23. Exercise machines in the streets

Nowadays fitness rooms appear in many cities directly in the streets. It is impossible to install standard facilities, with which the training halls are usually equipped, in the streets. First of all, under severe conditions they will immediately be spoiled, and secondly, they are very expensive.

That's why the facilities should be as simple as possible (one shouldn't use electronics, engines, electricity, etc.), however, at the same time they should offer an opportunity to regulate the load.

What's to be done?

2.14 Problem 5.24. Collecting the scattered toys

Conditions of the problem

While playing, the children throw the toys around. Some of them are rather small, for example, parts of Lego.

How can one quickly collect the scattered toys without having lost a single one?

2.15 Problem 5.25. Burglars

Conditions of the problem

A flat in the outskirts of the city. The father of the family is at work till late at night, mother is always absent visiting shops, laundry and other places. Often only a little 6-year old girl stays at home.

A couple of local robbers learned about that and decided to rob such an easily accessible flat.

At the moment when the mother of the family went shopping, the burglars rushed into the flat. Prior to breaking the lock they decided to do the control ring at the door. Thin voice of a little girl answered:

- Who is there?
- This is a postman. Is there anybody of the grown-ups at home?
- No, nobody is at home. Only me and Vasska.
- Hmmmmm, well, I see, and how old are you and Vasska?
- I am six and he recently became three years old.

The burglars were pleased with such a situation. They quickly broke an old lock and rushed inside the flat. However, a six-year old girl and Vasska, who was three, managed to arrest them.

How did they manage to do it?

2.16 Problem 5.26. Bringing a Steam Engine across the River

Conditions of the problem

In the middle of the nineteenth century one Russian industrialist bought a steam engine. When it was rather near the installation site, it appeared that the road is blocked off by the river. The bridge across the river could not withstand the weight of the steam engine. What's to be done?

2.17 Problem 5.27. Biocorrosion

Conditions of the problem

Oil platforms are installed on piles—steel pipes, hammered into the bottom of the sea. In water medium the surface of the pipes is populated with different micro-organisms

(bacteria, water weeds, microscopic fungi). Microorganisms emanate acids, alkali and gasses, which contribute to biocorrosion and pitting of pipes.

The load carrying capacity of piles decreased thereby. What's to be done?

2.18 Problem 5.28. The Pit

Conditions of the problem

In order to arrange for reliable electrical earthing, it is necessary to dig a pit in the ground. It can be thought that solving this problem won't offer any difficulties. Yes, it is simple to do it in summer. However, in winter, especially in the North, the ground gets frozen. And in this case, it is difficult to dig a pit in frozen ground is utterly difficult—it is necessary to batter the ground, for example, with a pinch-bar. What's to be done?

2.19 Problem 5.29. Construction in the North

Conditions of the problem

Propose a certain construction material, which would be reliable under the conditions of North, which would enable to create different structures out of it, including water ducts and which would demand minimum expenditures thereby.

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Chapter 6

Modeling



In dark ages people full of light are easily distinguished.
Erich Maria Remarque
The most difficult thing in life is to understand, which bridge
should be crossed and which should be burnt.
Erich Maria Remarque

6.1 Kinds of Modeling

A model is always a simplification. It describes an object being modeled incompletely, but at correctly chosen parts and connections, for accurately restricted areas and conditions of application; the model describes the object sufficiently completely and correctly. Inaccurate source model yields inaccurate results and vice versa.

The model may be *graphical* (technical drawing, picture, plan, diagram—convenient for modeling spatial, temporal, quantitative and functional relations), *mathematical* (formulae), *theoretical* (laws and theories), *verbal* (verbal description), *substantial* (a globe is a model of the Earth, playthings are models of a human being and objects...) or *image-bearing, virtual* (nowadays computer modeling is used still more extensively).

One and the same object, depending upon the goals of research can have different models.

Example 6.1 Play at Keeping House

In the game “Keeping a house” the model of the human being is a doll or the children themselves.

Example 6.2 Testing a Parachute

In testing a parachute, the model of the human being is a sack with sand (100 kg).

Example 6.3 Car testing

In testing anti-strike devices in a car, the model of the human being is a mannequin of the human being with a large number of sensors.

Now let us consider the model of the process.

Example 6.4 The process of buying

Let us model the process of buying:

1. Understanding of the problem (What do I want to buy?);
2. Search for the goods;
3. Evaluation of the variants of goods;
4. Selection of the goods and decision-taking;
5. Buying;
6. Reaction to the buying.

This is a verbal sequence of the operations.

Example 6.5 Improvement of the product

In the company OTIS the modeling aimed at identification of the appearance of its would-be products is performed in the following way.

The company invites people with different life experience and from different stratus of society for conducting special sessions. One invited person and several employees of the company take place in the session. During these sessions, implying the use of mockups of buildings, cars, etc., the usual daily routine of the guest is being acted and he is questioned in greater detail, what he would like to improve or change. Thus, they try to find such places or events in everyday life, where he could “meet the company”. Special attention is given to people with restricted capabilities and to adolescents, since their way of life and mode of thinking “determine the place of the company in future”.

One of the known methods for modeling in different fields is *business games*, in which the participants consider a particular situation, distribute the roles and then enact this situation, identifying advantages and disadvantages of it and trying to solve the problems, which appear.

Apart from different methods of mental modeling it is desirable to create *simplest material models from paper, cardboard, wire, plasticine*, etc.

It is desirable to use different kinds of *computer modeling*.

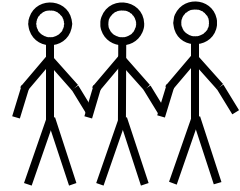
Another kind of modeling is trying to *break a complicated process into simpler ones* and to model each of them and then to gradually make the process more complicated.

Thus, the basis of **correct modeling** is the ability to find:

1. *Main parts;*
2. *Main connections between them.*

Not a single system *should be considered separately*, it will be an erroneous non-system approach.

Fig. 6.1 LMM—hard state of the substance



6.2 Modeling Tools in TRIZ

Use of inventive principles, for example, analogy, empathy and fantasizing, quoted in the work¹—is also a method of modeling.

TRIZ on the whole is a method of modeling objects and processes by constructing initial model and transforming it in such a way so that disadvantages should be eliminated or future systems would be forecasted.

Besides, there are special modeling tools in TRIZ, for example, identification of all kinds of *contradiction*, *functional approach*, *Su-Field analysis*, *LMM* (little men method), etc.

Modeling through identification of contradictions was partly considered by us above, as well as in [2–6].

Functional approach was partly shown in Sect. 3.5.3. “Analysis of identification of disadvantages” of textbooks on TRIZ [2].

Modeling with the help of Su-Field analysis was demonstrated in textbook [2, 7].

Modeling with the help of LMM was quoted in [2, 8, 9].

6.3 Examples of Modeling

Problem 6.1 Phase transitions

Conditions of the problem

Demonstrate the process of different phase states of the substance using the little men method (LMM).

For example, water.

1. How to present it in hard state—ice?
2. How to present it in fluid state—water?
3. How to present it in gaseous state—vapor?

Solution

Hard state means that the little men should be closely bound together, for example, hold hands or embrace each other (Fig. 6.1).

¹ Petrov [1].

Fluid state means that they stand closely one to another, however don't touch each other. Another little man can pass between them and they will move apart and then once again will stand closely one to another (Fig. 6.2).

Gaseous state—the little men not only stand far one from another, but even run in different directions (Fig. 6.3).

Problem 6.2 Crossing

Conditions of the problem

Let us consider a well-known problem about transportation of wolf, goat and cabbage across the river in a small boat.

Main parts:

Hunter, wolf, goat, cabbage, small boat.

Main functions of main parts:

Hunter—ferryman.

Goat—can eat the cabbage and can be eaten by the wolf.

Wolf—can eat the goat and does not eat cabbage.

Cabbage—can be eaten by the goat and cannot be eaten by the wolf.

The little boat has enough place only for the hunter and one of the animals, who must be carried to another bank.

Main bonds between the parts:

One cannot leave unattended the goat and the wolf or the goat and the cabbage.

It appears that the object of special attention for the hunter should be the goat. In order that there should be no misfortune, the goat should be either alone, or under the supervision of the hunter. The goat is the main nuisance.

Fig. 6.2 LMM—fluid state of the substance

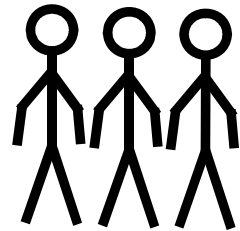
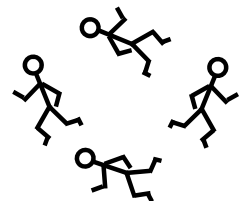


Fig. 6.3 LMM—gaseous state of the substance



Solution

At first the hunter takes the goat to another shore and leaves it alone. Then he carries the cabbage, then carries the goat back and leaves it alone. After that he carries the wolf and returns to take the goat. Thus, the hunter crosses the river seven times instead of five. And all this is because of the goat.

Problem 6.3 Three Scouts

Conditions of the problem

The situation was not less difficult with three unmounted scouts, who had to cross the river, where there was no bridge. It has to be mentioned that there were two boys, who were boating in a small boat and who were eager to help the soldiers. But the boat was so small that could take on only one soldier. Even one soldier and one boy could not get into the boat without the danger of drowning. None of the soldiers could swim.

It might seem that under such conditions only one soldier could get a chance to cross the river. However, soon all three scouts successfully found themselves on the opposite bank of the river. How did they manage to do it?

Answer

Six following crossings had to be made:

1st *crossing*. Both boys float to the opposite shore and one of them leads the boat to the soldiers (another boy is left on the opposite shore).

2nd *crossing*. The boy, who brought the boat, is left on this shore, while the first soldier gets into this boat and crosses the river. The boat returns back with another boy.

3rd *crossing*. Both boys cross the river and one of them gets back with the boat.

4-я *crossing*. The second soldier crosses the river. The boat gets back with the boy.

5th *crossing* is the repetition of the third one.

6th *crossing*. The third soldier crosses the river. The boat floats back with the boy and the children continues their interrupted boating. Now all three soldiers are on the opposite bank of the river.²

6.4 Independent Work

6.4.1 Questions for Self-check

1. Name the main kinds of modeling.
2. Enumerate the kinds of modeling, which are used in TRIZ.
3. Name the obligatory elements, which the model should contain.

² Perelman [10].

6.4.2 Topics of Reports and Reference Papers

1. Modeling methods in science, art and technology.
2. History of methods modeling development.

6.4.3 Fulfill the Tasks

In fulfilling the tasks necessarily single out all essential elements of the system, interconnections and influences of system elements between themselves, between system and supersystem, sequence of performing each step in the process. Analyze the model. Indicate useful and harmful effects. Try to find the solutions, which eliminate harmful effects, for example, using ARIZ logic, and use the resources to the utmost.

Any tools and methods could be used for modeling. Out of TRIZ tools try to use logic for solving non-standard problems, Su-Field analysis and LMM.

1. Give examples of modeling

Select any system or process out of the ones quoted below.

- 1.1. System modeling.
 - 1.1.1. Technical system.
 - 1.1.2. Natural object.
 - 1.1.3. Particular science as a system.
 - 1.1.4. Business system.
 - 1.1.5. Any object from life.
 - 1.1.6. Any system relating to your specialty, etc.
- 1.2. Modeling of processes.
 - 1.2.1. In developing new technological devices.
 - 1.2.2. In natural phenomena.
 - 1.2.3. In different sciences.
 - 1.2.4. In business processes.
 - 1.2.5. In life processes, etc.

2. Create the models

- 2.1. System modeling.
 - 2.1.1. Problem 6.4. Ball-point pen.
Create the model of a ball-point pen using Su-field analysis.
 - 2.1.2. Problem 6.5. Toothbrush.
Create the model of a toothbrush, using Su-Field analysis and LMM.
 - 2.1.3. Problem 6.6. Table.
Create the model of a table, using Su-Field analysis and LMM.

- 2.1.4. Problem 6.7. Glass with water.
Create the model of a glass of water, using Su-Field analysis and/or LLM.
- 2.1.5. Problem 6.8. Shelf on the wall.
Create the model of shelf, hanging on the wall, using Su-Field analysis and/or LLM.
- 2.1.6. Problem 6.9. Incandescent lamp.
Create the model of incandescent lamp, using Su-Field analysis.
- 2.1.7. Problem 6.10. Saucepan with food stands on the functioning burner.
Create the model of a saucepan with food on the cooker, using Su-Field analysis.
- 2.1.8. Problem 6.11. A conversation between two people.
Create the model of a conversation between two people, using Su-Field analysis.
- 2.1.9. Problem 6.12. System from your field.
Create the model of a system relating to your specialty, using Su-Field analysis and/or LMM.
- 2.1.10. Problem 6.13. System according to your choice.
Create the model of a system according to your choice using any tool or method.
- 2.2. Processes modeling.
 - 2.2.1. Problem 6.14. Archimedes law.
Model the Archimedes law with the help of LMM.
 - 2.2.2. Problem 6.15. Electric current.
Using the little men method, create the model of motion of electric current along the conductors. What does the resistance depend on?
 - 2.2.3. Problem 6.16. Model of the workday.
Create the model of the workday using any tool or method.
 - 2.2.4. Problem 6.17. Making tea.
Create the model of making tea in a teapot, using any tool or method.
 - 2.2.5. Problem 6.18. Washing hands.
Create the model of the process of washing hands, using any tool or method.
 - 2.2.6. Problem 6.19. Writing on a sheet of paper.
Create the model of the process of writing on a sheet of paper, using any tool or method.
 - 2.2.7. Problem 6.20. Business process.
Create the model of business process, using any tool or method.
First create the model of business-process, as it exists now in the company, then refine this model and restructure it in such a way, that business process should be more ideal. At that one can identify and resolve the contradictions, formulate IFR, etc.

- 2.2.8. Problem 6.21. Any process according to your choice.
Create the model of any process according to your choice, using any tool or method.

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Chapter 7

Complex Use



Imagination control is one of the qualities of well-organized thinking. And since there are no limits of improvement of organization of thinking, there are no limits in improvement of imagination.

G. S. Altshuller

7.1 Review

Main components of inventive thinking were quoted above.¹

In real life, as a rule, we have to use not one of the components of inventive thinking, but their combinations or even all of them taken together.

That's why after practicing each individual component of inventive thinking it is necessary to practice in a complex way, to use combinations of individual components. This is the goal of this chapter.

The next paragraph will contain the conditions of the problems, for solving of which it will be necessary to use several constituents of inventive thinking.

Possible variants of solving problems will be quoted in Appendix 1.

7.2 Practice

Conditions of the problems are quoted below.

Problem 7.1 Spies

During the cold war period hundreds of American spies were caught in the USSR out of a silliest possible reason.

Falsified passports were the main tools for the spies. That's why during the process of falsifying it was necessary to give attention to any details, even the smallest ones.

What detail betrayed the American spies?

¹ Altshuller (1980).

Problem 7.2 Theater

1920-ies, one of the cities of Europe. The opening of the theater, which was long awaited for and the building of which took too much time. The first performance. The curtain opens... But what happens? The spectators are silent trying to hear, what they are saying there, on the stage, and once again, having heard nothing, inadvertently started to murmur unsatisfied at the inability to hear anything. Wherever the spectator was sitting, in the stalls or in the balcony, it was necessary for the spectator to hear every word. While in this case not only words, whole phrases were lost! No, there was no fault with the actors. Maybe, the fault was with the builders: they could ignore the laws of architectural acoustics? No, the architect, who had designed the theater hall, knew all the subtleties of behavior of sound waves.

Then, where is the reason for that? And what's to be done in the given situation?

Problem 7.3 In the Swamp

A detachment of French soldiers during the march in Algeria once found themselves in a place, which was completely devoid of plants and with soil, which was so swampy, that, though it was possible to step on it, to sit on it was completely impossible. The tired detachment was moving forward in search of appropriate place for rest, however, this swampy soil spread in all directions for dozens of miles. How to arrange for having rest, when there isn't a single dry place around and nothing, which one could lay on the ground as a blanket or on which one could sit?

And still one happy thought dawned upon one soldier, which helped the detachment to get out of a trying situation. The soldiers sat down and had their rest.

How did they manage that? Please, guess!

Problem 7.4 The Hedgehog and the Hare (a tale)

The hare and the hedgehog once had a dispute, who would quicker run along the trench and backwards. The hedgehog came home and told his wife about that. She scolded him a lot, but agreed to help.

How did the hedgehog manage to win the dispute?

Problem 7.5 Customs

A customs official of the customs, where the control over the goods sent abroad was effected, became suspicious about plastic bowling balls produced by one of the companies. They had exactly the same weight as the wooden balls of the same size. The balls were not massive, but the walls were equally hard in all places. The customs official thought that there is a cavity inside each ball, where one could hide the goods being smuggled. And really, with the aid of a most simple experiment, which did not involve special devices, the customs official stated that the smuggled goods are hidden in one of 12 spheres. When the ball was opened, there was a diamond article of jewelry there. How did they manage to identify this sphere?

Problem 7.6 A Barrel of Wine

Once a landlord called a poor man and said that he would richly reward him in case he would be able to find out how much wine there is in the barrel (more or less than a half) and will not use either stick or rope; in general he will not lower down anything into the barrel. The poor man simply slanted the tub and told the landlord that the volume of wine amounts to a little bit less than half of the barrel.

How did he learn about that?

Problem 7.7 Robert Surcouf

Famous corsair Robert Surcouf during his youth was engaged in slave-trading, however, during the French Revolution slavery was abolished. He understood that it was more profitable, and (what was the most important thing) still more noble to be a corsair than to carry slaves from Africa in the lower holds of the ships.

However, in order to get a corsair patent,² it is necessary to post a bail, which Surcouf did not have and he decided to earn some money by slave-trading. He went to Africa for the last time to get slaves, all the more for the fact that after the prohibition of slave-trading their cost grew quit a lot, since the plantation owners badly needed slaves for growing sugar-cane.

Once the governor was angry with Surcouf because of something. Hardly had the ship gone into the sea, when the order followed: upon returning the ship had to be detained and the captain had to be arrested as slave-trader and repeat violator of Human Rights Declaration.

A friend of Surcouf told him about the order for arrest. Another man would give up the idea of a trip to Africa and of buying slaves, but Surcouf stuffed the lower holds of the ship with slaves and calmly went to the destination port. Although, it has to be noted that he had taken precautions by sending the slaves to the shore in boats and next day the ship boldly entered the port.

The police was already waiting for Surcouf. Hardly had the crew tidied the lower holds, when the commissary of the police and his helper went on board of the brig and, having examined it, proposed that the captain should follow him to the prison.

What should Surcouf do, in order not to be arrested and not to get into prison?

Problem 7.8 Capture of a ship

Surcouf captured a caravan of three English ships (two merchant vessels toed the line of the pilot brig), having raised the British flag. On the way they grappled with one more merchant vessel. He changed for pilot brig with the team of less than twenty people and he sent his own ship and the remaining crew to escort the captured ships.

² Corsairs (Fr.corsaire)—pirates, who have the documentary permission of French government to capture enemies' ships. The corsairs were officially at the service of the King's Army. A certain part of the booty was given to the King's treasury. If the corsairs attacked the ships of their own country, they were prosecuted and sentenced to death.

Next day the weather was hot and sunny. Surcouf saw an anchored ship under the British flag, which was armed with numerous guns. It might seem that Surcouf had to hurry to the open sea, but he decided to extract profit from this obviously unprofitable situation and captured a battleship.

How did he manage to do that?

Problem 7.9 The Spanish Armada

During dozens of years the British privateers³ robbed and sank Spanish ships. Thus, only in 1582 the losses of Spain amounted to a fantastic sum exceeding 1 900 000 ducats.

Spanish monarch Philipp II considered it to be his duty to help the British Catholics in their struggle with protestants.

The Spanish Armada was formed according to his order. It was a great navy. It consisted of 130 ships, 2430 guns, 30 500 men, out of which 18 973 soldiers, 8050 sailors, 2088 rowing slaves, 1389 officers, gentry, priests and doctors. The main forces of the fleet were subdivided into 6 squadrons. The armada also included: 4 Neapolitan galleasses (big galleys)—35 men, 50 guns, 4 Portuguese galleys—320 men, 20 guns, many light ships for scouting, dispatch boats and vessels with supplies.

The Spanish king gave the order to the Armada to reach the Channel, align with the 30-thousand army, located in Flanders (Spanish Netherlands), cross the Channel and to march to London. Phillip II expected that the British Catholics will leave their protestant Queen and will pass to his side. The plan of the Spaniards did not take into account two most important circumstances: the power of the British fleet and the shallowness of waters, which did not allow the ships to approach the shore and to take the army on board.

The British, in their turn, also hoped for the decisive victory, which would open for Britain the way to free use of the sea, would break the monopoly of Spain for trade with the New World and would contribute to dissemination of Protestant thought in Europe.

The Spanish fleet consisted of unwieldy ships with numerous guns of small range of fire adapted to boarding fight. The British ships were more maneuverable and were equipped with a large number of guns with long range of fire. The Englishmen expected that they would not approach the enemy closely and will destroy him at the distance.

The Spanish admiral arranged his fleet as a crescent, placing at the edges the strongest battleships with guns characterized by long shooting range.

³ Privateers—British corsairs—private persons, which had the permission of the authorities (the license) used armed ships for capturing merchant marines of the country, which was in the state of war (based on the materials of Wikipedia).

The British could not manage to violate the fighting order of the Spanish fleet. Waiting for the army, the Spanish admiral ordered the fleet to anchor the ships. How did the Englishmen manage to violate the fighting order of the Spanish fleet?

Reference

1. Altshuller, G. S. 1980. Sun beam of imagination, No. 7. Nauka I tekhnika. <http://www.altshuller.ru/triz/triz21.asp>. (In Russian).

Conclusion

*We are here in order to contribute to
this world. Otherwise, why are we here?*
Steve Jobs

Recommendation on Optimization of Skills

The only way to master the skills of inventive thinking is the constant training of these skills.

We advise you to practice these skills every day both separately and in clusters.

Look at the objects and processes surrounding you. Analyze, to what extent they are systematic. What is non-systematic about them? Identify, what creates hindrances for them to be systematic and thereby identify contradictions. Resolve these contradictions. Use resources to the utmost thereby.

Constantly identify regularities and tendencies in the development of events, which take place around us. What is good and what is bad about them? Identify and resolve contradictions. Don't forget to use resources. Ask a question "How will these objects and processes look in future?" and use trends of system evolution.

Try (at least mentally) to model the objects and processes researched by you. Of course, it is better to do it in written form. At least, use the LMM and Su-Field analysis. If necessary, create simplest models of paper, wire, etc. Those, who have an opportunity, do computer modeling.

It is very important to develop creative imagination every day. Select some principle or method for developing creative imagination (DCI) and train using it, for no <5 min a day. Constantly change these principles and methods.

As additional material we recommend you the book "Exciting world of phantazizing"¹ by Pavel Amnuel and his web-site.²

¹ P. Amnuel. Fascinating World of Fantasizing.—Jerusalem, Milky Way, 342 p. ISBN: 978-1537571805. <http://litgraf.com/detail.html?book=106> (In Russian).

² URL: <http://milkyway2.com/amnuel>.

After some time such training sessions will be a habit with you and then they will become an absolute must. Later on, you will do it automatically, constantly noticing what in this object or process is non-systematic, how to improve it. You will automatically formulate contradictions and ways of resolving them. Resources will be seen in everything, what surrounds us. You will automatically prolong the tendencies of development, using trends of system evolution, etc.

We wish you success!

Please, send your questions, remarks and proposals to:

Vladimir Petrov, E-mail: vladpetr@013net.net.

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1. Altshuller, Genrich. 1979. *Creation as an exact science. Theory of inventive problem solving*, 184 p. M.: Soviet radio. Cybernetics (in Russian).
2. Altshuller, Genrich. 1986. *Find the idea. Introduction to theory of inventive problem solving*, 209 p. Novosibirsk: Nauka (in Russian).
3. Altshuller, G. S., B. L. Zlotin, A. V. Zussman, and V. I. Filitov. 1989. *Search for new ideas: From insight to technology (Theory and practice of inventive problem solving)*, 381 p. Kishinev: Cartia moldoveniaske (in Russian).
4. Petrov, Vladimir. 2018. *Trends of System Evolution: TRIZ/Vladimir Petrov*, 894 p. [s. l.]: Publishing Solutions. ISBN 978-5-4490-9985 (in Russian).

Appendix

Analysis of Problems

In this section we shall show you our variant of analysis of certain problems.

A.1 Evolutionary Thinking

A.1.1 Identification of Regularities of Evolution

Problem 3.1 Find the regularity

Conditions of the problem

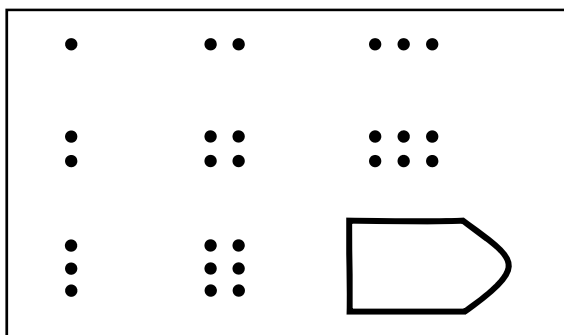
It is necessary to find the regularity in the arrangement of figures (Fig. A.1) and to select the missing figure out of the presented variants (Fig. A.2).

Answer

Figure 3.

Figure A.3.

Fig. A.1 Find the regularity



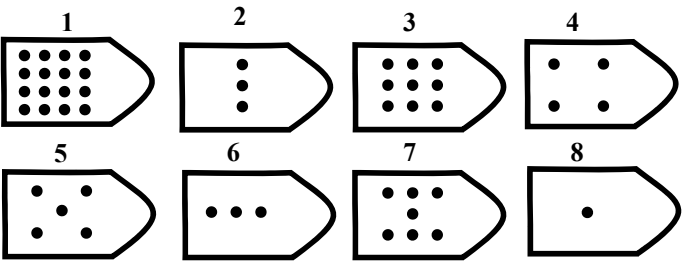
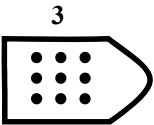


Fig. A.2 Variants of answers

Fig. A.3 Answer



Problem 3.2 Find the regularity and continue the row of figures

Conditions of the problem.

101, 112, 131, 415, 161, 718 ...

Answer

Figure 192.

Commentary

Try to place commas after each two figures...

It is clear now that the next number will be 192.

The sequence consists of Figs. 10, 11, 12, etc., but the commas are placed after each three figures.

Problem 3.3 Find the regularity

Conditions of the problem

It is necessary to identify the regularity in the arrangement of figures inside the squares (Fig. A.4) and to select the variant of answer from the tablets (A–E) in Fig. A.5, in order to continue the sequence.

7	3	8	3	2	4	3	1	2	
1	2	1	2	7	4	6	7	4	
5	9	6	8	5	7	4	6	9	

Fig. A.4 Find the regularity

<table><tr><td>6</td><td>2</td><td>1</td></tr><tr><td>4</td><td>3</td><td>7</td></tr><tr><td>2</td><td>9</td><td>8</td></tr></table>	6	2	1	4	3	7	2	9	8	<table><tr><td>6</td><td>4</td><td>7</td></tr><tr><td>4</td><td>5</td><td>2</td></tr><tr><td>3</td><td>5</td><td>6</td></tr></table>	6	4	7	4	5	2	3	5	6	<table><tr><td>2</td><td>6</td><td>2</td></tr><tr><td>3</td><td>5</td><td>3</td></tr><tr><td>8</td><td>4</td><td>7</td></tr></table>	2	6	2	3	5	3	8	4	7	<table><tr><td>3</td><td>2</td><td>1</td></tr><tr><td>8</td><td>4</td><td>7</td></tr><tr><td>1</td><td>8</td><td>7</td></tr></table>	3	2	1	8	4	7	1	8	7	<table><tr><td>2</td><td>2</td><td>8</td></tr><tr><td>3</td><td>3</td><td>8</td></tr><tr><td>6</td><td>9</td><td>5</td></tr></table>	2	2	8	3	3	8	6	9	5
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Fig. A.5 Variants of answers

10	3	6	7	?
1	?	5	4	9

Fig. A.6 Find the regularity

Answer

Variant B.

Commentary

Sum of the figures in each of the columns of the tablets is equal correspondingly to: 13–15.

Problem 3.5 Find the regularity

Conditions of the problem

Find the regularity in the sequence, presented in Fig. A.6, and place the necessary figures instead of interrogation marks.

Answer

In the first row = 2, in the second row = 8.

Commentary

One can notice that there are two sequences here and both of them have the shape of a zigzag. The first is as follows: 1, 3, 5, 7, 9. It means that the second is as follows: 2, 4, 6, 8, 10—only in reverse order.

Problem 3.6 Find the regularity

Conditions of the problem

Arrange arithmetic signs (+ – x:) in the empty cells (Fig. A.7) in such a way that the equality would be correct. In this case you can indicate only one sign two times. Arithmetic counting implies that figures go one after another from left to right, without taking into account the priority of signs.

6		3		5		7		4		8	=	13
---	--	---	--	---	--	---	--	---	--	---	---	----

Fig. A.7 Find the regularity

10	11	9	10	8
?	5	6	4	9
11	?	6	8	7
13	12	?	13	15

Fig. A.8 Find the regularity

10	11	9	10	8
12	5	6	4	9
11	7	6	8	7
13	12	14	13	15

Fig. A.9 Answer

Answer

$6 \times 6 - 3 - 5 + 7 : 4 + 8 = 13.$

Problem 3.8 Find the sequence
Conditions of the problem

Figure A.8 presents one definite sequence of numbers (not necessarily only from left to right). Define it and insert the missing numbers instead of interrogation marks.

Answer

The sequence is as follows (Fig. A.9): it begins with the Fig. 4 and goes in spiral order counterclockwise, alternating the variations of numbers by +2 and -1.

Problem 3.9 Find the regularity
Conditions of the problem

Arrange 12 cards in Fig. A.10. The rules are as follows: the sum of all 12 cards is equal to 84 points; all 12 cards are of different value; the points of junior cards are equal to their value; ace = 1, jack = 11, queen = 12, king = 13; the cards of equal color cannot be one near another either vertically, or horizontally, there should be 4 different suits in each horizontal row; each vertical column should contain 3 different suits. Additional conditions are as follows: the six is near and higher than the ten, which is nearby, and higher than the two of spades; the value of card C is 3 points less than F, which, in its turn, is 3 points less than the card L, which is higher in value than the card A; ace of clubs is placed near and higher than the card, the value of which is 3 points less than the card H, which coincides in suit with the card C; jack of diamonds is in one horizontal row with the card of hearts' suit, the value of which is 2 points higher than the card B.

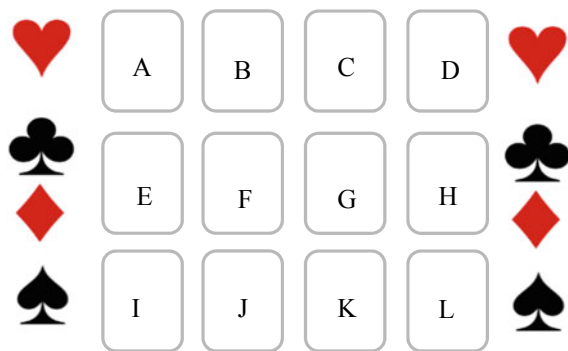


Fig. A.10 Find the regularity

Answer

(♠) – spades, (♥) – clubs, (♦) – diamonds, (♣) – hearts

A – ace, K – king, Q – queen, J – jack.

3♠	4♥	6♣	J♦
A♥	9♠	10♦	5♣
8♣	K♦	2♠	Q♥

Problem 3.10 Find the regularity (Fig. A.11) and continue the sequence

Conditions of the problem

Answer

One Two Three Four Five Six Seven Eight Nine Ten.

Problem 3.11 Find the regularity

Conditions of the problem

Find the regularity in presented numbers and insert the missing number instead of the interrogation mark.

97.263
25.298
13.452
?
3.420

Answer

Number 5,670.

O	T	T	F	F	S	S	E	N	T
---	---	---	---	---	---	---	---	---	---

Fig. A.11 Find the regularity

Commentary

The sequence is as follows: each next number, starting from the top, is calculated by multiplication of the number, consisting of all even figures of the above number, by number consisting of all odd figures of the upper number. The figures in numbers are read from left to right. The dot is placed after the third sign from the right. For example: 13.452 are two numbers 135 and 42, if they are multiplied, we shall get the number 5670 or 5.670.

Problem 3.12 Define the sequence

Conditions of the problem

Guess the sequence (Fig. A.12) and add the required letter instead of the interrogation mark.

Answer

Letter S.

Commentary

All letters of the alphabet correspond to numbers from 1 to 26, in keeping with their numerical order. The numeric value of the central letter is equal to the sum of numeric values of two upper letters minus sum of numeric values of two lower letters. Thus, in the last group of letters we obtain number 19, which corresponds to letter S.

Problem 3.13 Define the sequence

Conditions of the problem

Guess the sequence (Fig. A.13) and define, which number should close the sequence?

Fig. A.12 Sequence

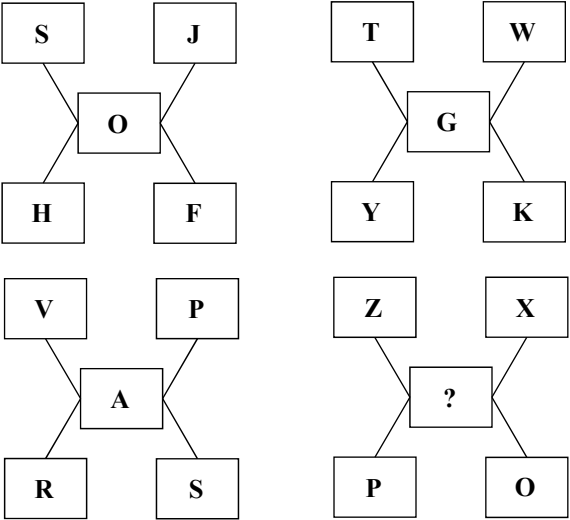


Fig. A.13 Sequence

3	8	15	22
32	42	51	58
70	79	88	?

Answer

Number = 102.

Commentary

The essence of the sequence is that each next number is equal to the preceding number plus the number of segments, of which this number consists. I.e., the number 88 consists of 14 segments, therefore, the next number should be = 102.

Problem 3.15 Find the regularity**Conditions of the problem**

It is necessary to define the regularity in the arrangement of figures (Fig. A.14) and to select the missing figure out of the presented variants (Fig. A.15).

Answer

Variant 1.

Commentary

The figure, appearing in the right cell of each row, is a sum of all components of two figures of the same row, without a part, which is common for them. For example, in the upper cell of the left row there is a square and four points in it. In the middle square there are four points and a cross between them. The sum of all components of these two cells constitutes a square, a cross in the middle and four points. Four points is a component, which is common for two figures and therefore they are not depicted in the right cell. Only a square and a cross are depicted in the right cell. Exactly the same regularity can be extended to columns. Let us check the left column. The sum of all components of two upper cells in this column constitutes a square, four points and a cross (exactly like in the upper row). Therefore, the explanation, which was given above, is true for this case as well, and the figure in the lower cell will be a square and a cross in the middle. The sum of all components of the left and the middle cells in the lower row is a square, a cross in the middle and a circle with a cross. The figure, which is common for these two cells, is absent in the right cell of this row, therefore, the figure, which should be located in the right column is a square and a circle in the middle.

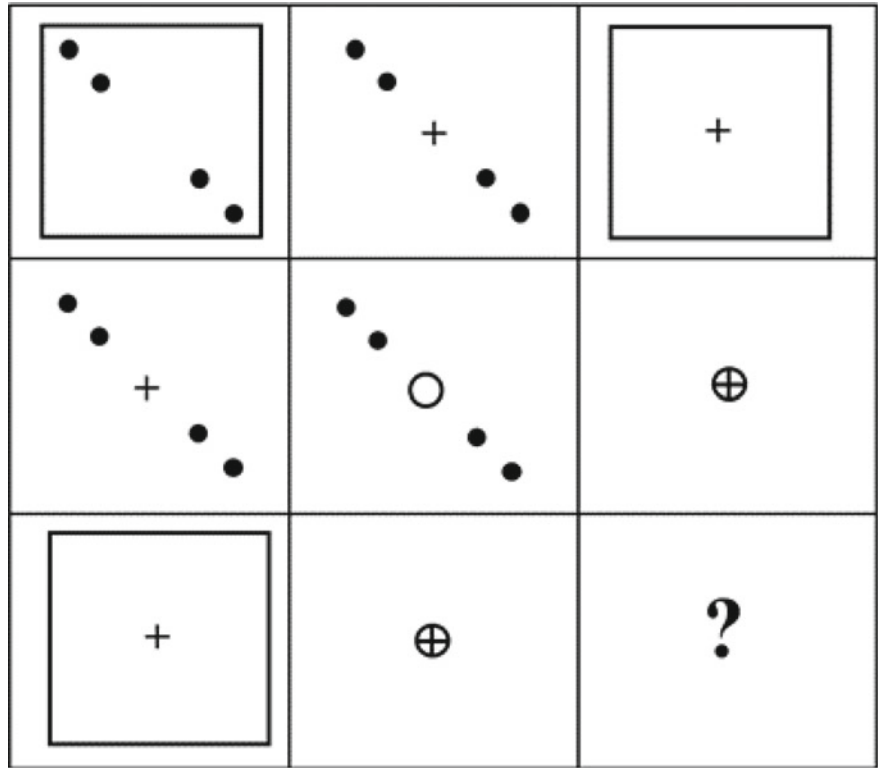


Fig. A.14 Sequence

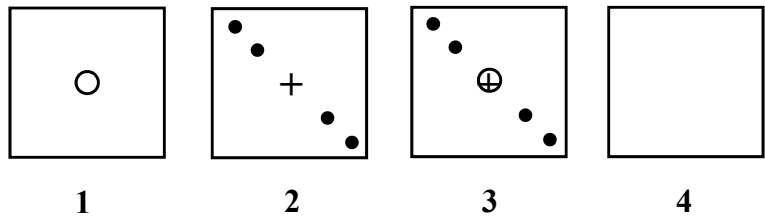


Fig. A.15 Variants of answers

Problem 3.16 Find the regularity

Conditions of the problem

It is necessary to find the regularity in the arrangement of figures (Fig. A.16) and to select the missing figure out of the presented variants (Fig. A.17).

Fig. A.16 Regularity

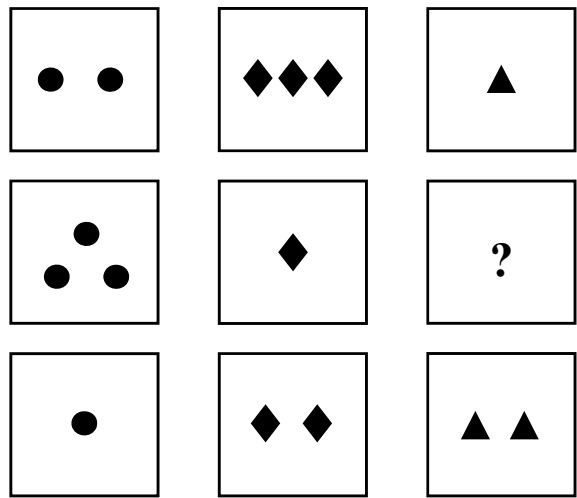
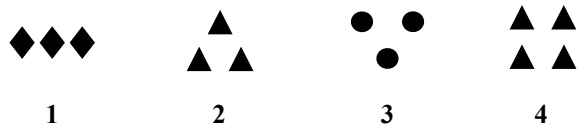


Fig. A.17 Variants of answers



Answer

Variant 2.

Commentary

A definite geometrical figure is depicted in each column. Each geometrical figure is depicted in each column in three different variants: a single figure, two similar figures, three similar figures. Consequently, the cell with three triangles in the right column is empty.

A.1.2 Use of Trends of System Evolution

Fulfill it independently.

A.2 Thinking Through Contradictions

Problem 4.11 The castle of Montaigne

Conditions of the problem

France, sixteenth century. Civil war. Philosopher Michel de Montaigne had a castle of his own. His neighbors—noblemen locked their castles and armed their peasants. Montaigne had neither servants, nor arms ...

What trick did he use, so that his castle would not be robbed?

Analysis of the problem

SC: A

SC: How to protect the castle. Positive effect (PE)—protection of the castle (**A**).

RC: A – anti-B

RC: Protection of the castle (A) requires the presence of servants and availability of arms, while Montaigne hasn't got any (**anti-B**).

IFR: A, B

IFR: The castle is protected (**A**) without servants and without arms (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: *There should be* servants (**P**), in order to *protect the castle* (**A**), and *there shouldn't be servants* (**anti-P**), since *Montaigne does not have them* (**B**).

Resolution of CP

Resolution of contradiction **in structure**.

Solution

Montaigne held the gates widely opened. Seeing this, the robbers were sure—the castle had already been robbed.

Problem 4.12 Artificial leg

Conditions of the problem

In prosthetics it is very important that the artificial leg should have the same shape as the actual leg. It might seem to be simple: to make a plaster cast of an actual leg and then to cast an artificial leg. However, it does not work, because nobody needs two left or two right legs.

What's to be done?

Analysis of the problem

SC: A

SC: How to create an artificial leg, which would be identical in shape to the real leg. Positive effect (PE)—accurate copy of the actual leg in terms of shape (**A**).

SC: A – anti-B

RC: Producing a model, which is like a live leg (**A**) leads to obtainment of two identical (two right or two left) legs (**anti-B**).

IFR: A, B

IFR: Prosthesis is similar to the live leg (**A**), They form a pair (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: Prosthesis *should be* an exact copy of the live leg (**P**), in order to be identical to it (**A**), and *should not be* an exact copy of the live leg (**anti-P**), in order to form a pair (**B**).

Resolution of CP

Resolution of a contradiction **in structure**.

Solution

The cast of a healthy leg is embodied as a thin elastic film, which after that is turned inside out like a stocking.

Problem 4.13. Taking off the veil in Iran

Conditions of the problem

The last shah of Iran Mohammed Reza Pehlevi was a progressive man, and he did not like that the women wear paranjas only in his country, that is the first point. The second and the most important point: he wanted to weaken the influence of Islamic clergy in the country, since wearing a paranja is a commandment of Islam and to take it off means to dishonor the woman.

But how is it possible to make the women take the paranjas off? If an order is issued for the women to take the paranjas off their faces, it will be an abuse of religious feelings of the majority of population. It cannot be ordered, because civil disturbances could start.

Analysis of the problem

SC: A

SC: How to make the woman not to put on the paranja? Positive effect (PE)—the women don't put on paranjas (**A**).

RC: **A – anti-B**

RC: The wish of the shah that the women should not wear a paranja (**A**) leads to the violation of customs of the country, which could lead to civil disturbances (**anti-B**).

IFR: **A, B**

IFR: The paranjas are not worn (**A**) and there are no civil disturbances (**B**).

CP: $P \rightarrow A, \text{anti-}P \rightarrow B$

CP: The veil *should be taken off* (**P**), in keeping with the wish of shah (**A**), and *should not be taken off* (**anti-P**), so that there *should be no civil disturbances* (**B**).

Resolution of CP

Resolution of contradiction **in structure**.

Solution

According to the decree of shah, the woman herself decided, if she would wear the paranja, but the prostitutes were obliged to wear it.

Problem 4.14. Conveyer of Ford

Conditions of the problem

In due time, when Henry Ford created his conveyor, he was very proud of it and organized guided tours to the workshops, demonstrating the functioning of the conveyor and acting as a guide himself. However, crowds of visitors created hindrances for the workers, distracting them from their work, which decreased productive efficiency. The workers complained to Ford about that and he invented a trick, which eliminated the problem. What did he do?

Analysis of the problem

SC: **anti-B**

SC: Decrease of efficiency because of excursions. Undesirable effect (UE)—decrease of efficiency (**anti-B**).

RC: **A – anti-B**

RC: Ford's wish to organize guided tours (**A**) leads to decrease in efficiency (**anti-B**).

IFR: A, B

IFR: The excursions are organized (A), while the efficiency of production is not decreased (B).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: Excursions *should be organized* (C), since *Ford wants it* (A), and *should not be organized* (anti-C), since *the efficiency of production decreases* (B).

Resolution of CP

Resolution of contradiction **in time** and **in structure**.

Solution

Ford increased the intensiveness of excursions, so that the workers should more quickly get used to the work under the conditions of “disturbance”.

Problem 4.15 Ancient sculpture**Conditions of the problem**

Michelangelo wanted to prove his geniality and to falsify the sculpture “Sleeping Cupid”.

According to the story by the artist and historian of Renaissance epoch Giorgio Vasari, the sculpture was acknowledged to be an authentic Ancient statue and sold to the Cardinal San Giorgio Raffaello Riaroso for 200 ducats.

How did Michelangelo manage to do that?

Analysis of the problem**SC: A**

SC: How can one prove that the created statue is an ancient one? Positive effect (PE)—confirming the authenticity of the fake with an authentic thing (A).

RC: A – anti-B

RC: Confirming the authenticity of the fake (A) and actual age of the statue (anti-B).

IFR: A, B

IFR: The authenticity of the fake is confirmed (A), in spite of the actual age of the statue (B).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The statue *should look like an ancient one* (**P**), since *its authenticity was confirmed* (**A**), and *should not look like an ancient one* (**anti-P**), since it was *just made* (**B**).

Resolution of CP

Resolution of contradiction **in structure and in time**.

Solution

According to the story of artist and historian of Renaissance period Giorgio Vasari the sculpture by genial Michelangelo “The sleeping Cupid” was dug into earth and then dug out and passed off as an ancient statue.

Problem 4.16 Chinese braids

Conditions of the problem

In 1641 the Manchurians established their ruling in China (Tsing dynasty). As a sign of faithfulness, the Chinese had to shave their heads and to leave a tress on the top of the head. First the barbers were commanded to execute the order of the Emperor. With a razor in one hand and sword in the other they patrolled their territories, offering each Chinese a choice: either to sacrifice the head or the haircut. The majority agreed to undergo an operation, though it was shameful, but some people were indignant and even rebelled. However, the second Emperor of Manchurian dynasty Kan Shi chose a wittier and more efficient measure, as a result of which the Chinese themselves, without any compulsion started to shave their heads and carry braids on the tops of their heads.

What did the wisdom of Kan Shi’s decree consist in?

Analysis of the problem

SC: anti-B

SC: What is the way to prevent riots in connection with the order to shave heads and to leave a braid on the top of the head. Undesirable effect (UE)—riots (**anti-B**).

RC: A – anti-B

RC: Shaving heads (**A**) and riots of the population (**anti-B**).

IFR: A, B

IFR: Shaving heads (**A**) and absence of riots (**B**).

CP: $P \rightarrow A$, $\text{anti-P} \rightarrow B$

CP: The Chinese *should shave their heads* (**P**), in order to *fulfill the order* (**A**), and *should not shave their heads* (**anti-P**), in order that *there should be no riots* (**B**).

Resolution of CR

Resolution of contradiction **according to condition.**

Solution

Kan Shi strictly forbade monks and criminals to wear tresses as well as those who belonged to the lowest social stratus. And thus, the tress, which was so much hated before, became a distinction mark of each honorable and respected man. The resistance stopped immediately and then it was impossible to imagine a Chinese without a tress. On the contrary, there was no greater shame for a Chinese than to be devoid of his braid.³

Problem 4.17 Evacuation of manufacturing plant

Conditions of the problem

During the Great Patriotic War one of the most urgent tasks of industry was the evacuation of manufacturing plants located near the front line. One of such plants was the metallurgical combine “Severonickel”. A decision was taken concerning the dismantling of the combine and transporting it to Norilsk. Trying to capture a strategically important manufacturing plant of non-ferrous industry as a single whole and intact, the Germans did not bombard it, which, of course, made the evacuation of equipment easier. However, the employees of Severonickel understood that it was enough for the Germans to know about the dismantling of the combine, which was in full swing, and hundreds of bombs will be thrown on the territory of the manufacturing plant. It means that one had to deceive the enemy.

But how?

Analysis of the problem

SC: A

SC: The way to evacuate manufacturing plant. Positive effect (PE)—evacuation of the plant (**A**).

RC: A – anti-B

RC: Evacuation of the plant (**A**) and the bombardment of the plant (**anti-B**).

IFR: A, B

IFR: Evacuation of the plant (**A**) and absence of the bombardment of the plant (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

³ V. Krivtsov. Father Iakinthus.—L.: Khudozhestvennaya literature publishers, 1988, p. 212 (In Russian).

CP: The equipment of the plant *should be dismantled* (**P**), in order to *carry out the evacuation* (**A**), and *should not be dismantled* (**anti-P**), in order that *there should be no bombardment* (**B**).

Or:

The plant **should operate** (**anti-P**), in order that the Germans *should not bombard it* (**B**), and **should not operate** (**P**), in order that it *could be evacuated* (**A**).

Resolution of CP

Resolution of contradiction **in structure**.

It is necessary to arrange for such a situation that the Germans should consider a dead plant to be operating. How can the Germans define at a distance, if the plant is functioning or not? Of course, by the smoke rising from the chimney stacks!

Solution

Smoke barrels were brought to the plant and ignited.⁴

Principle 26 is used. Copying.

Problem 4.18 The Bat

Conditions of the problem

As it is known, the bats find their way in space through ultrasound. In order “to see” small objects by hearing them, the bats should have a rather sensitive hearing apparatus, while in order to be oriented at long distances, they should issue an ultrasound of large amplitude. However, in this case the bat would deafen itself: the sound issued by it would damage the auditory nerve endings.

What outcome did nature find?

Analysis of the problem

SC: anti-B

SC: What’s to be done, so that the bat wouldn’t deafen itself? Undesirable effect (UE)—to deafen itself (**anti-B**).

RC: A – anti-B

RC: The bat must have a very sensitive hearing system, in order to locate small objects (**A**) and a possibility to deafen oneself (**anti-B**).

IFR: A, B

IFR: The bat locates small objects (**A**) and does not deafen itself (**B**).

CP: P → A, anti-P → B

⁴ In the World of Metals.—M.: Metallurgia publishers, 1988, p. 80 (In Russian).

CP: The bat *should have sensitive hearing system* (**P**), in order *to locate the smallest objects* (**A**), and *shouldn't have sensitive hearing system* (**anti-P**), in order *not to deafen oneself* (**B**).

Resolution of CP

Resolution of contradiction **in time**.

While issuing ultrasound the bat hears badly, while during the signal reception he hears well.

Solution

The bat becomes “deaf” during signal transmission, while during the remaining time he is capable of perceiving signals. It is possible due to peculiarities of internal structure. The larynx is connected with the ear bones through a special ligament, larynx ligament pulls the hammer from the incus, and the bat doesn't hear.

Problem 4.19 Children from neighboring houses

Conditions of the problem

One American farmer was annoyed at the children of his neighbors. They horsed about on a lawn in front of his house. They shouted and knocked with sticks at the tops of rubbish cans.

Persuasions or menaces won't help and the children will continue to disport themselves.

What's to be done?

Analysis of the problem

SC: anti-B

SC: How to stop the roistering of the children? Undesirable effect (UE)—roistering of children (**anti-B**).

RC: A – anti-B

RC: Roistering of children on a lawn in front of the farmer's house is an amusement for them (**A**), but is a nuisance for the farmer (**anti-B**).

IFR: A, B

IFR: The children don't want to roister on the lawn in front of the farmer's house (**B**) and don't bring any nuisance to the farmer (**A**).

CP: $P \rightarrow A$, $\text{anti-P} \rightarrow B$

CP: The children *should misbehave in front of the farmer's house (P)*, since it *gives them pleasure (A)*, and *shouldn't misbehave in front of the farmer's house (anti-P)*, in order *not to bring any nuisance to the farmer (B)*.

Resolution of CP

Resolution of contradiction **according to condition.**

Solution

The farmer went out, approached the children and said cordially: "I am so glad, boys that you are playing in front of my house. I am already old and it is so pleasant for me to look at you. Please, come to my house every day and play. I will pay 20 cents to each of you for that". The boys gladly agreed and their playing continued next day. However, in a day the farmer announced the following to the children: "My business is unsuccessful, therefore, I will pay you 10 cents". The boys did not like that, but they came to work off their payment. Though they did it without the previous enthusiasm. After that the farmer told the boys that he was short of money and therefore he could pay them 5 cents only.

– Oh, no!—said the boys. For such chickenfeed you yourself run about your lawn!

Since that day silence reigned at the farm.

Principle 23 is used. Other way round.

Problem 4.20 Fishing

Conditions of the problem

Since old times the citizens of the southern France fished in the mouth of the Rhone river. With the tide shoals of fish come into the river, while with the low tide they go back into the sea. It is not possible to place a fishing net on their way: it will be carried away by the low tide.

What's to be done?

Analysis of the problem

SC: A

SC: Fishing with fishing nets. Positive effect (PE)—fishing with a fishing net (**A**).

RC: A – anti-B

RC: Fishing with a fishing net (**A**) and the net is carried away by the low tide (**anti-B**).

IFR: A, B

IFR: Fishing with a fishing net (**A**) and the net is not carried away by the low tide (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The net *should necessarily be set* (**P**), in order to catch fish (**A**), and *shouldn't be set* (**anti-P**), so that it *should not be carried away by the low tide* (**B**).

Resolution of CP

Resolution of contradiction **in time**.

Solution

The net is set only for the time of high tide, and then it is taken away together with the caught fish.

Resolution of CP

Resolution of contradiction **in structure**.

Prompt

It is possible to make use of resources of the sea.

Solution

The fishermen called dolphins for help. They made a din with rattles and shouted: “Snub-nosed! Snub-nosed!”. The dolphins swam to the shore and, as if shuddered to a halt, struggled with the strong low tide. The fishermen rushed to the boats—the dolphins blocked the way to the fish. It tossed about between “live fence” and the beach. After hunting the cheerful and satiated dolphins swam to the boats and disported themselves in shallow waters. Many historians of ancient times wrote about joint hunting with dolphins. They also mentioned the nickname, which was given to these animals in some coastal countries. Everywhere the fishermen and “Snub-nosed” were friends. And the inhabitants of the islands of the Southern part of the Pacific Ocean even organized a “dolphin” holiday. Of course, the heroes of the day were also invited. However, they were called not with shouts and rattles, but with calm melodious song.⁵

Problem 4.21 Wise Ali

Conditions of the problem

There lived a khan, who was very angry, all people were afraid of him. The Khan loved only his son Khusain. Khusain every day went hunting. It happened once that he did not return home by night. The khan got angry and called all his servants. He ordered them to seek for his son, but the man, who would bring bad news will have a melted lead thrown into his throat. After long rambling through the woods, the servant finally found Khusain, but he was dead: the wild animal killed him.

How can the servants tell about the death of the beloved son, but at the same time remain alive?

They went to the wise shepherd Ali, who rescued them.

What did Ali do?

⁵ A. Vladimirov. The Golden Strings.—M.: Detskaya literature publishers, 1977, p. 19 (In Russian).

Analysis of the problem

SC: A

SC: Shall the servants execute the order of khan to find his son? Positive effect (PE)—fulfillment of khan's order (**A**).

CR: A – anti-B

RC: Fulfillment of khan's order (**A**) is associated with the promise to pour melted lead into the throat of a servant who would bring the news about the son's death (**anti-B**).

IFR: A, B

IFR: The order of khan is fulfilled (**A**), and he doesn't pour lead into the throat of the courier (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The servants *should inform khan about the death of his son* (**P**), in order to fulfill his order (**A**), and *shouldn't inform khan about the death of his son* (**anti-P**), in order that he *shouldn't fulfill his to pour melted lead into the throat of a servant who would bring the news about the son's death* (**B**).

Resolution IIC

Resolution of contradiction **in structure**.

Solution

The shepherd Ali created a new musical instrument—dombra. Next day Ali, surrounded by frightened servants entered the tent of the khan and started playing. Sharp whistling of wind mixed with the roaring of the wild animal. The strings screamed, as if it was a human voice, begging for help. And once again the roaring of the animal and again the sorrowful noise of the wood...

The khan understood everything, got angry and threatened Ali with fulfilling of promise to punish the man, who would bring sad news.

— My khan,—calmly answered the old shepherd,—I did not tell you anything. I did not pronounce a single word. If you are angry, punish this instrument, which I manufactured and named dombra.

Thrown into a rage, the khan ordered to throw some melted lead on the dombra. The lead burnt a round whole in the wood under the strings.

Thus, old Ali, due to his inventiveness and skill saved the life of khan's servants. And the inhabitants of steppes since then received a new musical instrument—dombra.⁶

Principle 24 is used. The intermediary.

Problem 4.22 False picture

Conditions of the problem

One of the museums of Germany acquired a picture by Rubens in a museum of France. The German experts, who examined the canvas in situ, acknowledged its authenticity and confirmed it with their signatures on the back side of the picture. After that the picture was packed and sent to Germany. When it was delivered to the indicated address, another examination was made (the picture might have been replaced during transportation) and discovered that it was a fake, while on the reverse side of fake picture there were authentic signatures of experts, certifying its authenticity!

Analysis of the problem

SC: A

SC: Evaluation of authenticity of the picture is confirmed by signatures. Positive effect (PE)—confirming the authenticity of the picture (**A**) in France.

RC: A – anti-B

RC: Confirming the authenticity of the picture (**A**) in France and confirming in Germany the falsification of the picture, signed in France (**anti-B**).

IFR: A, B

Let us analyze IFR from the standpoint of the forgers.

IFR: confirming the authenticity (**A**) of a falsified picture (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The signatures *should be put on the authentic picture* (**P**), since the experts confirmed the authenticity of the picture (**A**), and *should be put on the falsified picture* (**anti-P**), since the German experts confirmed that it is a fake (**B**), so that the forgers could realize their plan.

Resolution of CP

Resolution of contradiction **in time**.

The pictures were authentic, when the experts were putting their signatures and became fake, when they arrived to the place of destination.

⁶ Master Ali (Kazakh folk tale) (In Russian).

How to realize that—resolution of contradiction **in structure**.

Solution

A fake picture was put under an authentic one, being connected to it by one frame. The experts saw the authentic picture, while the signatures were put on the reverse side of the fake picture. The authentic picture was stolen, while a false picture with real signatures of experts was sent to Germany.

Principle 5 was used. Merging.

Problem 4.23 Lorgnette

Conditions of the problem

Catherine the Second did not like aristocrats carrying lorgnettes and long white gloves.

But how could one prohibit these things?

What's to be done?

Analysis of the problem

SC: anti-B

SC: Catherine cannot prohibit the aristocrats to wear lorgnettes and long white gloves. Undesirable effect (UE)—impossibility of prohibition (**anti-B**).

RC: A – anti-B

RC: Contradiction between the wish of Catherine II, that the aristocrats wouldn't wear lorgnettes and long white gloves (**A**), and the impossibility to prohibit it (**anti-B**).

IFR: A, B

IFR: The aristocrats themselves don't want to wear lorgnettes and long white gloves (**B**), thereby satisfying the wish of Catherine (**A**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: The aristocrats *want to wear lorgnettes* (**P**), in order to *obey the order* (**A**), and *don't want to wear lorgnettes* (**anti-P**), so that *there would be no riots* (**B**).

Resolution of CP

Resolution of contradiction **according by condition**.

Solution

Catherine the Second issued an order to hand out lorgnettes and white gloves to *budochniks* (policemen on duty),⁷ whose posts are located at the crossings.

Problem 4.24 Potatoes

Conditions of the problem

In France potatoes were treated with incredulity for long time.

How to disseminate potatoes in France?

Analysis of the problem

SC: A

SC: How to disseminate potatoes in France? Positive effect (PE)—dissemination of potatoes in France (**A**).

RC: A – anti-B

RC: Dissemination of potatoes (**A**) in France is associated with strong incredulity of people, especially of peasants (**anti-B**).

IFR: A, B

IFR: Dissemination of potatoes (**A**) – wish of peasants (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: The potatoes *should be disseminated* (**P**), since *there is a wish to disseminate it* (**A**), and *should not be disseminated* (**anti-P**), since it causes incredulity among the peasants (**B**).

Resolution of CP

Resolution of contradiction **according by condition and in structure**.

Solution

Significant role in the dissemination of potatoes was played by the Parisian apothecary Antoine Auguste Parmantier. He asked the king Louis 16th to allot him a field of sandy land near Paris and planted potatoes there. During the daytime Parmantier ordered a detachment of armed watchmen guard his kitchen-garden. At night there were no watchmen. But some curious peasants appeared, who decided to check, what the freaky apothecary guards so zealously. They came there at night, secretly stole the

⁷ Budochnik (policeman on duty)—since the first half of the eighteenth century the lowest position of the municipal police in the Russian Empire, a police guard, whose post was near a special *budka*—guard booth (*Material from Russian Wikipedia*).

bulbs and planted them in their kitchen gardens. That was exactly what Parmantier wanted to achieve. Very soon the French peasants started to love the new plant.

Principle 13 is used. Other way round.

Problem 4.25 Sharks

Conditions of the problem

A pneumatic gun is used for protection against sharks, which shoots explosive bullets. Such bullets are sure to kill the shark, however, the blood, which is discharged in large amount, attracts other flesh-eaters. What's to be done?

Analysis of the problem

SC: anti-B

SC: Killing a shark is associated with pouring a lot of blood, which attracts other sharks. Undesirable effect (UE)—attraction of sharks (**anti-B**).

RC: A – anti-B

RC: Safety of the diver (**A**) and attraction of flesh-eaters (**anti-B**).

IFR: A, B

IFR: Safety of the diver (**A**)—doesn't attract flesh-eaters (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The diver *should shoot the shark* (**P**), in order to *render it harmless and achieve safety* (**A**), and *should not shoot the shark* (**anti-C**), in order *not to attract other sharks* (**B**).

Resolution of CP

Resolution of contradiction **in structure**.

Solution

Divers of US Naval forces use pneumatic guns for protection from sharks. The bullets for this gun are made of hard carbon dioxide. When such a bullet hits the shark, it gets inflated like a balloon, rises to the surface, becomes absolutely safe for the human and does not attract other sharks. Such guns, in particular (according to the data of the magazine "Science Digest", v. 72, No. 6), were used, by the divers, in particular, during the splashdowns (water landings) of "Apollo" spaceships.

Problem 4.26 Curing the bees

Conditions of the problem

The bees sometimes fall ill. It is possible to cure the bees from a number of illnesses with the aid of medicines, applied over their bodies. However, to catch the

bees one by one and to apply the medicine over the body of each of them takes a lot of time and is expensive, while to spray the medicine inside the entire bee-hive is impossible, since the quality of honey in this case is degraded (it can be entirely spoiled!), and this is inadmissible.

What's to be done?

Analysis of the problem

SC: anti-B

SC: How to avoid spoiling honey in curing the bees? Undesirable effect (UE)—spoiling honey (**anti-B**).

RC: A – anti-B

RC: Curing the bees (**A**) is associated with spoiling honey (**anti-B**).

IFR: A, B

IFR: The bees are cured (**A**)—honey is not spoiled (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The medicine *should act* (**P**), in order to cure the bees (**A**), and *shouldn't act* (**anti-P**), in order not to spoil honey (**B**).

Resolution of CP

Resolution of contradiction **in structure**.

Solution

Light spheres placed inside the bee-hive are impregnated with medicine. The bees constantly maintain tidiness in their beehives. Pushing alien objects from their lodgings, they get the necessary portion of medicine.⁸

Problem 4.27 Burial in Tibet

Conditions of the problem

There are neither cemeteries, nor individual graves in Tibet. An exception is made only for Dalai Lama and Panchen-Lama. Their bodies after their death are blamed and covered with gold. It is very difficult to dig graves in such rocky soil and the religious philosophy dictates non “committal to earth”, but “committal to sky”. However, since there is no fuel of any kind, cremation is impossible.

What do they do in Tibet in order to perform the ritual of transcending of the dead man to a different level of existence?

⁸ Das Tier, 1987, No. 12.

Analysis of the problem

SC: A

SC: How to carry out the funeral of Tibet citizens? Positive effect (PE)—funeral (**A**).

RC: A – anti-B

RC: Carrying out the funeral (**A**) in Tibet is associated with the absence of place and means for cremation (**anti-B**).

IFR: A, B

IFR: Carrying out the funeral (**A**) no necessity to cremate (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: Carrying out the funeral (**A**) (cremation) *there should be certain combustible materials* (**P**), and *in connection with the absence of them* (**B**) they *should be absent* (**anti-P**).

Resolution IIC

Resolution of contradiction **in structure**.

Solution

“Celestial funeral” ((jhatör (Wiley: bya gtor)—the main kind of funeral in Tibet and in a number of regions, which are situated close to Tibet. It is also called “giving alms to the birds”. According to Tibetan beliefs the soul leaves the body at the moment of death, and the man should bring use to other people at all stages of his life. Therefore, the dead body is fed to the birds as the last manifestation of charity.

Until today many Tibetans consider exactly this way of burial to be the only possible type of funeral.⁹

Use of resources.

Principle 10 is used. Preliminary execution.

Problem 4.29 The Genie

Conditions of the problem

A peasant once unearthed a jug on his field. He immediately opened it—he expected to find golden coins inside... However, an angry Genie rushed out of the bottle.

- For thousand years I prayed to God that somebody would let me out,—shouted he, shaking his fists.—I promised heaps of gold and diamonds to my savior... But

⁹ Celestial burial in Tibet, URL: <http://masterok.livejournal.com/1533064.html> (In Russian).

nobody even listened to my prayers! Then I decided to kill the man, who would disturb my peace. One more thousand years passed and you decided to look for hidden treasures in your own kitchen garden!!! You are in your dying hours, poor man!

However, the peasant was not a coward...

- Let it be like this, Genie. But fulfill my last request.
- Speak it out,—said the Genie reluctantly.—I will fulfill it. But bear it in mind: I will not let you go, I will not do any harm to myself, and I'm not going to get back into the jug. I also read fairy tales and I will not let anyone deceive me.

The peasant expressed his request—and remained alive. What in particular did he ask for?

Analysis of the problem

SC: anti-B

SC: The genie promised to kill the man, who would uncork the jug. Undesirable effect (UE)—killing the peasant (**anti-B**).

RC: A – anti-B

RC: Contradiction between the wish of the peasant to remain alive (**A**) and the promise of the genie to kill the man who would uncork the jug (**anti-B**).

IFR: A, B

IFR: The promise of genie is fulfilled (**B**) and the wish of the peasant to remain alive is granted (**A**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: The genie *should kill* the peasant (**P**), since he *promised to do that* (**A**), and *he shouldn't kill* the peasant (**anti-P**), since *the peasant wants to remain alive* (**B**).

Resolution of CP

Resolution of contradiction **in time**.

Solution

- OK, kill me,—but in a hundred years!

Problem 4.30 Vaccine

Conditions of the problem

Influenza is an infectious disease. It can be eliminated through vaccination. However, the virus of influenza is able to mutate quickly. By the time a serum is

developed against a particular strain of virus, the next variant of it already exists in the world. In connection with this the struggle with epidemics of this illness is rather difficult. According to the World Health Organization, from 250 to 500 thousand people die annually. However, people die not from the very presence of virus, but from complications. Because of its activity inflammation of vessels takes place in the organism, they start to conduct fluid, which penetrates to the alveolus, which leads to such state, when the human suffocates.

What's to be done?

Analysis of the problem

SC: A

SC: Fight against virus with the aid of vaccine. Positive effect (PE)—vaccine struggles with the virus of influenza (**A**).

RC: A — anti-B

RC: Fight against virus of influenza (**A**) and quick mutation of influenza virus (**anti-B**).

IFR: A, B

IFR: Vaccine fights (**A**) irrespective of mutation (**B**) with any strain of influenza.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: *There should be* vaccine (**P**), in order *fight with the virus of influenza* (**A**), and *there should be no* vaccine (**anti-P**) because of *its quick mutation* (**B**).

Resolution of CP

Resolution of contradiction **in structure**.

Solution

Not to fight with the virus, but to strengthen the organism, in particular, vessels.

Research workers from medical institutions of Canada—Sunnybrook hospital and Saint-Michael hospital developed a medicine, which fights with influenza, helping organism. Developed experimental medicine Vasculotide strengthens the walls of the vessels. As a result of experiments with mice, which got the medicine, 80% survived, all mice, which did not get the medicine, died.

In this case the medicine acted even being given several days after infection and did not tell negatively on antibody response of the organism.¹⁰

Problem 4.31 Ferry-boat

Conditions of the problem

Ferry-boats are intended for carrying people and means of transportation by water between two shores (of the river, lake, straight and sea). One of the most widely spread kinds of ferry-boat moves along the rope, stretched between two shores of a river. The ferry-boat may take the form of a boat, several connected boats or a pontoon. Either an engine or the force of the human muscles are used for motion. In the latter case the ferryman rotates the pulling hoist, applying significant force.

How to make the labor of ferryman easier, if there is no opportunity to use the engine?

Analysis of the problem

SC: anti-B

SC: Amount of effort of the ferryman on rotating the hoist. Undesirable effect (UE)—manual labor (**anti-B**).

RC: A — anti-B

RC: Travel of the ferry-boat (**A**) and the amount of manual labor (**anti-B**).

IFR: A, B

IFR: The amount of manual work is zero (**B**), while the tugboat travels (**A**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: *It is necessary to apply certain amount of effort (**P**), in order to move the tugboat (**A**), and no amount of effort is needed (**anti-P**), in order to make the labor of ferryman easier (**B**).*

Resolution of CP

Resolution of contradiction **in structure**.

Efforts needed for travel of the ferry-boat have to come from outside, i.e., resources should be used.

Solution

What resources, able to generate effort for traveling, exist? One can use the flowing of the river.

¹⁰ URL: <https://geektimes.ru/post/251570>.

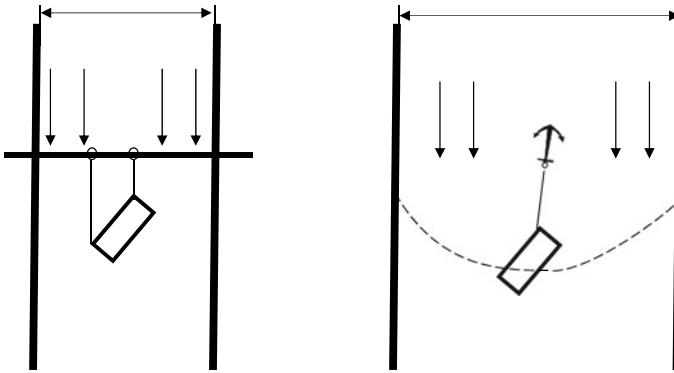


Fig. A.18 Diagrams of ferry-boats

Such solutions were used back during the times of Peter the Great. They were called the airplane of Peter the Great.

If the river was narrower than 40 m (Fig. A.18a), a strained rope was used, while the ferry-boat was turned round at an angle to the flow. The force of the flow pressed upon the wall of the ferry-boat and moved it. The ferry-boat was turned around with the aid of two wire ropes or with the aid of the steer.

If the river was wider than 40 m (Fig. A.18b), then an anchor was installed in the middle of the river and a wire rope was fastened to it, while another end of this wire rope was fastened to the ferry-boat. The ferryman positioned the ferry-boat at an angle of about 55 degrees to the flow with the aid of a steer, and the flow moved it along the curvy line from one river bank to another.¹¹

Problem 4.32 Chess-player

Conditions of the problem

One weak chess-player proposes a strong chess-player to have a chess game with him through letter-exchange for money. The condition is that in case of a loss the weak player pays the strong one a sum, which is two times less, because he plays worse. If the match ended in a draw, the strong player pays the full sum to his partner, but in case a weak player wins, he gets a double sum.

He has no chances to win, but nevertheless he gets at least a half of the sum. How did he manage to do it?

Analysis of the problem

SC: A

SC: A weak chess-player wants to win a game with a strong player. Positive effect (PE)—wish to win the game (A).

¹¹ URL: http://sneg5.com/load/tekhnika/samoljoty_petra_pervogo/9-1-0-111.

RC: A – anti-B

RC: Wish to win the game (**A**) and the impossibility to do it in connection with the difference in the level of skills of both players (**anti-B**).

IFR: A, B

IFR: In spite of the difference in the level of skills of both players (**B**), the weak player wins the game (**A**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: *It is necessary for the weak player to have an equal level of skills with the strong player (**P**), in order to win the game (**A**), and impossibility to win the dispute (**anti-P**) because of the difference in the level of skills (**B**).*

Resolution of CP

Resolution of contradiction **in structure**.

Solution

The weak player agrees on the same conditions with another strong player of the same level, but with one player he agrees to play with white chess pieces and with another—with black chess pieces.

In the process of the game he simply copies the moves of the first player and sends them to the second player, and vice versa. Finally, he either “wins” the game with one of them or gets a tie with both partners. In any case he takes half of the gain.

Principle 24 is used. Intermediary and resources.

Problem 4.33 Blinding the driver

Rays of the sun blind the driver. The drivers use protective visor, but it restricts the vision. Neither is the use of protective sunglasses always comfortable.

What's to be done?

Analysis of the problem**SC: anti-B**

SC: Protective visor restricts the vision. Undesirable effect (UE)—restriction of vision (**anti-B**).

RC: A – anti-B

RC: Sun-screening protective visor prevents the driver from blinding (**A**), but restricts the vision (**anti-B**).

IFR: A, B

IFR: Visor does not restrict the vision (**B**) and prevents the driver from blinding (**A**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: Visor *should be transparent* (**P**), in order *not to restrict the vision of the driver* (**B**), and *shouldn't be transparent* (**anti-P**), in order *to prevent the driver from blinding* (**A**).

Resolution of CP

Resolution of contradiction **in structure**.

Visor should dissipate the sun rays.

Solution

The visor Clear View made of acryl has been developed. It dissipates sun beams without blinding the driver and does not restrict the vision of the driver. Besides, this visor has a special glass of yellow color for protecting the driver from blinding by the lights of the car coming from the opposite direction during the night, as well as improves the visibility in the fog, during the rain and snowfall.

Problem 4.34 Foldable umbrella**Conditions of the problem**

Invention of a foldable umbrella significantly simplified its use. However, there are disadvantages:

- The umbrella creates a hindrance in entering the building or in getting into a car. It is necessary to close it prior to entering the building or getting into a car. During this time the owner of the umbrella will get wet under the rain.
- The outward surface of the umbrella gets wet under the rain and everything, which is touched with the umbrella, also gets soaked.

What's to be done?

Analysis of the problem**SC: anti-B**

SC: The human gets soaked under the rain, while the umbrella is closed in front of the door. Undesirable effect (UE)—soaking (**anti-B**).

RC: A – anti-B

RC₁: Closing the umbrella in front of the door (**A**) leads to soaking of the human under the rain (**anti-B**).

It is also possible to formulate another RC.

RC₂: The human passed through the door, therefore, he doesn't get soaked under the rain (**A**), but it is practically impossible to close the umbrella (**anti-B**).

IFR: A, B

IFR: The human doesn't get soaked under the rain (**B**) and the umbrella close freely (**A**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The umbrella *should be opened* (**P**), in order *not to get soaked under the rain* (**B**), and *should be closed* (**anti-P**), in order *to pass through the door* (**A**).

Resolution of CP

Resolution of contradiction **in structure**.

The umbrella should close and open in the opposite direction.

Principle 13 is used. The other way round.

Solution

Japanese design studio H Concept belonging to Mr. Hiroshi Kajimoto developed a reversible umbrella and called it Unbrella (Fig. A.19). The umbrella opens (inside) without creating any hindrance in opening or closing it, when a person is in the room or in the car. It is two times stronger than the previous umbrella structures and cannot be bent in the opposite direction as a result of strong wind.

Besides, in a folded state the soaked side remains inside the umbrella, while the dry side of tissue appears to be turned outwards, without causing any trouble for those, who touches it.

In a folded state such an umbrella can be simply stood on the floor.

A similar umbrella is manufactured by the company Suprella (Fig. A.20). In contrast to Unbrella, it has two layers of tissue (external and internal). The external layer is made water-resistant, therefore water rolls off it and the tissue does not get soaked.

A.3 Resource Thinking

Problem 5.11 Treasures of the Temple

Conditions of the problem

In Lhassa, in the prayer hall of the Jokan temple, which was erected by the Tibet King Sronzan Gambo (founder of Tubo dynasty) in 646, there is a statue of Shakyamuni, brought from China by the wife of the king Princess Veng-Cheng. The majestic figure of Shakyamuni steps out from the darkness, lighted with huge lamps arranged



Fig. A.19 Reverse umbrella—Unbrella



Fig. A.20 Reverse umbrella—Suprella

around it. The statue of Shakya-muni is cast of gold and decorated with countless precious stones. It has facial resemblance with the young Buddha. During the entire period of existence of the temple not a single precious stone was carried out of this hall, though the temple is open not only for pilgrims. Nobody ever mentioned even to touch the stones unnoticed.

How do the priests of the temple provide for good preservation of the temple's treasures?

Analysis of the problem

SC: A

SC: How do they provide for good preservation of treasures? Positive effect (PE)—preservation of treasures (**A**).

RC: A — anti-B

RC: Preservation of treasures (**A**) requires guarding (**anti-B**).

IFR: A, B

IFR: Preservation of treasures (**A**) without guarding (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: *It is necessary to have the guards (P), in order to preserve treasures (A), and it is not necessary to have the guards (anti-P), since they are absent (B).*

Resolution of CP

Resolution of contradiction **in structure**.

Solution

The treasures are preserved by black cats, who noiselessly appear from the darkness and clutch by the throat all those, who dare to stretch their hands towards treasures.¹²

Problem 5.12 Tent

Conditions of the problem

A tent is a tissue, which is stretched by wires (stretchers). Usually a tent is fastened by attaching it to trees or to other structures standing nearby. There are also tents with supporting pillars. Imagine that you come to the beach and you have nothing to fasten the edges of the tent with. There are no trees around or other means for fastening the tent edges.

How can one fasten the stretches?

¹² V. Ovchinnikov. Ascension to Shambala.—M.: Drofa, 2003, p. 11 (In Russian).



Fig. A.21 Tent

Analysis of the problem

SC: A

SC: How can one fasten the stretches? Positive effect (PE)—fastening of stretches (**A**).

RC: A – anti-B

RC: Fastening of stretches (**A**) requires improvised means (**anti-B**).

IFR: A, B

IFR: Fastening of stretches (**A**) without improvised means (**B**).

CP: $P \rightarrow A$, $\text{anti-P} \rightarrow B$

CP: *It is necessary to have improvised means (P), in order to fasten the stretches (A), and not to have improvised means (anti-P), since they are absent (B).*

Resolution of CP

Resolution of contradiction **in structure**.

Solution

Otentik company created a tent, which is fastened with the aid of sacks, filled with sand or with pebbles (Fig. A.21).

Resources are used, which are available on the beach—sand or pebbles.

Problem 5.13 Fire on the island

Conditions of the problem

Imagine that one man is staying on the island. Because of long-time drought the grass and the bushes on the island have become extremely dry. All of a sudden a fire broke out at one end of the island and the wind drove the fire towards the man.

He cannot escape to the sea, since there are numerous sharks near the shore. And there are no shores without plants on the island. What should this man do in order to escape death?

Analysis of the problem

SC: A

SC: How to escape from the fire? Positive effect (PE)—saving from the fire (**A**).

RC: A – anti-B

RC: Saving from the fire (**A**) requires the presence of place on the island, where no plants grow, while there is no such place (**anti-B**).

IFR: A, B

IFR: Saving from the fire (**A**) a place appears on the island, where no plants grow (**B**).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: *It is necessary to have a place on the island, where no plants grow (P), in order to save from the fire (A), and the absence of place, where no plants grow (anti-P), since the entire island is covered with plants (B).*

Resolution of CP

Resolution of contradiction **in time, in space and in structure.**

Solution

The man has to start a fire on the leeward side (opposite to the wind direction) from himself (Fig. A.22a) and to step forward a little in the direction of the main fire. The wind will drive the fire started by the man to the leeward end of the island. When this section is burnt out (Fig. A.22b), the human will be able to return there and calmly wait until the main fire will reach this part and be extinguished, since there will be nothing to burn.¹³

Problem 5.14 Copyright protection Conditions of the problem

Some publishers of dictionaries and atlases copy them from other publishers as pirated editions. How to prove that the given book is a pirated edition?

Analysis of the problem

SC: anti-B

¹³ URL: <http://www.potehechas.ru/zadachi/zadachi.shtml>.

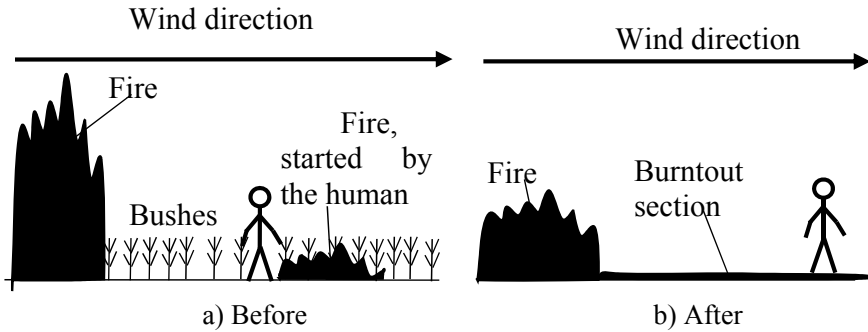


Fig. A.22 The fire

SC: Pirate copying of dictionaries and atlases. Undesirable effect (UE)—identification of pirate copying (**anti-B**).

RC: A – **anti-B**

RC: Publishing a dictionary and an atlas (A) and impossibility to identify pirate copying (**anti-B**).

IFR: A, B

IFR: Identification of pirate copy of dictionary and atlas (B) and publishing them (A).

CP: $P \rightarrow A$, **anti-P** $\rightarrow B$

CP: *Dictionaries and atlases should be distinguished (P)*, so that *it would be possible to identify a pirate copy (B)*, and *should not be distinguished (anti-P)*, in order to *publish them for future use (B)*.

Resolution of CP

Resolution of contradiction **in structure** and **according to condition**.

Solution

Usually the publishers include with the dictionary a non-existing word, and place a non-existing island inside the atlas. If they are found in some other publication, the fact of copying becomes obvious.

Problem 5.15 Saucer

Conditions of the problem

A saucer, plate or cup should evenly stand on horizontal surface. For this purpose, their bottom is polished, but it will be expensive to polish the entire surface of the bottom and it will take too much time.

What's to be done?

Analysis of the problem

SC: anti-B

SC: To polish the surface of the bottom is a long and expensive task. Undesirable effect (UE)—polishing the surface of the bottom is long and takes much time (**anti-B**).

RC: A – anti-B

RC: The saucer should evenly stand (A), and polishing the surface of the bottom is long and takes much time (**anti-B**).

IFR: A, B

IFR: Quick and inexpensive polishing of the bottom of the saucer (B), and the saucer evenly stands on the horizontal surface (A).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: *The bottom of the saucer should be polished (P)*, so that *it should evenly stand on the horizontal surface (A)*, and *shouldn't be polished (anti-C)*, so that *it would not be so long and expensive (B)*.

Resolution of CP

Resolution of contradiction **in space** and **in structure**.

Solution

The saucer is embodied with a ring-shaped hedge from the lower part. Only this ring-shaped hedge is polished.

Problem 5.16 A cucumber in the bottle

Conditions of the problem

Everybody knows that there is a method to place a ship model inside the bottle. But how can one arrange for such a situation, when a whole ripe cucumber is placed inside the bottle and the bottle remains undamaged?

Analysis of the problem

SC: anti-B

SC: A big cucumber cannot pass through the bottleneck. Undesirable effect (UE)—the cucumber does not pass through the bottleneck (**anti-B**).

RC: A – **anti-B**

RC: It is necessary to place a big cucumber inside the bottle (**A**), but it does not get into the narrow bottleneck (**anti-B**).

IFR: A, B

IFR: A big cucumber is placed inside the bottle (**A**).

CP: $P \rightarrow A$, **anti-P** $\rightarrow B$

CP: The cucumber *should be small* (**P**), in order *to be able to squeeze through the narrow bottleneck* (**B**), and *should be big* (**anti-P**), since *it is big inside the bottle* (**A**).

Resolution of CP

Resolution of contradiction **in time**.

Solution

The kukes, which appeared on the stem, is placed inside the bottle through a narrow bottleneck. The cucumber ripens and grows inside the bottle. As it is known, the cucumbers ripen very quickly and in a few days the cucumber will grow inside the bottle.

Problem 5.17 Radiator

Conditions of the problem

One of the most widely spread methods for creating thermal comfort in the room is heating with the aid of radiator, in which hot water circulates.

It is possible to increase the efficiency of the radiator functioning through enlarging the area of contact with the ambient air or through enhancement of radiating capacity of the radiator.

It is possible to increase the area of contact with the air by increasing the number of ribs of the radiator up to creating needle-type ribs, like it is done with radiators for cooling electronic devices, however, it requires changes in the design of radiators.

Another method of increasing the area of contacts with the ambient air is to increase the number of radiators, for example, by filling all walls with the radiators or by placing them in several rows.

If all walls are filled with radiators, there will be a hindrance with hanging something on the walls, for example, shelves. Arrangement of the radiators in several rows will occupy a lot of place and will decrease the area of the room.

It is possible to enhance the radiating capacity of radiators by conferring the shape of a parable to it, as it is done in electric heaters. It is difficult to manufacture radiators of parabolic shape and it is possible only, when new radiators are produced.

How to increase the efficiency of heat release of available radiators without increasing their number?

Analysis of the problem

SC: A

SC: How to increase the efficiency of heat release of available radiators? Positive effect (PE)—increase of heat release (**A**).

RC: A – anti-B

RC: Increase of heat release (**A**) requires the increase of the number of radiators or the variation of shape of a radiator (**anti-B**).

IFR: A, B

IFR: Heat exchange intensifies (**A**) without the increase in the number of radiators and without any modification of shape (**B**) of the radiator.

It is possible to specify IFR. Since the stated problem implies that the heat release is intensified without increasing the number of radiators, we get:

IFR 1: Heat release is intensified (**A**) without modifying the shape of the radiator (**B**).

Thus, we have to increase thermal release only due to increasing the radiating capacity of the radiator.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The radiator *should increase its radiating capacity* (**P**), in order to increase thermal release (**A**), and *should not increase its radiating capacity* (**anti-C**), in order not to modify the shape (**B**) of the radiator.

Resolution of CP

Resolution of contradiction **in structure**.

Let us use resources.

Solution

It is necessary to fasten aluminum foil on the wall after the radiator so that thermal emission would be reflected from the battery and directed inside the room. In doing this, leave a gap between the foil and the wall or place there a sheet of material with

low heat conductivity. It will decrease thermal losses through the outward wall and create a directed flow of thermal energy into the room.

Problem 5.18 Ice in the pipes

Conditions of the problem

Because of condensation of moisture in the warm air leaving the room, Ice crust is formed on the internal surface of ventilation pipes in winter. Ice crust can clog the entire cross-section of the pipe.

Many mechanical devices have been invented, which eliminate the ice crust from the internal surface of the air ducts. All of them are unreliable (they get iced themselves) and require servicing.

What's to be done?

Analysis of the problem

SC: anti-B

SC: Ice crust can seal all cross-section of the pipe of the air duct. Undesirable effect (UE)—sealing of the pipe (**anti-B**).

RC: A – anti-B

RC: Ventilation pipe is necessary for the ventilation of the room (**A**), however, it is sealed by ice during winter time (**anti-B**).

IFR: A, B

IFR: Ventilation pipe ventilates the room (**A**) and is not sealed by ice during winter time (**B**).

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: Ventilation pipe *should be of variable shape—should be dynamic* (**C**), in order *not to be sealed with ice or to be able to drain ice* (**B**), and *should be of unchangeable shape—static* (**anti-P**), in order *to conduct air for ventilation* (**A**).

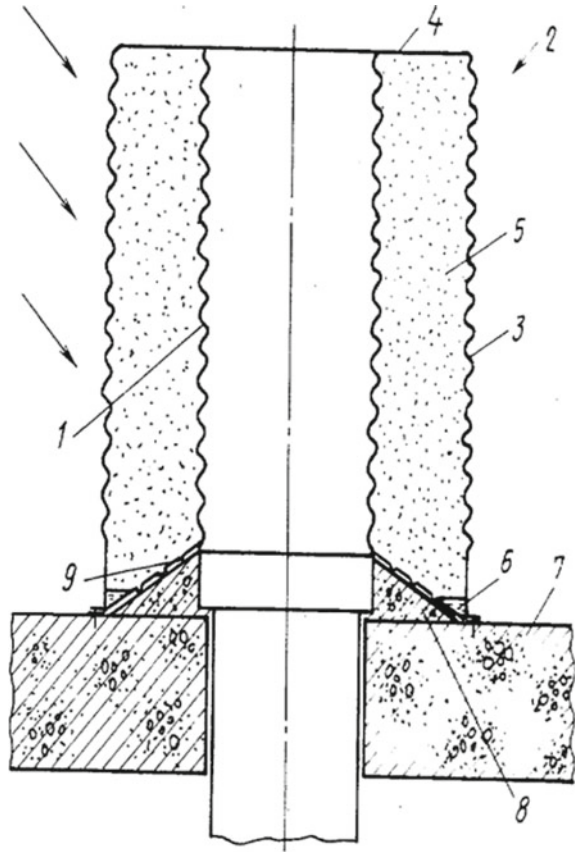
Resolution of CP

Resolution of contradiction **in time**.

Solution

Air duct is embodied as two coaxial corrugated pipes made of elastic material and hermetically sealed from two butts. The cavity formed between the pipes is filled by easily evaporating fluid, the outward wall being transparent, and the inner wall—blackened.

Fig. 1.23 Air duct. Author's certificate 1 298 488. 1—air duct; 2—device for ice removal; 3—pipe made of light-conducting film material; 4—pipe end; 5—cavity, filled with low-boiling fluid 6; 7—roofdeck; 8—ring nozzle; 9—gutters for freon leak-off



The sunlight heats the fluid 5 (Fig. A.23), which evaporates, increasing pressure, which causes the deformation of the air duct. Thereby it causes deterioration and exfoliation of ice. Condensation of fluid vapors additionally heats the air duct, improving the process of ice exfoliation.¹⁴

Problem 5.19 Repair of pipes

Conditions of the problem

Leakages in water pipes take place due to different reasons. As a result, water leaks out. In order to do the repairing, one has to shut off water for long time and to drain it through pipes. The repairing could be done without shutting off water, however, much water will leak out and there may be a deluge in rooms. How to do repair work without shutting off water and without water leaking out?

Analysis of the problem

¹⁴ Author's certificate 1 298 488.

SC: anti-B

SC: Large losses of water. Undesirable effect (UE)—losses of water (**anti-B**).

RC: A – anti-B

RC: Opportunity to repair the pipe (A) leads to large losses of water (**anti-B**).

IFR: A, B

IFR: Repair of pipe (A) without losses of water (B).

CP: $P \rightarrow A$, anti-P $\rightarrow B$

CP: In the place of repair *there should be no water* (P), in order *to do the repair work* (B), and *there should be water in the pipe* (anti-P), so that *there should be no losses of water* (A).

Let us deepen the contradiction.

CP₁ : $P_1 \rightarrow P$, anti-P $\rightarrow$ anti-P₁

CP₁: Water should *become non-water* (P₁), so that *there should be no water in the place of repair* (P), and should **remain to be water** (anti-P₁), in order *to remain in the pipe* (anti-P).

Resolution of CP

Resolution of contradiction **in time, in space and in structure**.

Water should be non-fluid (hard) during the time of repair, in the place of repair and should be fluid all the remaining time.

Solution

Let us use the first-order phase transitions.

Water is frozen in the place of repair.

Sleeve gaskets are put on the pipe, which embrace the pipe and “dry ice” (hard carbon dioxide) is poured into these gaskets. Water in the place of repair is frozen, and the ice plug enables to perform the work without water shutoff.

Problem 5.20 Water evaporation Conditions of the problem

Institute for Air Dynamics and Hydro Dynamics (TsAGI) was created according to the initiative of Nickolay Yegorovich Zhukovsky, in December 1918.

At the institute, which started functioning in the building of the Technical school, the laboratory was heated with a small tile stove, but the amount of heat was insufficient. The employees of the laboratory put on the cooker a water boiler in order to obtain more heat. A stove with a water boiler on it more or less heated the laboratory. However, the boiler produced a lot of vapor, i.e., a lot of moisture was released into

the air. Moisture in the class was worse than cold and it was utterly unpleasant to study. What's to be done?

Analysis of the problem

SC: anti-B

SC: A boiler with hot water creates dampness in the room. Undesirable effect (UE)—dampness (**anti-B**).

RC: A – anti-B

RC: A boiler with hot water heats the room (A), however, generates dampness (**anti-B**).

IFR: A, B

IFR: Heating the room (A) without dampness (B).

CP: $P \rightarrow A$, $\text{anti-}P \rightarrow B$

CP: Vapors of water *shouldn't penetrate into the room* (P), so that *there would be no dampness* (B), and *should penetrate into the room* (**anti-P**), in order to *heat it* (A) and due to physical properties of vapor to rise upwards.

Resolution of CP

Resolution of contradiction **in space** and **in structure**.

Water vapors should remain in the boiler with water.

Solution

N. Ye. Zhukovsky proposed to pour some engine oil over water. Water did not evaporate and the heat remained for a long time.

Problem 5.21 Sail

Conditions of the problem

Strong wind can overturn the yacht.

How to create a sail, which will itself, automatically, without any electronics regulate the traction force?

Analysis of the problem

SC: anti-B

SC: Strong wind can overturn the yacht.

Undesirable effect (UE)—overturning of the yacht (**anti-B**).

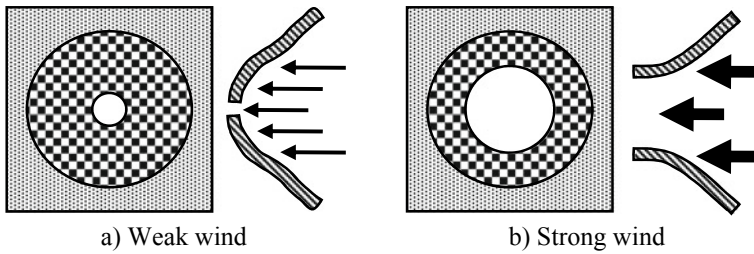


Fig. A.24 The sail

RC: A – anti-B

RC: Wind moves the yacht (A), but can overturn it (**anti-B**).

IFR: A, B

IFR: Wind moves the yacht (A), but does not overturn it (B).

CP: P → A, anti-P → B

CP: Surface area of the sail *should be small* (P), in order *not to overturn the yacht* (B), and *should be large* (**anti-P**), in order *to move it* (A).

Resolution of CP

Resolution of contradiction **according to condition** and **in structure**.

The surface area of the sail varies depending upon the force of wind.

Solution

“A schoolboy from the Moscow region Lyosha Kostenko invented a ‘holed’ sail with an automatically adjustable ‘traction’ force. To a certain extent this sail reminds of an aperture of the photographic camera—an elastic disc with a hole in the middle. When the wind is weak (Fig. A.24a) a small hole in elastic material (or in an insert) is sufficient. In case of a strong wind (Fig. A.24b), able to overturn the yacht, the sail gives in to pressure, increasing the hole and the resistance to wind is decreased”.¹⁵

Problem 5.22 The Buoy

Conditions of the problem

A buoy is floating on the surface of fluid in a vessel. It is necessary that with the cooling of fluid, the portion of buoy immersed in it, should increase. What’s to be done?

¹⁵ Elshansky I. I. I want to become a Kulibin.—M.: “Drofa publishing house”, 2008, 208 p: ill. ISBN: 978-5-3580-1557-9, 978-5-358-04282-7 (In Russian).

Analysis of the problem

SC: A

SC: It is necessary that the buoy should be immersed in the liquid differently depending upon the variations in its temperature. Positive effect (PE)—immersion of the buoy depends upon the temperature of fluid (A).

RC: A – anti-B

RC: The depths of immersion of buoy in the liquid varies depending upon the variation of its temperature (A), however, usually the buoy does not do (**anti-B**) that

IFR: A, B

IFR: The depths of immersion of buoy in the liquid varies depending upon the variation of its temperature (A).

The depths of immersion depends upon the weight and volume of the buoy.

To change the weight depending upon temperature is much more difficult than to change volume.

Weight may be varied, for example, by vaporizing the substance or by burning it, therefore, it is easier to use changing the volume depending upon temperature.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The buoy *should change its weight and volume depending upon the temperature of fluid, in which it is immersed (P)*, in order to change the depths of its immersion (A), and *shouldn't change its weight and volume (anti-C)*, since it does not do it (B).

Resolution of CP

Resolution of contradiction **in structure**.

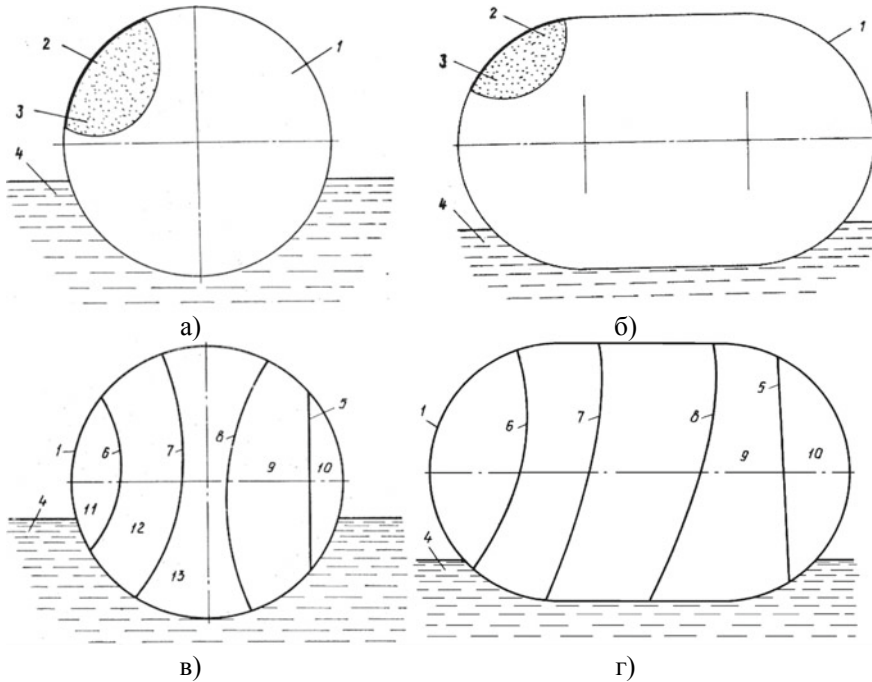
Let us use resources.

Solution

The shell of the buoy is embodied of tight elastic material, which is subdivided into sections (Fig. A.25c and d). Each of the sections is filled with low-melt and low-boiling medium with different temperature of melting and vaporization. The medium may take the form of a hard, fluid, powderlike and gaseous substance.

When the temperature of fluid, into which the buoy is immersed, changes, the substance inside it becomes hard, fluid or gaseous. The volume changes, and, accordingly changes the Archimedes force, acting upon the buoy. The buoyancy of the buoy will vary depending upon the temperature of the fluid.

For example, in case of using water as medium 3, filling the volume 2 of the shell, its freezing will lead to the increase in the volume of the buoy by 0.09, while in



1 – elastic shell; 2 – restricted volume; 3 – low-melt and low-boiling medium; 4 – fluid, into which the buoy is immersed; 5 – 8 – partitions; 9 – 13 – parts (sections).

Fig. A.25 Buoy. Author's certificate 1 314 175. 1—elastic shell; 2—restricted volume; 3—low-melt and low-boiling medium; 4—fluid, into which the buoy is immersed; 5–8—partitions; 9–13—parts (sections)

case of its vaporization will lead to the increase in the corresponding volume several hundred times.

Figure A.25a and c shows the buoy at the assigned temperature of fluid, into which the buoy is immersed; it is of spherical shape. Figure A.25b and g the size of the buoy is increased and it has the shape of ellipsoid. The volume of the buoy in Fig. A.25c and g is separated by interstices

This buoy may float or drown depending upon temperature of fluid, into which it is immersed.¹⁶

Problem 5.23 Training facilities in the streets

Conditions of the problem

Nowadays training halls appear in many cities directly in the streets. It is impossible to install standard facilities, with which the training halls are usually equipped,

¹⁶ A. c. 1 314 175.

in the streets. First of all, under severe conditions they will immediately be spoiled, and secondly, they are very expensive.

That's why the facilities should be as simple as possible (one shouldn't use electronics, engines, electricity, etc.), however, at the same time they should offer an opportunity to regulate the load.

What's to be done?

Analysis of the problem

SC: A

SC: It is necessary to install controllable exercise machines in the streets. Positive effect (PE)—controllable training facilities (**A**).

RC: A – anti-B

RC: Controllable exercise machines (**A**) require electronic and electric equipment (**anti-B**), it is not reliable in the street and expensive.

IFR: A, B

IFR: Controllable exercise machines (**A**), which don't use expensive electronic and electric equipment (**B**), in order to be reliable.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: An exercise machine *should use* electric and electronic equipment (**P**), in order to control the loads (**A**), and *shouldn't use* electric and electronic equipment (**anti-P**), in order to be reliable (**B**) under any weather conditions in the street.

Resolution of CP

Resolution of contradiction in structure.

Let us use resources.

The easiest thing is to use the force of gravity.

The equipment should use the weight of the human.

Solution

The exercise machines are designed in such a way that a human can raise its own weight (Fig. A.26).

Problem 5.24 Collecting the scattered toys

Conditions of the problem

While playing, the children throw the toys around. Some of them are rather small, for example, parts of Lego.

How can one quickly collect the scattered toys without having lost a single one?



Fig. A.26 Street exercise machines

How to quickly collect scattered toys without having lost a single one?

Analysis of the problem

SC: A

SC: How to quickly collect scattered toys? Positive effect (PE)—collecting toys (**A**).

RC: A – anti-B

RC: Collecting all scattered toys (**A**) requires a lot of time (**anti-B**), and there is probability that a certain part will get lost.

IFR: A, B

IFR: All toys are collected (**A**) quickly (**B**) and without losses.

CP: $P \rightarrow A$, anti- $P \rightarrow B$

CP: The toys *should be collected all in one time (P)*, in order *not to spend too much time on that (B)*, and *should not be collected one after another (anti-P)*, in order to be collected entirely (A).

Resolution of CP

Resolution of contradiction **in structure**.

Use resources.

Solution

The children play on the fabric, which is arranged on the floor in the form of a carpet. A rope is run through the edges of the fabric and when the playing is finished, the ends of the rope are pulled together, transforming the fabric into a sack, inside which all toys are placed.

Problem 5.25 Burglars

Conditions of the problem

A flat in the outskirts of the city. The father of the family is at work till late at night, mother is always absent visiting shops, laundry and other places. Often only a little 6-year old girl stays at home.

A couple of local robbers learned about that and decided to rob such an easily accessible flat.

At the moment when the mother of the family went shopping, the burglars rushed into the flat. Prior to breaking the lock, they decided to do the control ring at the door. Thin voice of a little girl answered:

- Who is there?
- This is a postman. Is there anybody of the grown-ups at home?
- No, nobody is at home. Only me and Vasska.
- Hmmmmmm, well, I see, and how old are you and Vasska?
- I am six and he recently became three years old.

The burglars were pleased with such a situation. They quickly broke an old lock and rushed inside the flat. However, a six-year old girl and Vasska, who was three, managed to arrest them.

How did they manage to do it?

Analysis of the problem

IFR: The burglars wait for the coming of police.

Solution

A huge body jumped at one of the burglars from the darkness of the corridor. It pressed the thief to the floor and clutched him with its teeth by the neck, grabbing it in a squeeze. The second burglar stopped dead in horror, looking at a Caucasian shepherd dog, which fell his crime accomplice to the floor. The thing was that Vasska was a three-year old dog, the representative of this particular dog breed, about the presence of which the burglars did not know.

Having received from the housemistress the usual command “Guard!”, the dog lay down in the corridor and hardly had the aliens come into the flat, fulfilled his duty. The fluffy guard caught on bandit with a choking clasp and he became immobile, as a statue, being afraid even to move his finger, neither dead, not alive, captured by the teeth of the dog, and the second simply hid in the corner of the corridor and after the first wild roaring gave up all attempts to run away. They remained in these positions not only till the return of the mother, but also till the arrival of the police.

Problem 5.26 Bringing a steam engine across the river

Conditions of the problem

In the middle of the nineteenth century one Russian industrialist bought a steam engine. When it was rather near the installation site, it appeared that the road is blocked off by the river. The bridge across the river could not withstand the weight of the steam engine. What's to be done?

Analysis of the problem

IFR: The steam engine moves across the river itself.

Solution

History has it that one of the workers cut with his axe a small birch tree, which was growing not far from the place of crossing, made a large number of plugs of it and plugged with them all holes on the heaviest part of the machine—the steam boiler. After that the boiler was pushed down to the water and, as a buoy, tugged to another shore.

Problem 5.27 Biocorrosion

Conditions of the problem

Oil platforms are installed on piles—steel pipes, hammered into the bottom of the sea. In water medium the surface of the pipes is populated with different microorganisms (bacteria, water weeds, microscopic fungi).

Microorganisms emanate acids, alkali and gasses, which contribute to biocorrosion and pitting of pipes.

The load carrying capacity of piles decreased thereby. One can assign the divers to do the removal of microorganisms, however this work is labor intensive and expensive.

What's to be done?

Analysis of the problem

IFR: Microorganisms remove themselves due to natural forces.

Solution

It is necessary to use resources.

What natural phenomena could be used for removal of microorganisms? What phenomena in the sea could generate a force for removal of microorganisms from the piles?

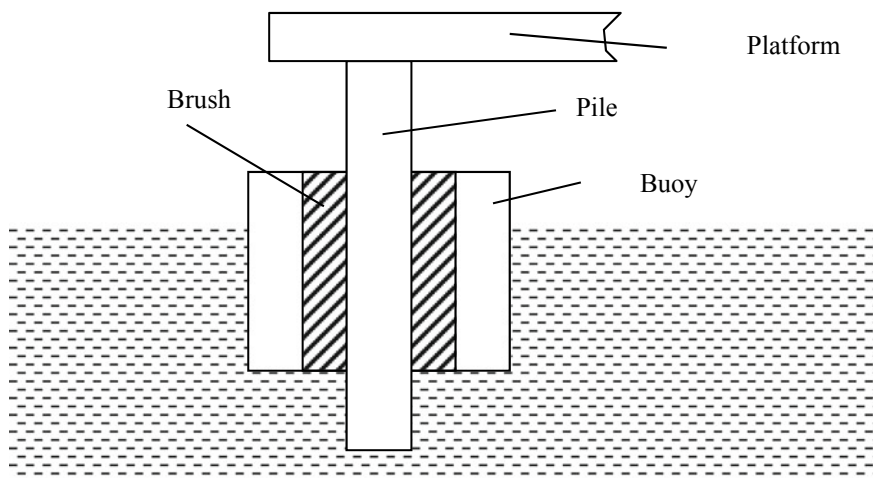


Fig. A.27 Ring brush on the buoy

There are waves, high tides and low tides in the sea.

A ring-shaped brush was invented, which embraces the pipe from the outward side, which is located on the buoy (Fig. A.27).

Oscillation of water raises and lowers the brush, which does not allow the formation of microorganisms.

Problem 5.28 The Pit

Conditions of the problem

In order to arrange for reliable electrical earthing, it is necessary to dig a pit in the ground. In winter, especially in the North, it's very difficult, since the ground gets frozen.

What's to be done?

Analysis of the problem

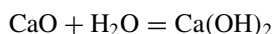
Perform the complete analysis of this problem independently.

IFR: The ground gets softened itself.

For this purpose, it is possible to use corresponding physical and chemical effects.

In the North during winter time it is possible to warm permafrost layer with sunbeams. For this purpose, a sack of polyethylene film is made in the shape of the lens and then it is filled with water. Water gets frozen and a lens is obtained. In summer the Sun shines for 24 h. The lens concentrates the energy of sun beams, warming the soil.

The simplest method is to use chemical exothermic reactions with the release of a large amount of heat. For example, through interaction of calcium oxide (burnt lime) with water.



This is an exothermic reaction, it develops with the release of 16 kcal (67 kJ) per mole.

Solution

The arrangement for electrical earthing in winter will be significantly easier, if on the eve of the day, when the work is performed (in the evening), the ground in the selected place is cleared from snow, covered with burnt lime and then again covered with snow. Integrating with snow (water) burnt lime releases heat, which is sufficient for the ground to be softened by morning and to become accessible for digging even at 20° frost.¹⁷

Problem 5.29 Construction in the North

Conditions of the problem

Propose a certain construction material, which would be reliable under the conditions of North, which would enable to create different structures out of it, including water ducts and which would require minimum expenditures thereby.

Analysis of the problem

Perform the analysis of this problem independently.

Solution

In very cold regions ice is the most profitable building material. It is proposed to use it for manufacturing of: pipelines by placing a light inflatable hose (sleeve) in the trench and pouring a layer of water over it (author's certificate 1,146,360), different structures (author's certificate 861,164), including sea-going ships.

In 1942 an idea appeared in England to create an aircraft-carrier from a drifting iceberg. Such an aircraft-carrier should be at bottom inexpensive. Because of the fact that it is a continuous block of ice, it is not afraid of torpedoes and bombs. As a result of joint effort of England and Canada such an ice ship with the deadweight of 2 million tons was built. It had the shape of a parallelepiped with the wall thickness of 9 m and its height over water was 15 m. In its upper part there was a flight strip the area of which was $600 \times 500 \text{ m}^2$. Sixteen refrigeration installations, which maintained the temperature of the walls of about -15°C . were mounted on the ship. Owing to the functioning of 20 engines, the power of each of which was 20 thousand horse powers, the iceberg could travel at a speed of 7 knots. All superstructures on it were built of a mixture of ice with wooden sawdust: this material is 4 times stronger than ice, is characterized by ductility and by approximately the same anti-explosivity as concrete.

¹⁷ O. G. Verkhovtsev, K. P. Liutov. Practical advice for amateur electricians and lovers of gadgetry.— L: Energoatomizdat. Leningrad department, 1984, 200 p., ill, p. 5–11 (In Russian).

A.4 Modeling

Perform the analysis of problems independently.

A.5 Complex Use

Problem 7.1 Spies

Conditions of the problem

During the cold war period hundreds of American spies were caught in the USSR out of a silliest possible reason.

Falsified passports were the main tools for the spies. That's why during the process of falsifying it was necessary to give attention to any details, even the smallest ones.

What detail betrayed the American spies?

Analysis of the problem

Perform the analysis of this problem independently.

Solution

The Americans neglected something, which was at a glance unimportant. However, it led to disclosure of hundreds of American spies, already inserted in the USSR. This "something" was such an unimportant thing as a paper clip pinning together the pages of the passport!

Paper clips, which were used in soviet passports, began to corrode rather early. At the same time in the USA they used paper clips made of stainless steel for creating forged passports. It was evident that such paper clips could not corrode. And it became a kind of a sign for Soviet officials, which told them that the passport could be a fake.¹⁸

Problem 7.2 Theater

Conditions of the problem

1920-ies, one of the cities of Europe. The opening of the theater, which was long awaited for and the building of which took too much time. The first performance. The curtain opens... But what happens? The spectators are silent trying to hear, what they are saying there, on the stage, and once again, having heard nothing, inadvertently started to murmur unsatisfied at the inability to hear anything. Wherever the spectator was sitting, in the stalls or in the balcony, it was necessary for the spectator to hear every word. While in this case not only words, whole phrases were lost! No, there was no fault with the actors. Maybe, the fault was with the builders: they could ignore the laws of architectural acoustics? No, the architect, who had designed the theater hall, knew all the subtleties of behavior of sound waves.

¹⁸ URL: <https://facte.ru/history/21592.html#ixzz37SJGZ11O>.

Then, where is the reason for that? And what's to be done in the given situation?

Analysis of the problem

Perform the analysis of this problem independently.

Solution

The architect could not foresee the changes in the fashion. During the years when the theater was being built, the ladies wore bouffant skirts with crinolines. When the theater was opened the bouffant folds of silk, which absorbed the sound, were outmoded. They were replaced with short and also narrow clinging dresses. Since the very first performance both the actors and spectators started to complain: the acoustics was bad! One could talk about rebuilding the hall! However, a much more simple outcome was found: in several rows of the stalls the space between the legs of the chairs was laced up with velvet. The spectators stopped to complain about bad acoustics, about the words "eaten up".¹⁹

System approach is used, as well as resources and modeling.

Problem 7.3 On the Swamp

Conditions of the problem

A detachment of French soldiers during the march in Algeria once found themselves in a place, which was completely devoid of plants and with soil, which was so swampy, that, though it was possible to step on it, to sit on it was completely impossible. The tired detachment was moving forward in search of appropriate place for rest, however, this swampy soil spread in all directions for dozens of miles. How to arrange for having rest, when there isn't a single dry place around and nothing, which one could lay on the ground as a blanket or on which one could sit?

And still one happy thought dawned upon one soldier, which helped the detachment to get out of a trying situation. The soldiers sat down and had their rest.

How did they manage that? Please, guess!

Analysis of the problem

Perform the analysis of this problem independently.

Solution

The soldiers sat... on the knees of each other! They formed a circle and each soldier sat on the knees of his neighbor. Maybe, you think that the first soldier had to sit on the swampy soil? Nothing of the kind—with the circular arrangement there is no such "first" soldier: each of them leans on the knees of his neighbor, and the ring of the sitting soldiers closes. If this seems dubious for you, try with several dozens

¹⁹ A. Vladimirov. The Golden strings.—M.: Detskaya literatura publishers, 1977, p. 38, 41 (In Russian).

of friends to sit in such a way, forming a ring. You will be really convinced that an inventive soldier really found a way out.²⁰

Resources, modeling, geometrical effect.

Problem 7.4 The Hedgehog and the Hare (a folk tale)

Conditions of the problem

The hare and the hedgehog once had a dispute, who would quicker run along the trench and backwards. The hedgehog came home and told his wife about that. She scolded him a lot, but agreed to help.

How did the hedgehog manage to win the dispute?

Analysis of the problem

Perform the analysis of this problem independently.

Solution

Next day the Hedgehog stood at the beginning of the furrow, while the Hedgehog's wife occupied the position at the end of the furrow, only with her back to the Hare, so that the Hare wouldn't recognize her. The Hare reached the end of the furrow, and the wife of the hedgehog shouts: "I came here first!". While the Hedgehog was quietly waiting for the Hare at the beginning of the furrow. Thus the hedgehog with the aid of his wife won the dispute.

Principles: 5. Merging. 24. Intermediary. 26. Copying.

Trend of transition to supersystem. Bi-system is used.

Problem 7.5 Customs

Conditions of the problem

A customs official of the customs, where the control over the goods sent abroad was effected, became suspicious about plastic bowling balls produced by one of the companies. They had exactly the same weight as the wooden balls of the same size. The balls were not massive, but the walls were equally hard in all places. The customs official thought that there is a cavity inside each ball, where one could hide the goods being smuggled. And really, with the aid of a most simple experiment, which did not involve special devices, the customs official stated that the smuggled goods are hidden in one of 12 spheres. When the ball was opened, there was a diamond article of jewelry there. How did they manage to identify this sphere?

Analysis of the problem

Perform the analysis of this problem independently.

²⁰ Ja. I. Perelman. Cheerful problems. Two hundred puzzles (in Russian).

Solution

The customs official put the spheres into a bucket of water. One of the spheres was unsteadily dancing on the water surface—its center of gravity was not in the center of the sphere. The jewels were hidden exactly in that sphere.

Problem 7.6 A tub of wine**Conditions of the problem**

Once a landlord called a poor man and said that he would richly reward him in case he would be able to find out how much wine there is in the tub (more or less than a half) and will not use either stick or rope; in general he will not lower down anything into the tub. The poor man simply slanted the tub and told the landlord that the volume of wine amounts to a little bit less than half of the tub.

How did he learn about that?

Analysis of the problem

Perform the analysis of this problem independently.

Solution

When the poor peasant tilted the barrel in such a way that the level of fluid touched the edge of the barrel, he saw a small section of the barrel bottom; it meant that the amount of wine was a little bit less than the half of the barrel. If the amount of wine was more than half of the barrel, it would not be possible to see the bottom. If you don't believe me, check it yourself ...

Resources. Effects. Geometrical effects.

Problem 7.7 Robert Surcouf**Conditions of the problem**

Famous corsair Robert Surcouf during his youth was engaged in slave-trading, however, during the French Revolution slavery was abolished. He understood that it was more profitable, and (what was the most important thing) still more noble to be a corsair than to carry slaves from Africa in the lower holds of the ships.

However, in order to get a corsair patent,²¹ it is necessary to post a bail, which Surcouf did not have and he decided to earn some money by slave-trading. He went to Africa for the last time to get slaves, all the more for the fact that after the prohibition of slave-trading their cost grew quit a lot, since the plantation owners badly needed slaves for growing sugar-cane.

Once the governor was angry with Surcouf because of something. Hardly had the ship gone into the sea, when the order followed: upon returning the ship had to be detained and the captain had to be arrested as slave-trader and repeat violator of Human Rights Declaration.

²¹ Corsairs (øp.corsaire)—pirates, who have the documentary permission of French government permission to capture enemies' ships. The corsairs were officially at the service of the King's Army. A certain part of the booty was given to the King's treasury. If the corsairs attacked the ships of their own country, they were prosecuted and sentenced to death.

A friend of Surcouf told him about the order for arrest. Another man would give up the idea of a trip to Africa and of buying slaves, but Surcouf stuffed the lower holds of the ship with slaves and calmly went to the destination port. Although, it has to be noted that he had taken precautions by sending the slaves to the shore in boats and next day the ship boldly entered the port.

The police was already waiting for Surcouf. Hardly had the crew tidied the lower holds, when the commissary of the police and his helper went on board of the brig and, having examined it, proposed that the captain should follow him to the prison.

What should Surcouf do, in order not to be arrested and not to get into prison?

Analysis of the problem

Perform the analysis of this problem independently.

Solution

Surcouf did not argue, but invited the commissary to have breakfast together with him, since the guests and the host were in the beginning of a long and difficult day. The commissary saw that the food consisted of the pleasures of the table, rare wines and delicious dishes.

While Surcouf was treating the guests to the wines, his assistants sent the commissary's boat, on which the police officials reached the ship, to the shore, pretending that it was the will of the commissary, raised the anchor, set the sails and floated off into the sea. The commissary understood everything, when the rocking of the ship became tangible. The commissary demanded that he would be immediately brought back to the port, but the sailors with pistols, who were standing round him, cooled him off.

Surcouf explained to the commissary that he had to return to Africa to pick up another gang of slaves. He would leave commissary and his assistants there, since they care for the freedom of the Negroes and it would be pleasant for them to spend the remaining days of their life in their company. And meanwhile Surcouf would gladly be at the disposal of his dear guests, who could use his kitchen, wine cellar and other services. At the same time the ship was heading to the open sea.

Surcouf waited until dark and unnoticed turned back. In the evening the storm broke out and it made the commissary more compliant. He tore the already prepared accusation of Surcouf in slave trading and kidnapping of an official and compiled a document, in which he informed the governor that detailed examination of the ship proved complete groundlessness of accusation of the citizen Surcouf of slave-trading. Moreover, when the anchor rope was torn by pure chance and the ship was carried into the sea, the commissar spent several days in the company of Mr. Robert Surcouf and can confirm that he is characterized by high professional and moral features.

Surcouf set his prisoners free only after a week. He stood near the shore and traded with the government of the island until he got complete pardon. Only after that he parted with the commissary.²²

Several constituents of inventive thinking were used in this problem:

1. System thinking.

First all the slaves were let out of the ship. After that Surcouf invited the police commissary to have breakfast with him and sent the policemen to the shore. Then he pretended that he was heading for Africa. After that he made the commissary write a letter about his being innocent and faithfulness and, finally, he conducted negotiations with the government of the island until he got complete pardon.

2. Thinking through contradictions.

Identify the contradictions independently.

3. Resource thinking.

Surcouf told the policemen that they could go to the shore. He took commissary as a hostage, suggested that he would stay in Africa, made him write a letter, stating that there were no slaves on the ship, as well as no signs that there had been slaves and that Surcouf was a faithful citizen. Through demands he got complete pardon for himself from the government of the island.

4. Modeling.

Surcouf in advance modeled the entire situation, described in item 1. System thinking.

5. Development of creative imagination.

It is necessary to have good creative imagination, in order to plan and perform this operation. He also used the principle “Other way round”. Instead of rejecting the idea of traveling to Africa to take slaves, he took the maximum number of them. Instead of being arrested by the police commissary, he arrested the commissary himself. Instead of a situation, when the island’s government dictated to him their conditions, he himself dictated his conditions to them.

Problem 7.8 Capture of the ship

Conditions of the problem

Surcouf captured a caravan of three English ships (two merchant vessels toed the line of the pilot brig), having raised the British flag. On the way they grappled with one more merchant vessel. He changed for pilot brig with the team of less than twenty people and he sent his own ship and the remaining crew to escort the captured ships.

²² The problem is compiled based on the following works: *I. Muromov*. 100 great adventurers.—M.: Veche, 1999. ISBN: 5-7838-0437-1, *T. Archangelskaya*. Surcouf, the Terror of the Seas—M: Nedelya, 1968, No. 35 and *V. Potapov*. Surcouf, Robert (In Russian).

Next day the weather was hot and sunny. Surcouf saw an anchored ship under the British flag, which was armed with numerous guns. It might seem that Surcouf had to hurry to the open sea, but he decided to extract profit from this obviously unprofitable situation and captured a battleship.

How did he manage to do that?

Analysis of the problem

Perform the analysis of the problem independently.

Solution

They immediately recognized the pilot's brig at the British battleship, therefore there wasn't even any necessity to raise the British flag for camouflage. Using the signals, the brig was asked to come closer, in order to impart the news. It was the middle of the day and the majority of the team of the battleship were downstairs, hiding from the blinding midday sun. There was practically no wind. Surcouf understood that his daring plan is proving to be more than successful. The pilot's brig approached the very board of the battleship and Surcouf, leading the group of twelve pirates unexpectedly jumped to the deck of the British ship.

First the pirates banged down the hatches, imprisoning the team downstairs and disarmed the watchmen. One hundred and fifty persons became prisoners of twenty men.²³

Several constituents of inventive thinking were used in this problem:

1. System thinking.

At first the pilot's brig was captured. Then it was recognized by the sailors from the British battleship. Using this opportunity, they came close to the board of the battleship. Before that the attack was planned taking into account hot sunny weather. People of Surcouf first banged down the hatches, leaving the entire team downstairs. Only the watchman were on the deck and it was easy to arrest them.

2. Thinking through contradictions.

Identify the contradictions independently.

3. Resource thinking.

Using the camouflage in the form of the pilot's brig, hot sunny weather and the factor of unexpectedness.

4. Modeling.

Surcouf modeled in advance the entire situation, which is described in item 1. System thinking.

²³ Problem is compiled based on the motives of the book: *I. Muromov. 100 great adventurers.*—M.: Veche, 1999. ISBN: 5-7838-0437-1 (in Russian).

5. Development of creative imagination.

One must have good creative imagination, in order to plan this operation and to carry it out. He also used the principle “The other way round”. Instead of escaping to the sea, far from the battleship, he attacks it.

Problem 7.9 The Spanish Armada **Conditions of the problem**

During dozens of years the British privateers²⁴ robbed and sank Spanish ships. Thus, only in 1582 the losses of Spain amounted to a fantastic sum exceeding 1,900,000 ducats.

Spanish monarch Philipp II considered it to be his duty to help the British catholics in their struggle with protestants.

The Spanish Armada was formed according to his order. It was a great navy. It consisted of 130 ships, 2430 guns, 30,500 men, out of which 18,973 soldiers, 8050 sailors, 2088 rowing slaves, 1389 officers, gentry, priests and doctors. The main forces of the fleet were subdivided into 6 squadrons. The Armada also included: 4 Neapolitan galleasses (big galleys)—635 men, 50 guns, 4 Portuguese galleys—320 men, 20 guns, many light ships for scouting, dispatch boats and vessels with supplies.

The Spanish king gave the order to the Armada to reach the Channel, align with the 30-thousand army, located in Flanders (Spanish Netherlands), cross the Channel and to march to London. Phillip II expected that the British Catholics will leave their protestant Queen and will pass to his side. The plan of the Spaniards did not take into account two most important circumstances: the power of the British fleet and the shallowness of waters, which did not allow the ships to approach the shore and to take the army on board.

The British, in their turn, also hoped for the decisive victory, which would open for Britain the way to free use of the sea, would break the monopoly of Spain for trade with the New World and would contribute to dissemination of Protestant thought in Europe.

The Spanish fleet consisted of unwieldy ships with numerous guns of small range of fire adapted to boarding fight. The British ships were more maneuverable and were equipped with a large number of guns with long range of fire. The Englishmen expected that they would not approach the enemy closely and will destroy him at the distance.

The Spanish admiral arranged his fleet as a crescent, placing at the edges the strongest battleships with guns characterized by long shooting range.

The British could not manage to violate the fighting order of the Spanish fleet.

Waiting for the army, the Spanish admiral ordered the fleet to anchor the ships.

How did the Englishmen manage to violate the fighting order of the Spanish fleet?

²⁴ Privateers—British corsairs—private persons, which had the permission of the authorities (the license) used armed ships for capturing merchant marines of the country, which was in the state of war (*based on the materials of Wikipedia*).

Analysis of the problem

Perform the analysis of the problem independently.

Solution

Having made use of the fact that the Spanish fleet was anchored, the Englishmen at night directed to Armada eight ignited vessels with combustible materials and explosives (branders).²⁵ The majority of Spanish captains cut the anchors and desperately tried to escape from the danger. After that powerful wind and strong tide carried them northwards. They were already unable to return back to the place of meeting the army.

Several constituents of inventive thinking were used in this problem:

1. System thinking.
Absence of system thinking with the Spaniards.
 - The fleet consisted of high-sided hulking ships with numerous cannons with short hitting range, adjusted to boarding fight.
 - The Spaniards did not take into account the power of the British fleet and shallowness of waters, which did not allow the ships to approach the shore and to take the army on board.
 - The Spanish admiral ordered the fleet to anchor waiting for the army.
2. Thinking through contradictions.
Identify contradictions yourselves.
3. Resource thinking.
 - Use of maneuverability of the British ships and the hitting range of the cannons.
 - They used the opportunity that the Spanish fleet anchored its ships and at night directed branders at the Spanish fleet. It led to panic.
 - Powerful wind and strong tide carried the ships northwards and did not allow them to return to the place of meeting with the army.
4. Modeling.
The entire situation, described in item 1. System thinking was modeled in advance.

²⁵ Brander—a ship loaded with easily combustible materials or explosives, intended for igniting or destroying the enemy ships (*material from Wikipedia*).

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